

The Impact of Social Insurance on Household Debt *

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Abstract

This paper investigates how the expansion of social insurance affects households' accumulation of debt. Insurance can reduce reliance on debt by lessening the financial impact of adverse events such as illness and job loss. But it can also weaken the motive to self-insure through savings, and households' improved financial resilience can increase access to credit. Using data on 10 million people and a quasi-experimental research design, we estimate the causal effect of expanded insurance on household debt, exploiting ZIP-level heterogeneity in exposure to the staggered expansions of one of the largest US social insurance programs: Medicaid. We find that a 1 percentage point increase in a ZIP code's Medicaid-eligible population increases credit card borrowing by 0.38%. Decomposing this effect in a model of household borrowing, we show that increased credit supply in response to households' improved financial resilience drives the rise in borrowing and contributes 30% of the net welfare gains of expanding Medicaid.

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1 Introduction

Unsecured debt is an important tool that households rely on to smooth their consumption, not only across time but also across states of the world. For example, of the 4 in 10 US adults that report anticipating difficulty in meeting an unexpected \$400 expense, credit cards are the most-cited tool that they expect to rely on.¹ And when experiencing an income shortfall, 43% of US households report turning to borrowing, including credit cards.² When unsecured debt functions as a de facto source of insurance, changes to social insurance could significantly affect households' desire and ability to access credit.

Whether social insurance crowds in or crowds out household debt is theoretically ambiguous. The total effect depends on the strength of competing channels. On the one hand, improved access to social insurance can reduce households' reliance on debt to cope with adverse shocks such as job loss and illness, crowding out debt. On the other hand, credit demand and supply channels may also work in the opposite direction. Social insurance can increase the demand for debt by weakening the precautionary savings motive. Additionally, when social insurance makes households less likely to default, this reduction in risk can make creditors more willing to lend. Expanded insurance can improve households' financial resilience, which can in turn increase access to credit.

This paper sheds new light on how credit markets shape the impact of social insurance on household debt and welfare. We first estimate the causal effect on credit card borrowing of the expansion of Medicaid under the Affordable Care Act (ACA), one of the largest changes to the US social safety net in recent years. We find that a 1 percentage point increase in the share of a ZIP code's population eligible for Medicaid increases revolving credit card balances by 0.38%. Next, we develop a heterogeneous-agent model in which households face both income and medical expenditure shocks while having the ability to borrow and default on unsecured debt. Simulating the expansion of Medicaid in the model, we find that increased credit supply more than accounts for the overall increase in household debt. Moreover, the credit supply response to expanding Medicaid accounts for 30% of the net welfare gain.³ Our findings demonstrate that expanding social insurance can *crowd in* the supply of unsecured debt and that this improvement in credit access can lead to first-order welfare gains. We conclude that policymakers may significantly underestimate the benefits of social insurance programs if they overlook potential increases in credit access spurred by improved financial resilience.

Our focus on health insurance is motivated by the importance of medical expenses as a source of financial distress for US households. Unexpected medical bills recently ranked first among expenses that US residents report worrying most about affording, ahead of rent or mortgages, transportation, and food.⁴ Unanticipated medical expenses are indeed common and large: 27.7% of US households

¹Source: 2019 SHED.

²Source: 2016 SCF.

³Our measure of net welfare reflects improvements stemming from the financial benefits of the Medicaid expansion (net of the tax costs of expansion). We do not model either mortality or utility as a function of health. Therefore, our welfare estimates do not reflect gains from improved health or reduced mortality.

⁴Source: Kaiser Family Foundation Health Tracking Poll (February 2020). Link: <https://www.kff.org/health-costs/poll-finding/data-note-public-worries-about-and-experience-with-surprise-medical-bills/>.

experience a major out-of-pocket medical or dental expense shock each year, with a mean cost of 8.5% of household income (Fulford and Low, 2024). And among bankruptcy filers, 29% named medical bills as the reason for their bankruptcy (Himmelstein, Thorne, Warren and Woolhandler, 2009). Health expenditures also constitute a large and growing share of US GDP, reaching 18.3% in 2021.⁵ Despite the importance of health expenses, they have received less attention in the macroeconomics literature compared to other sources of risk such as job loss and housing.⁶

We begin by estimating the causal effect of Medicaid eligibility on a variety of credit outcomes using credit bureau data on 10 million US borrowers over 2010-2021 in a continuous difference-in-differences (DiD) analysis. We exploit granular variation in the *size* of eligibility changes induced by states' staggered expansions of Medicaid under the ACA. This variation arises from two sources. The first comes from states' pre-expansion Medicaid policies. Expanding Medicaid under the ACA required states to implement a common set of eligibility requirements. As a result, ZIP codes with similar distributions of income saw larger increases in eligibility if their state previously had stricter requirements. The second source of variation comes from within-ZIP differences in the distribution of income among low-income households. Eligibility increased more in ZIP codes that had more low-income households clustered between the pre- and post-expansion income eligibility thresholds.

Our DiD estimator compares credit outcomes before and after expanding across ZIP codes with bigger versus smaller changes in eligibility. The within-state variation in the impact of expansions makes it possible to use county-time fixed effects in our analysis, which help account for other unobserved state- or county-level policy changes that may have coincided with expansions and also affected credit outcomes. Identification requires that the change in eligibility is not correlated with other shocks coinciding with the expansions.

We find that expanded Medicaid eligibility increases credit card borrowing. A 1 percentage point increase in eligibility leads to a 0.08 percentage point increase in the share of households with at least one credit card and raises revolving (i.e., unpaid) credit card balances by 0.38%. Consistent with a positive credit supply response, we document that expanded eligibility also increases credit limits. On the demand side, we find that the number of credit card inquiries rises. In line with our financial resilience hypothesis, we show that expanding Medicaid also reduces both severe forms of default (debt in collections) and minor forms (delinquency), and increases credit scores. Using individual-level data and imputed effective credit card APRs, we also show that becoming eligible reduced average APRs by 4.9 percentage points.

In order to speak to welfare and the aggregate impacts of expanding Medicaid, we construct a heterogeneous-agent model in which households have access to credit card debt. The need for a model to quantify aggregate impacts arises from the "missing intercept" problem affecting our DiD estimates. These estimates identify the impact of the expansion on cross-sectional differences in outcomes across ZIP codes (the "slope" with respect to the change in eligibility), but do not capture any component of the policy's effect that is common across ZIP codes (the "intercept"), as such an effect is absorbed by the time-varying fixed effects.

⁵Source: Centers for Medicare and Medicaid Services NHE Fact Sheet.

⁶Notable exceptions include Livshits, MacGee and Tertilt (2007); Chatterjee, Corbae, Nakajima and Ríos-Rull (2007); Attanasio, Kitao and Violante (2010); De Nardi, French and Jones (2010).

In the model, households face idiosyncratic income shocks as well as idiosyncratic medical expenditure shocks. They can save using a risk-free asset or borrow via credit card debt, which they can decide not to repay. Households incur debt both by choosing credit card borrowing and as a result of experiencing medical expenditure shocks that they are unwilling or unable to cover on impact. The interest rate paid on credit card debt is endogenous and depends on the probability households do not repay their debt.

The nature of credit card debt in our model is a hybrid of one-period and long-term debt used in the literature on consumer bankruptcy and sovereign debt.⁷ When households are not delinquent, they rollover their credit card and repay it in full. However, households have the option not to repay their debt and become delinquent. When delinquent, households cannot take on more credit card debt before paying their current debt in full, but uninsured medical expenditure shocks can continue to add to their total debt. Delinquent households get a stochastic haircut on their debt, which can help them exit the delinquency state. Since financial intermediaries who hold claims on delinquent debt are not repaid immediately, the pricing of debt has a long-term component to it.

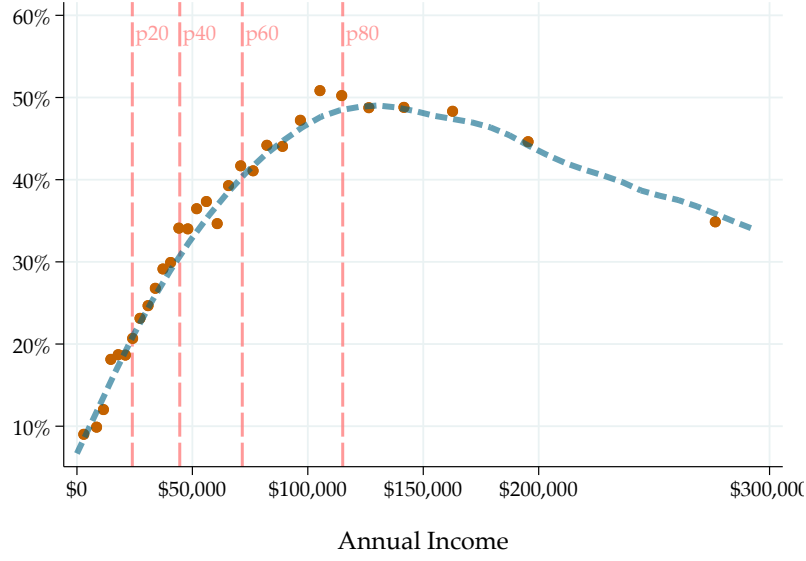
The delinquency option on debt allows us to capture a key aspect of the data. In a standard incomplete-markets model, low-income households sit at the bottom of the asset distribution and borrow the most. In the data, the opposite holds at the bottom: the relationship between having unpaid credit card balances and income follows an inverse U-shape, as displayed in Figure 1. Less than 25% of households with annual income below \$25,000 have any credit card debt. This share rises to 50% for households with an annual income of \$100,000 and declines as household income rises. In our model, low-income households borrow little not because they do not want to, but because they cannot: delinquency risk, together with endogenous credit supply, restricts their access to credit. This matters for studying policies that target low-income households, such as Medicaid. By reducing default risk, social insurance can improve credit access for the households it targets.

We model Medicaid as a policy that covers a fraction of households' expenditure shocks, where eligibility depends on household income. In the model, expanding health insurance affects the aggregate level of credit card debt through three channels. First, the direct effect of more generous health insurance is increasing households' disposable income. Households can achieve the same consumption levels while borrowing less. At the same time, lower medical bills reduce delinquency, which raises credit card debt: households that avoid delinquency finance their medical bills with credit rather than accumulating medical debt, and less of their debt is written down. A priori, the direct effect of health insurance on credit card debt is ambiguous.

The second channel operates through credit demand. Health insurance affects the demand for credit even if credit terms remain unchanged. Lower medical expenditure risk weakens households' precautionary savings motive, which raises borrowing. It also makes households more likely to repay their debt, so that at unchanged interest rates, debt becomes more expensive in expectation, which discourages borrowing. The effect of the credit demand channel is therefore also theoretically ambiguous.

⁷Notable examples of such models include Chatterjee et al. (2007); Livshits et al. (2007); Arellano (2008) for short-term debt and Hatchondo and Martinez (2009); Chatterjee and Eyigungor (2012) for long-term debt.

Figure 1: Share of Households with Credit Card Debt



Notes: This figure plots a binscatter of the share of households with a non-zero amount of unpaid (i.e., revolving) credit card balances. Data come from the PSID (2011–2019). Annual income is in real 2019 USD.

The third channel is the credit supply channel. The reduction in delinquency rates leads to lower interest rate spreads in equilibrium. Lower interest rates induce households to take on more credit, leading to an increase in the aggregate level of credit card debt.

While we ground our model in the context of health insurance, the economic channels we study can arise in many other contexts. Our model and approach can be readily adapted to study the impact of other types of social insurance, such as unemployment insurance or disability insurance. It could also be adapted to study how changes in risk exposure affect sovereign and corporate credit outcomes. Quantitatively, one could find that expanded insurance either increases or decreases borrowing, as the channels can work in opposing directions, and which dominates in equilibrium may vary with context.

We discipline the exogenous processes in the model directly with data. The income process is estimated from the Panel Study of Income Dynamics (PSID). Medical expenditure shocks, which vary with income, are estimated from the Medical Expenditure Panel Survey (MEPS), as are the out-of-pocket shares under different insurance types, including Medicaid. Medicaid coverage, as a function of income, is estimated from the American Community Survey (ACS). We then estimate the remaining parameters, which govern household preferences and the haircut process, to match the distribution of liquid savings and credit card debt across the income distribution in the PSID, together with the rate at which households enter credit card delinquency and its persistence in our credit bureau data. We simulate the expansion of Medicaid by replacing the pre-expansion relationship between Medicaid coverage and income with its post-expansion counterpart, as measured in the ACS.

We validate the model along three dimensions that are not targeted in the estimation. First, we replicate our empirical analysis within the model. The model reproduces our estimated response of

revolving credit card debt, with a coefficient of 0.42 against 0.38 in the data, and is consistent with the responses of credit limits and delinquency. Second, the distribution of interest rates in the model closely matches the distribution of effective credit card APRs documented by [Nelson \(2025\)](#). Third, the model's market value of delinquent debt, 13.4 cents per dollar, is close to the average price at which credit card debt was sold to debt buyers between 2021–2024.

The model predicts that expanding Medicaid led to a 6.7% overall increase in credit card debt. The direct channel has a small impact on the aggregate level of credit card debt, and the credit demand channel reduces it substantially. Together, these two channels decrease aggregate credit card debt by 5.9%. The overall increase is driven by the credit supply channel, which raises aggregate credit card debt by 12.6%. The increase builds gradually, as debt closes 36% of the gap to its new long-run level in the first year and 78% by year five. In terms of welfare, the unfunded expansion is equivalent to a permanent increase in consumption of 24 basis points. Financing the expansion with a uniform income tax yields a net welfare gain of 22 basis points, 30% of which results from the credit supply channel. Disregarding this channel could therefore substantially underestimate the welfare gains of social insurance programs.

Related Literature: This paper contributes to several strands of literature by bringing a new macroeconomic perspective to the effects of social insurance. First, our model builds on the macroeconomic literature on consumer bankruptcy and default (e.g., [Chatterjee, Corbae, Nakajima and Ríos-Rull, 2007](#); [Livshits, MacGee and Tertilt, 2007](#); [Mitman, 2016](#)). This literature focuses on the drivers of default—in particular, the role of bankruptcy policy. We study a policy that does not directly target bankruptcy or default: public health insurance. This policy can reduce default and increase both credit access and borrowing.

Second, we add to a large macroeconomic literature on heterogeneous-agent models with uninsurable risk ([Bewley, 1986](#); [Aiyagari, 1994](#); [Huggett, 1993](#)). This literature highlights how precautionary savings play a key role in shaping the macroeconomy. Our work studies how insurance provision can reduce the precautionary savings motive and traces the macroeconomic implications. We take a partial equilibrium approach, i.e., the risk-free rate in the economy is taken as given. In this sense, our approach is similar to the work of [Imrohoroğlu \(1989\)](#), [Zeldes \(1989\)](#), [Deaton \(1991\)](#), and [Hubbard, Skinner and Zeldes \(1995\)](#), who also study how insurance affects the precautionary savings motive. We build on these works by taking into account the impact social insurance has on the endogenous debt pricing schedule ("credit supply").

We also build on [Krueger and Perri \(2011\)](#), whose model features contingent contracts, resulting in no default occurring in equilibrium and credit supply *decreasing* when social insurance expands (financial autarky becomes less costly and thus default more tempting). Our model instead features non-contingent defaultable debt and leads to a positive credit supply response, which is consistent with our empirical estimates. Our paper also complements concurrent work showing that unsecured credit is a powerful substitute for unemployment insurance, and access to it lowers the optimal level of UI ([Braxton, Herkenhoff and Phillips, 2024](#)). We instead document a complementarity: expanded health insurance crowds *in* the supply of unsecured debt, leading to more consumption-smoothing

gains than the expansion of health insurance alone would suggest.

Third, we build on an applied microeconomics literature studying the consumer finance consequences of expanding insurance. We document first-of-its-kind evidence on revolving credit card debt and, using our model, quantify the welfare effects and decompose the underlying channels. Prior work on Medicaid expansions finds that they improve financial resilience, reducing bankruptcy, medical debt, missed debt and bill payments, reliance on payday loans, and debt in collections.⁸ On the other hand, [Kluender et al. \(2025\)](#) finds in two randomized experiments that forgiving old medical debt has essentially no effect on financial outcomes. Taken together, as [Kluender et al. \(2025\)](#) discusses, our findings suggest that *upstream* interventions, like expanding insurance, are more effective at improving financial resilience than downstream relief.

The earliest research on the financial health impacts of Medicaid on borrowing found results opposite to ours. [Finkelstein et al. \(2012\)](#) finds enrolling in Medicaid reduced participants' reported rates of either borrowing or skipping bills specifically to pay for medical expenses. We use administrative data instead of survey data and study all credit card borrowing, as opposed to only borrowing to pay for medical bills. We also build on [Hu, Kaestner, Mazumder, Miller and Wong \(2018\)](#), which estimates null effects of Medicaid eligibility on credit card balances in the first year following the 2014 expansions. We do so by studying *revolving* credit card balances (i.e., actual borrowing), using more granular (ZIP-level) variation in eligibility, and by studying a longer horizon—which proves to be important as the initial null effect turns into a large, positive response 2-6 years post-expansion. Our reduced-form results also complement those of [Gallagher, Gopalan, Grinstein-Weiss and Sabat \(2020\)](#), which finds that health insurance increases savings among households experiencing financial hardship. Our study of borrowing is important, as higher saving does not immediately imply lower gross or net borrowing.⁹ Lastly, consistent with credit supply improving, [Brevoort, Grodzicki and Hackmann \(2020\)](#) finds that expanded Medicaid lowered interest rates on credit offers sent via mail. This aligns with our findings of lower equilibrium effective APRs.

Research on unemployment insurance and job security regulations also finds evidence of improved financial resilience (lower mortgage default) and expanded credit access in the form of lower interest rates and higher application approval rates ([Hsu, Matsa and Melzer, 2018](#); [Matsa, Melzer and Zator, 2026](#)). Our paper also relates to [Aaronson, Agarwal and French \(2012\)](#), which documents strong and positive spending and collateralized debt increases among minimum wage workers following rises in the minimum wage. We build on their findings by documenting that revolving credit card debt also rises after a significant expansion of social insurance. We show that this rise is accompanied by a decline in default, improvements in credit scores, increases in credit limits, and lower effective APRs.

This paper is organized as follows. Section 2 presents background on the Medicaid expansions

⁸See [Gross and Notowidigdo \(2011\)](#); [Allen, Swanson, Wang and Gross \(2017\)](#); [Mazumder and Miller \(2016\)](#); [Miller, Hu, Kaestner, Mazumder and Wong \(2021a\)](#); [Gallagher, Gopalan and Grinstein-Weiss \(2019\)](#); [Goldsmith-Pinkham, Pinkovskiy and Wallace \(2025\)](#).

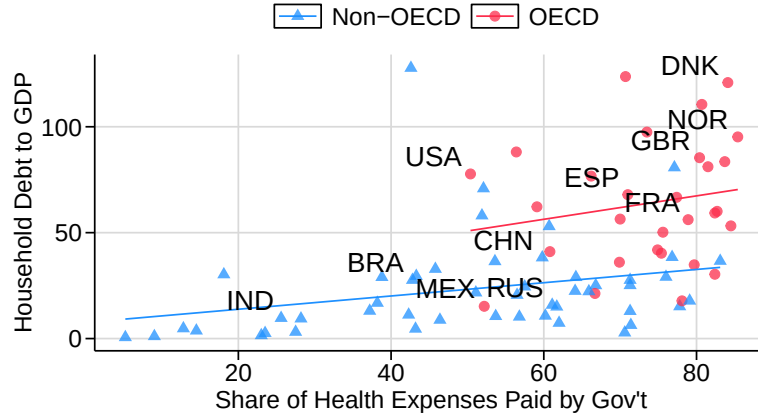
⁹In contrast to standard models of consumption and saving, household borrowing presents a "credit card puzzle" in that households tend to hold both high-interest credit card debt and low-interest savings simultaneously ([Gross and Souleles, 2002](#)). This behavior is consistent with a motive to maintain a liquidity buffer in the presence of incomplete markets ([Telyukova, 2013](#); [Druedahl and Jørgensen, 2018](#)).

and the empirical analysis of the impact of expansions on credit card debt. Next, Section 3 presents the model. Section 4 analyzes policy counterfactuals on debt (its distribution and aggregate level) as well as welfare. We decompose the effect of expanding health insurance on borrowing into its direct impact on debt for households experiencing adverse shocks and its general equilibrium impacts through credit demand and supply. Section 5 concludes.

2 Social Insurance and Debt: Evidence from Medicaid Expansions

At the macro level, cross-country comparisons show that the generosity of social insurance is *positively* related to household debt. Figure 2 plots the ratio of household debt to GDP versus the share of total health expenses paid by the government, a proxy for social insurance provision, across countries. Countries with more generous insurance have significantly higher levels of household debt. At the top right corner are the Scandinavian countries, which have both a high provision of social insurance and high levels of household debt. Viewed through the lens of our hypothesis, one reason Scandinavian households take on higher levels of debt is because they are more financially resilient—they are better insured against adverse events.

Figure 2: Household Debt and Social Insurance across Countries



Notes: Data come from the IMF Global Debt and WHO Global Health Observatory data repository.

This section estimates the effect of expanded social insurance on household debt. We study one of the largest changes to the US social safety net in recent decades: the expansion of Medicaid under the Affordable Care Act (ACA). We find a positive relationship between credit card debt and Medicaid eligibility, as well as patterns consistent with credit supply playing an important role in shaping the equilibrium response.

2.1 Institutional Background: Medicaid

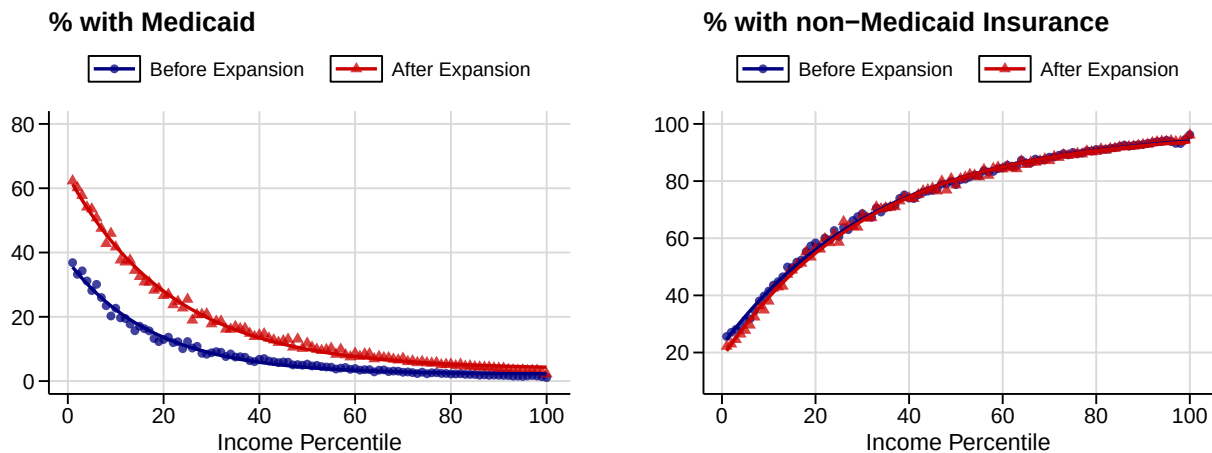
Medicaid is a program, administered jointly by state and federal governments, that offers low-income households free or low-cost health insurance. By the end of 2019, 64.3 million individuals—nearly 20% of the US population—were receiving health insurance through Medicaid. In the wake of COVID-19, the number of enrollees steadily climbed to 85.3 million as of December 2022.¹⁰ Med-

¹⁰We obtain enrollment count data from the Kaiser Family Foundation: <https://www.kff.org/other/state-indicator/medicaid-and-chip-monthly-enrollment>.

icaid spending is a large fraction of total US health care expenses, totaling \$597.4 billion in 2018, comprising 16% of aggregate health expenditures. To qualify for Medicaid, a household must have income below a specified threshold.¹¹

The ACA expanded Medicaid in participating states by requiring these states to set the income eligibility threshold to 138% of the federal poverty level (FPL) for *all* adults. Participating states receive federal funds to support the costs of the Medicaid expansion. Prior to the ACA, only a handful of states offered Medicaid to adults aged 64 or under without dependents. Among states that expanded, the uninsured share of the prime-age population (aged 25–55) fell 42% (from 18.0% to 10.4%) after expansion. Figure 3 plots insurance type by income both before and after expanding for this population. The share of these individuals enrolled in Medicaid rose 8.4 percentage points in expanding states.¹² The crowding out of private and other sources of government insurance appears limited; the fraction receiving non-Medicaid sources of insurance is similar before and after the expansions. Appendix Figure A.1 plots this enrollment growth in event time; enrollment rises steeply in the first two years after a state expands and plateaus thereafter.

Figure 3: Health Insurance Type by Income (Pre- and Post-Expansion)



Notes: The graphs plot the fraction of people aged 25–55 with either Medicaid (left panel) or other non-Medicaid sources of insurance (right panel), such as private insurance or Medicare. We report averages both before (blue circles) and after (red triangles) states expanded Medicaid. We include non-expanding states in the "before" group. Insurance type is measured at the end of the year, while income is measured over the entire year. Note that some "high-income" households may qualify for Medicaid if, for example, they lost their job during the year and were eligible by the end of the year. This, and aggregation across household sizes, time, and states, is likely why the share with Medicaid is smooth with respect to income (as opposed to being discontinuous at eligibility thresholds). Income percentiles are computed within the age-restricted sample. Source: American Community Survey (ACS), 2010 to 2021.

Participating in the expansion is optional, and the timing of adoption varied significantly across states (see Appendix Figure A.2). The first, and most common, year of adoption was 2014. As of 2021, 36 states (and D.C.) had expanded Medicaid eligibility under the ACA. The ACA was primarily financed by cuts to federal government spending on health care, taxes on insurers, and taxes on

¹¹Several states also use an asset-based means test in addition to the income threshold to determine eligibility. To expand Medicaid under the ACA, states were required to remove any asset-based means tests.

¹²These figures are calculated using American Community Survey (ACS) data from 2010 to 2021.

individuals.¹³

2.2 Data

This section describes the two datasets used in our reduced-form analysis. The first is credit bureau data. The second records Medicaid eligibility at the ZIP level. Throughout, we deflate nominal variables to real 2020 dollars.

Experian Data. Our credit bureau data come from Experian and covers a random sample of 10 million US residents. This Experian sample is an annual panel spanning the period 2010 to 2021 and contains over 90 million consumer-level observations. A subset of outcomes are measured at a quarterly frequency over 2012–2020. Our sample is geographically representative in the sense that individuals are randomly sampled from ZIP codes in proportion to their ZIP code’s population share. In our empirical analysis, we aggregate the consumer-level data to the ZIP level. Our analysis sample contains 11,537 unique ZIP codes over 12 years.¹⁴

Experian Summary Statistics. Table 1 presents summary statistics for our main credit outcomes of interest in the ZIP-level sample. On average, 86% of borrowers in a ZIP have at least one credit card, and 21% obtained a new credit card in the past year. Average consumer-level credit card balances are \$4,417. We measure revolving (i.e., unpaid) balances as the difference between average monthly balances and average monthly payments. These monthly averages are available at a quarterly frequency in the Experian data. In our sample, 61% of consumers have unpaid credit card balances at least once per quarter and average consumer-level revolving balances are \$3,776.

Our measure of unpaid balances may be subject to measurement error that depends on when balances are measured relative to the consumer’s scheduled payment date.¹⁵ Average use of revolving credit card debt in our sample is similar to estimates from regulatory data, suggesting that measurement error is close to mean zero.¹⁶ Importantly, this measurement error will *unlikely* bias the

¹³Congressional Budget Office (CBO) estimates projected the ACA to *reduce* budget deficits overall during the period 2013–2022 by \$109 billion (Congressional Budget Office, 2012). Over this period, the ACA’s projected costs to the federal government of expanding Medicaid, the Children’s Health Insurance Program (CHIP), health insurance exchanges, and other non-coverage provisions totaled \$1,455 billion. Health care-related government spending cuts were projected to save \$741 billion over this same time period, accounting for 50.9% of the total cost. The primary source of these cuts was reductions in payment rates for hospital services rendered to Medicare and Medicare Advantage patients and reductions in Disproportionate Share Hospital (DSH) payments. DSH payments help compensate hospitals for the cost of uncompensated care (i.e., missing or partial payments owed by patients).

¹⁴Before any filtering, our data cover 16,780 ZIP codes. We drop ZIP codes with fewer than 150 consumers reporting credit bureau data, ZIP codes that cannot be matched to our Medicaid eligibility data, and ZIP codes missing our control variables; these are mainly small ZIP codes, and 12,664 remain. Our analysis sample further excludes ZIP codes absorbed as fixed-effect singletons, leaving 11,537.

¹⁵Ideally, we would subtract payments from the *statement* balance at the end of the billing cycle. Average current balances will tend to understate statement balances when measured after the end of the billing cycle and overstate them when measured shortly before.

¹⁶Using regulatory data, Adams, Bord and Katcher (2022) report that around 68% of active credit card accounts are used to revolve in a given year and that approximately 75% of credit card balances are revolving at a given point in time. In our sample, 72% of people with cards are revolving at a given point in time (61.35% of people revolve while 85.53% have a card, Table 1). Alternatively, the share of revolvers is 52% in the 2019 Survey of Household Economics and Decisionmaking (SHED) and 45% in the 2019 Survey of Consumer Finances (SCF). However, Zinman (2009) argues that the SCF significantly under-reports credit card debt.

regression analysis. Revolving balances will be a response variable in our regressions, and the timing of a consumer's billing cycle is likely unrelated to their state's Medicaid expansion.

We consider several additional variables that reflect credit access. In our sample, average credit card utilization is 35%, and average limits are \$18,519. Credit limits vary significantly across ZIP codes; the standard deviation is \$8,760. The average ratio of new accounts per credit card inquiry is 0.51, indicating that typically two inquiries are necessary to obtain one credit card. Credit card inquiries per year average 0.43 per borrower.

Table 1: Summary Statistics for ZIP-Level Credit Outcomes

	Mean	SD	25th %	50th %	75th %	N
1[Has CC] (%)	85.53	8.419	80.8	87.14	91.91	129,386
1[New CC] (%)	21.24	4.837	18.02	21.36	24.51	129,386
CC Bal. (All)	4,417	1,756	3,176	4,123	5,342	339,785
CC Rev. Bal.	3,776	1,457	2,755	3,545	4,549	339,785
1[Revolver] (%)	61.35	11.27	53.7	61.81	69.82	339,785
CC Util. (%)	35.17	7.67	29.88	35.2	40.31	129,386
CC Lim.	18,519	8,760	12,125	16,624	23,252	129,386
New CC to Inq.	0.5078	0.1326	0.4208	0.5147	0.6002	129,386
CC Inq. (#)	0.4333	0.1348	0.3353	0.4277	0.5238	129,386
90+ Delinq. (%)	6.993	3.414	4.486	6.601	9.007	129,386
90+ Utilities Delinq. (%)	4.299	3.571	1.952	3.274	5.537	129,386
1[Any Col.]	23.22	11.82	14.12	21.38	30.41	129,386
Col. Bal.	661.2	435.2	348.1	561.7	869.4	129,386
Col. Bal./Income (%)	1.167	1.113	0.409	0.842	1.551	129,386
Vantage Score	686.4	37.75	661.4	687.7	713.5	129,386

Notes: This table reports ZIP-level summary statistics for the main credit outcomes of interest from the Experian data. All variables are reported annually (June of each year) except for credit card balances (total and revolving) and the percentage of people that have revolving balances ("revolvers"). The latter are reported quarterly. We calculate revolving (i.e., unpaid) balances by subtracting credit card payments from total credit card balances. Nominal variables are CPI adjusted to be in terms of 2020 dollars. Throughout, "CC" denotes "credit card." The "Has CC" variable indicates the fraction of borrowers with at least one credit card, while the "New CC" variable captures people that obtained at least one new credit card in the past year. Balances, limits, and utilization are calculated across all credit cards. Utilization and limit calculations include households with zero credit cards. The "inquiries" variables reflect total activity in the past year. The delinquency variables reflect whether borrowers had any debt 30+ or 90+ days past due (excluding collections). The collections variables indicate the fraction of households with some debt in collections (non-medical and medical debt are tabulated separately). Our credit score measure is the VantageScore.

For default, we examine both measures of delinquency and whether consumers have debt in collections. On average, 7% of consumers have some debt that is 90 or more days past due. Delinquency on utility bills is fairly common, with on average 4.3% of consumers being 90 or more days behind on at least one utility bill. These measures of delinquency exclude debt in collections, which is even more common. On average, 23% of consumers have at least some debt in collections.

Medicaid Eligibility Data. We use two data sources to impute Medicaid eligibility at the ZIP level. Adult household members are eligible for Medicaid if their income falls below a threshold that depends on household size and the number of dependents. We impute the Medicaid-eligible share of

the population at the most granular level possible with publicly available data: ZIP code. To do so, we fit a Pareto distribution to characterize the income distribution for each ZIP-year using data from the IRS Statistics of Income (SOI) for 2009–2021. Then, using the joint distribution of income and household size from the 2010 American Community Survey (ACS), we integrate over household size to calculate the eligible population share (see Appendix B for details). Measuring eligibility at the ZIP level enables us to include county-year fixed effects in our regressions, which absorb the effects of local economic trends on the credit outcomes we study. A limitation is that the IRS data only contain tax filers, and households with sufficiently low income are exempt from filing. However, many still file in order to obtain refundable credits or refunds of withheld taxes. We validate our IRS-based measure by comparing it to an ACS-based one.¹⁷ In addition to granularity, another advantage of the IRS data is that they are administratively reported rather than recalled in a household survey, as in the ACS. And the measure of income they report, adjusted gross income (AGI), is the primary component of Modified AGI (MAGI), which is used to determine eligibility for Medicaid.¹⁸

Eligibility Summary Statistics. Table 2 reports summary statistics for our eligibility and income data. In the median ZIP code, throughout our sample period, approximately 20% of households are eligible for Medicaid. Conditional on expanding, the newly eligible share of the population on average increased by 19.45 percentage points within ZIP codes.

Table 2: Summary Statistics for ZIP-Level Eligibility and Income Data

	Mean	SD	25th %	50th %	75th %	N
Elig. (%)	18.5	14.15	3.968	19.85	30.28	119,538
ΔElig.	19.45	10.96	11.38	20.56	27.45	7,272
Avg. HH Inc.	83.09	68.82	52.21	65.36	89.15	129,386

Notes: This table reports ZIP-level summary statistics for the eligibility measures and income. Eligibility is calculated as described in the text, with additional details provided in Appendix B. The first row reports average eligibility across all ZIP codes and years. The second row reports the average change in eligibility in the first year of expansion (this excludes ZIP codes in states that did not expand). Average household income is the average adjusted gross income (AGI) reported in the ZIP code. Nominal variables are CPI-adjusted to be in terms of 2020 dollars.

2.3 Empirical Strategy: Continuous Difference-in-Differences

Identifying the causal effect of Medicaid eligibility is challenging. Eligibility depends on household income, which is likely correlated with omitted variables. Appendix Figure A.3 shows that income is generally positively related to credit access and use at the ZIP level. Comparing credit outcomes of eligible and non-eligible people would likely *understate* the causal effect of Medicaid eligibility, as such comparisons would conflate the causal effect of eligibility with negative effects of low income.

¹⁷We aggregate our IRS-based measure to the public use microdata area (PUMA) level, the finest geography available in the public ACS. We construct an ACS-based measure by aggregating household-level imputations of eligibility to the PUMA level. These two measures have a correlation coefficient of 0.89. In a regression of the ACS measure on the IRS measure, with only an intercept, the R² is 0.79 and the slope is 1.01. The average levels are close as well, differing by less than 4 percentage points.

¹⁸MAGI is generally measured as AGI plus tax-exempt interest, excluded foreign income, and nontaxable Social Security.

To overcome these identification challenges, we exploit quasi-experimental variation in the *size* of Medicaid expansions in a difference-in-differences (DiD) analysis. We draw on two sources of variation. First, we exploit cross-state variation arising from states’ *pre-existing* Medicaid eligibility rules. Expanding under the ACA required states to raise the income eligibility limit to a common threshold of 138% of the federal poverty level for all adults aged 64 or less. States with previously stricter eligibility rules, such as lower income thresholds or rules precluding adults without dependents from eligibility, tended to experience larger changes in eligibility when expanding. Second, we exploit within-state variation due to differences in the *distribution* of income among low-income households. Specifically, ZIP codes where more households had incomes between the pre- and post-expansion income eligibility thresholds experienced a larger increase in eligibility.

Because we have a continuous measure of treatment intensity, we estimate a continuous DiD. The DiD compares ZIP codes with larger versus smaller shares of people that became newly eligible as a result of their state expanding Medicaid. This approach is similar in spirit to that of [Goodman-Bacon \(2018, 2021b\)](#), which use state-level variation in exposure to Medicaid expansions to estimate the impact on mortality. An advantage of using ZIP-level variation is that we can include county-time fixed effects to account for the impact of state- or county-level events coinciding with expansions. Motivated by similar concerns, [Dranove et al. \(2016\)](#) and [Garthwaite et al. \(2019\)](#) also use a continuous ZIP-level DiD to study the impact of Medicaid on health care utilization. We further refine their approach by constructing an eligibility measure that takes into account how Medicaid eligibility cutoffs vary with family size.

We estimate

$$Y_{zct} = \beta (\text{Post}_{st} \times \text{NewElig}_{zs}) + \phi_z + \phi_{ct} + \gamma X_{zct} + \varepsilon_{zct}. \quad (1)$$

Above, Y_{zct} is an average credit outcome in ZIP code z (located in county c of state s) in year t .¹⁹ Post_{st} equals one if state s has expanded Medicaid as of year t . We capture the intensity of treatment with a continuous variable, NewElig_{zs} , which corresponds to the fraction of ZIP z ’s population that became newly eligible for Medicaid over the first year of state s ’s expansion (relative to the year before its expansion). ZIP-code fixed effects, ϕ_z , account for persistent differences across ZIP codes. We also include county-time fixed effects, ϕ_{ct} , to account for the effects of other unobserved time-varying county- and state-level shocks. We include time-varying controls (X_{zct}); our main specification controls for logged ZIP-level income per household and deciles of the ZIP code’s income distribution.²⁰ The coefficient of interest is β on the interaction term, which reflects the causal effect of an expansion that induced a 1 percentage point increase in eligibility. Throughout, we cluster our standard errors at the state level because the decision to expand Medicaid occurs at the state level.

When does OLS estimation of the above equation identify the causal effect β ? The key identifying assumption is that household credit outcomes would have evolved in parallel—across locations

¹⁹Most of our outcome variables are measured annually, but a subset are observed at a quarterly frequency. The quarterly variables are credit card balances (both total and revolving).

²⁰Our results are robust to including fewer or more moments from the local income distribution, such as only controlling for logged average income or for 19 percentiles of the distribution. Appendix Table A.1 reports these results for our main variable of interest: revolving credit card balances.

with high versus low changes in eligibility—if Medicaid had not expanded. Phrased differently, we assume that the size of the newly eligible population is uncorrelated with other factors *changing* at the time of the expansion. This assumption would fail if other policies were implemented along with the expansions that also targeted the newly eligible population. Our design also mitigates this concern through the use of granular variation. County-time fixed effects help account for the impact of other state- or county-level policy changes on credit outcomes.

2.4 Empirical Results: The Effects of Medicaid Eligibility on Household Debt

Credit Card Debt. We first examine the response of credit card debt; Table 3 reports DiD estimation results. We start by documenting positive effects of increased eligibility on the extensive margin of credit card use. A 1 percentage point increase in a ZIP code’s Medicaid-eligible population leads to a 0.08 percentage point increase in the fraction of borrowers that have a credit card. The fraction of households that received a new credit card in the last year grows by 0.15 percentage points in response to the same shock. Per newly eligible person, these correspond to an 8.0 and 14.6 percentage point increase (respectively).

In terms of total credit card balances, average ZIP-level balances increase 0.69% in response to a 1 percentage point increase in eligibility. This corresponds to approximately a \$30.43 increase in average balances per capita.

Table 3: DiD Results for Credit Card Debt

	1[Has CC] (1)	1[New CC] (2)	log(CC Bal.) (3)	log(CC Rev. Bal.) (4)
$\text{NewElig}_{zs} \times \text{Post}_{st}$	0.080*** (0.016)	0.146*** (0.016)	0.689*** (0.071)	0.383*** (0.063)
Obs	129,386	129,386	339,785	339,785
R2	0.972	0.856	0.967	0.952
Mean	86%	21%	\$4,417	\$3,776

Notes: This table reports results from estimating the DiD in Equation (1). Each specification uses county-time and ZIP-code fixed effects and controls for logged ZIP-level average adjusted gross income (AGI) and logged deciles of the ZIP code’s income distribution (coefficients omitted). Note that data are annual in Columns (1)–(2) and quarterly in Columns (3)–(4) (due to data availability). Standard errors are clustered by state. Nominal variables are CPI-adjusted to be in terms of 2020 dollars. The dependent variable is labeled above the column number, and its mean is reported in the bottom row. Statistical significance: 10%*, 5%***, and 1%***.

Next, examining *revolving* balances, we find that a 1 percentage point increase in eligibility increases credit card borrowing by 0.38%. The statistical significance of this result is robust to a Fisher randomization test, which yields a p-value below 0.001.²¹ The borrowing response constitutes a \$14.46 increase in average balances. Per newly eligible person, this corresponds to a \$1,446 increase in credit card borrowing. Additional heterogeneity analyses reveal that the increase in revolving balances is strongest in below-median-income ZIP codes (0.44) while the response is small (0.09) and

²¹Because treatment varies both in timing and intensity, we randomize both of these aspects for our test. We permute the newly eligible share within counties across ZIP codes and the timing of adoption across states among those that ever adopted.

statistically insignificant in above-median ZIP codes (see Appendix Table A.6).

Interpreting Borrowing Magnitudes. While large, the estimated borrowing response is plausible. We next discuss magnitudes and benchmark our estimate through several observations. First, scaling coefficients per newly eligible person likely overstates the response of the newly eligible themselves. This is because other people, who do not immediately become eligible, are also impacted by the expansion. This includes consumers who resemble eligible consumers in the eyes of creditors’ underwriting algorithms (in terms of observables like credit score). It also includes consumers at risk of adverse shocks (like job loss) who would now be able to rely on Medicaid. Our ZIP-level design is meant to capture this total local effect.

Second, the DiD averages effects over seven post-expansion years, while NewElig_{zs} measures the share made eligible *at* expansion. Households flow into eligibility over those years through adverse events such as job loss, so the population ever covered exceeds the initially eligible share. The gradual growth in borrowing found in Figure 4 is consistent with an accumulating effect due to churn in and out of Medicaid over time.

Third, the response is concentrated among households with very low baseline borrowing, where large percentage effects imply smaller dollar effects. Revolving balances respond by 0.438 in below-median-income ZIPs, where they average \$2,870, versus an insignificant 0.086 above the median, where they average \$4,645 (Appendix Table A.6). A negative correlation between baseline borrowing and the response to the policy shock means that we should be cautious extrapolating our cross-sectional estimates to aggregate effects of increased eligibility. This is a key reason we rely on a structural model in Section 4 to derive implied aggregate effects.

Fourth, the magnitudes are in line with related estimates from the wider social insurance literature. Sullivan (2008) finds that a transitory unemployment spell raises unsecured debt by \$1,603 within a single year among households in the second and third deciles of the asset-to-earnings distribution—larger than our \$1,446 per newly eligible person, which accrues over up to seven years. Aaronson et al. (2012) finds that a \$1 minimum wage increase raises quarterly spending by \$1,080 against an income gain of \$318, a marginal propensity to spend of 3.4 that suggests a significant rise in borrowing.²²

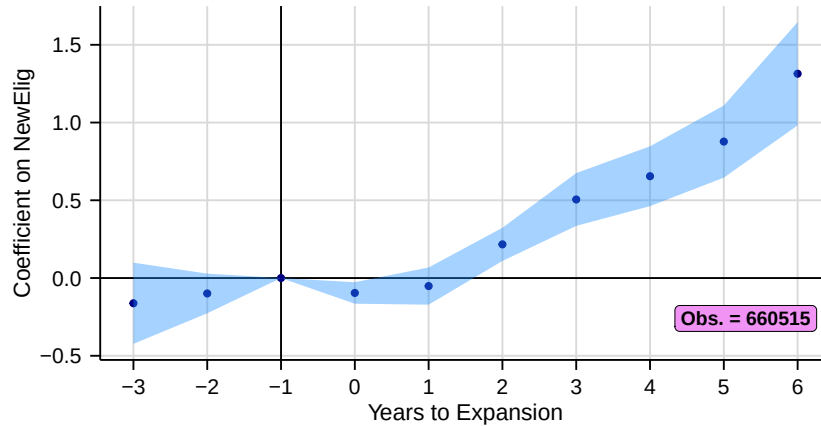
Finally, the response is modest relative to the insurance the expansion delivers. The Centers for Medicare & Medicaid Services (CMS) puts benefit costs at \$6,339 per newly eligible adult enrollee per year. With a take-up rate of 60%, a newly eligible person receives \$26,624 in gross benefits over the seven post-expansion years. Our \$1,446 then amounts to 5.4 cents of additional revolving balances on average per dollar of gross Medicaid benefits received.²³

²²Both converted to 2020 dollars. Sullivan’s estimate is the sum of the unemployment-spell coefficient and its interaction with the second and third asset-to-earnings deciles ($-454 + 1,568 = \$1,114$; his Table 4, Column (4)), in 2002 dollars. Aaronson et al. (2012) reports a \$815 quarterly spending response (their Table 2, Column (1)) against a \$240 income response (their Table 1, Column (4)), in 2005 dollars; the marginal propensity to spend is a ratio and requires no conversion.

²³CMS reports average federal and state benefit costs of \$6,339 per newly eligible adult enrollee per year in 2020 dollars, excluding administrative costs (*Actuarial Report on the Financial Outlook for Medicaid*). Scaling this by the average take-up rate among eligible adults over 2014–2019 of 60% (Abdus, 2025), over seven years, gross benefits per newly eligible person equal $0.60 \times \$6,339 \times 7 = \$26,624$, making \$1,446 equivalent to 5.4 cents per dollar of benefits received.

Pre-Trend Testing and Dynamic Estimates. We estimate a dynamic version of Equation (1), where we replace the post-treatment indicator with a set of categorical variables reflecting years relative to expansion. Figure 4 plots these estimates. Prior to expanding, we do not see economically or statistically significant pre-trends.²⁴ After expanding, revolving balances gradually increase over time. The gradual response is likely due to several factors. First, financial resilience can take time to improve, delaying the responses of credit supply and demand. Indeed, we find that on impact total balances rose while unpaid balances did not, indicating that repayment improved—an important determinant of credit scores. Credit scores begin improving one year post-expansion. The responses of these variables evolve gradually over time as well (see Appendix Figure A.5). Second, the CARD Act of 2009 made it difficult for credit card lenders to update existing credit cards’ interest rates. So most of the new and improved terms would come from new credit cards—delaying the reaction of credit card borrowing. Third, as more households experience medical expenditure shocks over time, the share of people that directly benefit from Medicaid continues to grow.

Figure 4: Dynamic Estimates for Revolving Balances.



Notes: This figure plots dynamic DiD estimates using the stacked estimator of Cengiz et al. (2019) for multiple horizons before and after Medicaid expansions. The shaded area is a 95% confidence interval for standard errors clustered at the state level. The standard two-way fixed effect estimator yields similar results (see Appendix Figure A.4).

The gradual response we find also aligns with results from Hu et al. (2018), which finds a null effect of the 2014 Medicaid expansion on total credit card balances in 2014–2015. Our analysis reveals that studying the longer-run impact of Medicaid expansions is important, as some impacts on financial outcomes emerge gradually.

Credit Demand and Supply Proxies. Next, to shed further light on the role of supply and demand, we examine proxies for these two forces in Table 4. Our proxy for demand is the average number of credit card inquiries per borrower. We estimate that inquiries increase by 0.364 per newly eligible person. This suggests that demand may be contributing to the equilibrium rise in credit card debt.

²⁴The pre-period estimates are small and both individually and jointly (p -value of 0.32) not statistically different from zero. Nonetheless, we further assess robustness to violations of parallel trends. We report confidence intervals under the “honest DiD” approach of Rambachan and Roth (2023) for our main outcome variables in Appendix Section A.4.

Table 4: DiD Results for Credit Card Demand and Supply Proxies

	Demand	Supply		
	CC Inq. (#) (1)	log(CC Lim.) (2)	CC Util. (%) (3)	New CC to Inq. (4)
$\text{NewElig}_{zs} \times \text{Post}_{st}$	0.364*** (0.032)	0.313*** (0.090)	0.011 (0.015)	0.029 (0.032)
Obs	129,386	129,386	129,386	129,386
R2	0.932	0.983	0.950	0.852
Mean	0.43	\$18,519	35%	0.51

Notes: This table reports results from estimating the DiD in Equation (1). Each specification uses county-year and ZIP-code fixed effects and controls for logged ZIP-level average adjusted gross income (AGI) and logged deciles of the ZIP code's income distribution (coefficients omitted). Standard errors are clustered by state. Nominal variables are CPI-adjusted to be in terms of 2020 dollars. The dependent variable is labeled above the column number, and its mean is reported in the bottom row. Statistical significance: 10%*, 5%** , and 1%***.

For our supply proxies, we consider credit limits, the ratio of credit card balances to credit limits (utilization), and the ratio of new credit cards to credit card inquiries (over the past year). Overall, we find evidence consistent with a positive credit supply response to expansion. A 1 percentage point increase in a ZIP code's eligibility leads to a 0.31% increase in credit limits, corresponding to a \$57.96 increase. With both limits and balances rising, the impact on average utilization is not obvious. We estimate a small, statistically insignificant change in utilization (0.011 percentage points). Thus, despite the increase in credit card borrowing, credit limits did not become more binding.

Lastly, we estimate a small and statistically insignificant effect of eligibility on the ratio of new credit cards to inquiries. However, when separately examining ZIP codes with below-median income, we find a statistically and economically significant increase of 0.061 percentage points in the likelihood of a credit card inquiry resulting in a new credit card per 1 percentage point increase in eligibility (Appendix Table A.7). These findings are also consistent with an expansion of credit supply. First, note that the flat, average response of credit card application success rates happens *despite* coinciding with a rise in both the number of inquiries and indebtedness. Additionally, if marginal applicants are less likely to be successful than the average applicant, the flat response of the average success rates would indicate increased success rates for marginal applicants (relative to what they would have experienced absent expanding Medicaid). Moreover, we do find that applicants from low-income ZIP codes do experience higher success rates in response to expanded Medicaid.

We note that each of these proxies reflects equilibrium outcomes. For example, inquiries can be influenced by marketing practices of credit card lenders and households' perceptions of lending standards, making them an imperfect proxy for demand. These limitations further motivate our model, in which we can separately quantify the role of supply and demand.

Financial Resilience. Finally, we turn to measures of default and find evidence indicating that increased eligibility improved households' financial resilience in Table 5. We first examine the likelihood a household is 90 or more days delinquent on any debt, finding that a 1 percentage point

increase in eligibility lowers it by 0.02 percentage points. This indicates that household "debt tolerance" has increased in the sense that, despite households increasing credit card debt overall, they become less likely to enter delinquency.

In order to better gauge the direct effect of insurance on delinquency, separate from indirect effects through changes in debt, we examine 90-day delinquency on utility bills (gas, electric, etc.). In response to expanded Medicaid, households are less likely to adjust consumption of utilities, compared to other goods. This makes utility delinquency a better measure of the direct effect of insurance on financial resilience. In line with improved financial resilience, we find that a 1 percentage point increase in eligibility lowers the rate of 90-day delinquency on utilities by 0.13 percentage points. The point estimate of this decrease is about three times larger in below-median-income ZIP codes (0.15 versus 0.05 percentage points), though the difference between the two subsamples is not statistically significant (Appendix Table A.8).

Next, we study more severe forms of default and find that Medicaid eligibility reduces debt in collections. A 1 percentage point increase in eligibility reduces the likelihood of having debt in collections by 0.12 percentage points. Similarly, the amount of debt in collections relative to average income (AGI) also shrinks by 0.029 percentage points in response to a 1 percentage point rise in eligibility. In relative terms, this constitutes a 2.5% reduction in this ratio.

Table 5: DiD Results for Default and Credit Scores

	1[Delinquency]		Debt in Collections		Credit Score
	1[90+ Days] (1)	1[90+ (Util.)] (2)	1[Any Col.] (3)	Col./Inc. (4)	VantageScore (5)
$\text{NewElig}_{zs} \times \text{Post}_{st}$	-0.022** (0.010)	-0.127*** (0.045)	-0.119*** (0.022)	-0.029*** (0.008)	0.214** (0.102)
Obs	129,386	129,386	129,386	129,386	129,386
R2	0.895	0.911	0.979	0.934	0.985
Mean	7.0%	4.3%	23%	1.2%	686

Notes: This table reports results from estimating the DiD in Equation (1). Column (1)'s delinquency measure reflects delinquency on any kind of debt, not only credit cards. Column (2)'s measure specifically reflects delinquency on utility bills. The collections measures also reflect outstanding collections for any kind of debt. Each specification uses county-year and ZIP-code fixed effects and controls for logged ZIP-level average adjusted gross income (AGI) and logged deciles of the ZIP code's income distribution (coefficients omitted). Standard errors are clustered by state. Nominal variables are CPI-adjusted to be in terms of 2020 dollars. The dependent variable is labeled above the column number, and its mean is reported in the bottom row. Statistical significance: 10%*, 5%** , and 1%***.

The improved financial resilience of borrowers translates into higher credit scores. We estimate an economically significant increase in credit scores in response to expanded Medicaid. Our point estimate implies that a 1 percentage point increase in eligibility leads to a 0.21 increase in consumers' average VantageScore. This corresponds to a 21 point score increase per newly eligible person.

To sum up, we find that expanding Medicaid eligibility increased both the prevalence of credit cards and revolving credit card balances. We find evidence suggesting both supply and demand contributed to the increase in equilibrium credit card usage. Lastly, we show that households become

more financially resilient, which could motivate expanded credit supply.

2.5 Robustness and Additional Results

2.5.1 Treatment Effect Heterogeneity Robustness

When the timing of treatment is staggered and treatment effects are heterogeneous, standard DiD estimators can identify a weighted average treatment effect (e.g., [Goodman-Bacon, 2021a](#)). These weights can sometimes be *negative*, biasing estimates in the opposite direction of the true (unweighted) average treatment effect. Settings like those in this paper, in which there are many *early* treated units, are especially vulnerable to having negative weights ([Roth et al., 2023](#)), which would result in our estimates understating the true treatment effects. To address this concern, we verify that our results are robust to using a "stacked" DiD estimator (as in [Cengiz et al., 2019](#)). The idea of this approach is to restructure the data and regression in order to estimate the treatment effect using only "good comparisons." See Appendix [A.1](#) for details on our implementation.

The stacked specifications yield quantitatively similar estimates. Appendix Tables [A.3](#), [A.4](#), and [A.5](#) report estimation results. The stacked regressions yield estimates within one to two decimal places of those obtained via TWFE estimation on the same subsample and with similar statistical significance. Notably, our stacked DiD estimate for revolving balances is 0.401 for the set of 2014 expanders, which is close to its TWFE counterpart obtained on the same subsample (0.362), and close to our original estimate of 0.383. This suggests that bias related to treatment effect heterogeneity is small in our setting. Our estimates become smaller (but remain positive and highly statistically significant) when adding additional expanders, as this shortens the horizon over which we can obtain stacked estimates, and treatment effects rise over time. We also find that our dynamic estimates are similar under either TWFE or the stacked estimator (Appendix Figure [A.4](#)).

2.5.2 Income Placebo Test

An important advantage of our granular ZIP-level measure of exposure to Medicaid expansions is that it allows us to control for county-time fixed effects, which absorb regional differences in time trends of credit card borrowing. Additionally, our exposure measure exploits cross-ZIP variation in a narrow range of the income distribution, while controlling for many moments of the ZIP's income distribution.²⁵ Together, these features allow us to plausibly control for other policies or local economic developments that tended to coincide with states' decisions to expand Medicaid, which may have also affected credit card borrowing. To better examine our success in avoiding bias from such confounders, we conduct a placebo test. We interact a post-expansion indicator with either one of two measures of ZIP-level income: a binary indicator for whether a ZIP had above-median average AGI or simply the ZIP's logged average AGI. The idea of the test is to see whether it is specifically expanded Medicaid eligibility that explains differences in credit card borrowing across ZIP codes, or is it due to another factor that is correlated with ZIP-level income.

²⁵For example, the pre-ACA limit in California was 106% of the FPL and 138% post-ACA. For a family of three in 2020, this corresponds to incomes of \$23,023 to \$29,974.

Appendix Table A.2 reports results from this test for revolving balances. Our findings here yield two useful insights. First, our estimate of the causal effect of increased eligibility remains similar (and highly statistically significant) even when including these additional interactions (0.376 and 0.452, versus our original 0.383). This suggests, in the sense of a mediation analysis, that our original findings are not simply an artifact from failing to control for other factors related to ZIP codes' differences in income. Second, we also find that the interactions of either income measure with a post-expansion indicator have a precisely estimated null coefficient. The point estimates imply that being in an above-median ZIP code predicts a (statistically insignificant) 0.1% lower level of revolving balances; a 1% increase in the ZIP's average AGI predicts a (statistically insignificant) 0.01% higher level. The small estimates here indicate that higher- versus lower-income ZIP codes did not generally have revolving balances start trending differently around states' Medicaid expansions (conditional on our controls). Taken together, these results are consistent with our measure of Medicaid exposure successfully using variation in a narrow income range that is unrelated to broader income-related confounders, and that the risk of such confounders is low given the absence of differential trends between high- and low-income ZIP codes around expansion timing.

2.5.3 Individual-Level and Interest Rate Results

A ZIP-level analysis enables us to plausibly estimate a broader effect of expanded Medicaid by including the responses of people who do not immediately become eligible but—as a result of the expansion—are nonetheless more likely to benefit from Medicaid in the future. However, it is still useful to estimate an individual-level variant. This more granular specification allows us to study an additional outcome: effective interest rates. We adapt the procedure of [Guttman-Kenney and Shahidinejad \(2023\)](#) and [Koşar et al. \(2025\)](#) to infer the average fee-inclusive annual interest rate on consumers' credit cards ("effective APR"), detailed in Appendix C.3. Individual-level data are more informative for this particular outcome, as ZIP-level changes in average rates can reflect changes in the composition of borrowers. For this analysis, we draw on a different data source to measure individual-level eligibility. The underlying data are imputed W2 income provided by Experian (see Appendix C.1). We only have income data for 2014 and therefore restrict this analysis to comparing people in states that either did not expand in our sample period or expanded in 2014. Because the source of eligibility data differs, this also creates an opportunity to test the robustness of our main results. Throughout this subsection, we report the estimates from Panel B of the relevant table, which includes ZIP-time fixed effects.

Appendix Table A.9 reports results from estimating an individual-level variant of Equation (1) that adds an individual fixed effect. The measure of treatment for this variant is an indicator for whether someone became newly eligible for Medicaid. In line with our ZIP-level estimates, we find large, positive effects of becoming eligible on the probability of having a credit card (23 pp) and on both total and revolving credit card balances (both 53 log points). The revolving balances estimate implies that eligibility increases revolving balances by \$1,597, relative to the pre-expansion mean of \$2,307. This is similar in magnitude to the increase per newly eligible person implied by our ZIP-level

estimates (\$1,446).²⁶

Appendix Table A.10 examines the responses of credit limits, credit scores, and APRs. As in our ZIP-level analysis, credit limits increase, rising 73 log points (\$3,930 relative to a pre-expansion mean of \$3,677). VantageScores increase by 20 points (relative to a pre-expansion mean of 581). APRs decrease 4.9 pp, which is a fifth of the pre-expansion mean for newly eligible consumers (23.8%). The decrease in APRs suggests that the Medicaid-induced expansion in credit card borrowing was supply-driven, rather than demand-driven.

Lastly, the individual-level data create an opportunity to evaluate the potential importance of identification confounders that vary at the ZIP-time level. To do so, we estimate our individual-level results both with county-time fixed effects (and the same ZIP-time-varying controls used in our ZIP-level specifications) in Panels A of Appendix Tables A.9 and A.10, and then again (in Panel B) with instead ZIP-time fixed effects. All coefficients remain nearly identical (differing by less than 3%). This suggests that such granular confounders are unlikely to be a threat to identification, building further confidence in the validity of the ZIP-level regressions.

2.5.4 Other Forms of Borrowing

We focus on card debt because of its widespread use and because it is a more natural form of credit to smooth consumption in response to shocks.²⁷ But given the positive impact of expanded Medicaid eligibility on credit scores, it is worth examining whether borrowing changes even for other forms of debt that are primarily used to finance the purchase of durables. To this end, we examine the impact of expanding Medicaid on auto and mortgage loans in Appendix Table A.11. We find statistically significant and economically meaningful effects on auto loans, while effects on mortgages are small and statistically indistinguishable from zero. A 1 pp increase in the eligible population share leads to a 0.15 pp increase in the fraction of people with an auto loan and a 0.92% increase in auto loan balances. Two features can account for the difference in responses between auto and mortgage loans. First, risk tends to be more strongly reflected in auto loan interest rates than in mortgage rates.²⁸ This may be due to the greater prevalence of negative equity for vehicles versus homes, and the tendency of borrowers to prioritize mortgage payments over auto loan payments (Experian, 2026). Thus, improvements in credit scores can translate into a larger reduction in auto and credit card rates, while mortgage rates react less. Second, the people most affected by the Medicaid expansions will tend to have lower income. These households may be closer to the margin of getting an auto loan (or a credit card), but not as close to the margin of getting a home loan.

²⁶Note that we should not expect perfect quantitative alignment in general, as the individual-level analysis uses a subset of expansions (2014 only) and captures only the direct effect on eligible people, and not indirect effects on people who also experience increased access due to either reduced risk or resemblance to eligible people in terms of underwriting observables.

²⁷For example, 86% of people in our sample have a credit card and 61% have unpaid balances, while 30% of our sample has a mortgage.

²⁸Auto loan rates vary much more with borrower credit scores than mortgage rates. For example, typical APRs on used auto loans recently ranged from 7.43% to 21.60% across the credit score distribution (Experian, 2025). In contrast, the typical mortgage rate varied from 7.20% to 7.85% across the credit score distribution (Bankrate, 2025).

3 Model

In this section, we develop a heterogeneous-agent model with health insurance and unsecured debt. Households are subject to income shocks as well as medical expenditure shocks and can save and borrow to smooth their consumption. The model serves two purposes. First, it allows us to study the different channels through which health insurance affects household borrowing behavior. Second, we use the model to analyze the macroeconomic effects of expanding health insurance. Our empirical estimates in the previous section exploit variation in treatment intensity across locations and recover the cross-sectional effect of the expansion of Medicaid. Because of a “missing intercept” problem, studying the macroeconomic response requires a model. We start by describing the economic environment, defining the equilibrium, and then analyzing the different channels through which health insurance shapes the borrowing decision of households.

3.1 Environment

The model is set in discrete time and there is no aggregate uncertainty.

Households. There is a continuum of households of measure one, indexed by $i \in [0, 1]$. Each household belongs to one of K types, indexed by $k \in \{1, \dots, K\}$, which differ permanently in their discount factors and in their utility cost of delinquency. All households face idiosyncratic income and medical-expenditure risk.

Income process. Household i ’s income at time t is denoted by y_{it} and evolves according to a compound Poisson process:

$$\ln y_{it} = \begin{cases} \rho \ln y_{it-1} + \epsilon_{it}^y & \text{w.p. } \lambda_y, \\ \ln y_{it-1} & \text{w.p. } 1 - \lambda_y, \end{cases} \quad (2)$$

where λ_y is the probability that an income shock arrives in a period. If such a shock does not arrive, the household’s income remains unchanged. Given an income shock, the household income follows an AR(1) process, where ρ is the degree of persistence and ϵ_{it}^y is an idiosyncratic income shock with mean zero and variance $\sigma_{\epsilon,y}^2$.²⁹

Medical expenditure shocks and health insurance. Households are also subject to stochastic medical expenditure shocks, denoted by m_{it} , which can be correlated with income. We denote the CDF of medical expenditure for a household with income y_{it} by $G(m_{it}|y_{it})$.

Medical bills are partially covered by health insurance. We allow the type of insurance to vary exogenously with household income. Health insurance covers a share of the medical bill due. We denote the share of medical bills a household has to pay out of pocket as $o(y_{it})$. Note that the out-of-pocket share depends only on the current household income.

²⁹The assumption that income shocks occur with probability λ_y allows us to match the high kurtosis of income changes observed in the data. The high kurtosis is quantitatively important for matching delinquency rates in the data.

Household problem. Each household has access to a non-state-contingent one-period debt security, denoted by b_{it} . Throughout, we adopt the convention that $b_{it} > 0$ denotes debt and $b_{it} < 0$ denotes savings. At the beginning of each period, the household can choose to repay or renege on its debt obligations. The household's decision to repay debt depends on its total debt, its level of income, and the size of its current medical bill. Households face an interest rate schedule for the amount of borrowing they choose, which we denote by $r_k(b', y)$. Note that the interest rate schedule depends on the household preference type, k . The price of debt is denoted by $q_k(b', y) \equiv \frac{1}{1+r_k(b', y)}$. The relevant state variable from the perspective of the household is its total repayment obligations after the medical expenditure shock, which we denote by $\tilde{b}_{it} = b_{it} + o(y_{it})m_{it}$.³⁰

The household's present discounted value is denoted by $V_k(\tilde{b}, y)$, where $\{\tilde{b}, y\}$ are its individual state variables. This value function is given by the max between the present discounted value of repaying debt obligations, denoted by $V_k^r(\tilde{b}, y)$, and the value of reneging and declaring delinquency, denoted by $V_k^d(\tilde{b}, y)$:

$$V_k(\tilde{b}, y) = \max \left\{ V_k^r(\tilde{b}, y), V_k^d(\tilde{b}, y) \right\}. \quad (3)$$

Conditional on the decision to repay its debt obligations, the recursive problem of a household of type k with total repayment obligations $\tilde{b}$ and income y is given by

$$\begin{aligned} V_k^r(\tilde{b}, y) &= \max_{c, b'} u(c) + \beta_k \mathbb{E} V_k(b' + o(y')m', y'), \\ \text{s.t.} \quad &c + \tilde{b} \leq y + q_k(b', y)b', \end{aligned} \quad (4)$$

where $u(\cdot)$ is the utility of the household from consumption, which is assumed to be strictly increasing, concave, and continuously differentiable. The discount factor of a household of type k is β_k .

When a household reneges on its debt obligations, the debt moves into a delinquency state. Modeling consumer default as persistent delinquency rather than formal bankruptcy follows [Drozd and Serrano-Padial \(2017\)](#), who document that persistent delinquency is the most prevalent form of consumer default. A household with debt in a delinquency state cannot save or borrow and suffers a utility cost ξ_k , which varies with household type k . At the end of the period, the household may receive a stochastic haircut on its debt obligations. The haircut, denoted by δ , is assumed to follow a compound Poisson process. With probability λ_δ , the household receives a positive haircut. Conditional on a haircut, the share of debt forgiven follows a Power distribution with shape parameter α . That is, conditional on receiving a haircut, the probability that the share of debt forgiven is less than δ equals δ^α . This haircut process may be interpreted as a reduced-form approximation of either bankruptcy or debt forgiveness obtained through negotiations with creditors.³¹

³⁰For repaying households, modeling medical expenses as an addition to debt obligations is equivalent to a reduction in income, as the budget constraint depends only on income net of total obligations. The two formulations differ in delinquency: delinquent households do not pay medical bills, which accumulate as medical debt.

³¹We assume that a haircut is applied to the household's outstanding debt regardless of whether the household subsequently repays the reduced balance or remains in delinquency. Thus, a household that remains delinquent may receive additional haircuts in subsequent periods, generating progressively more favorable repayment terms over time. This formulation provides a reduced-form way of capturing opportunities to settle delinquent debt that improve with the duration

The value of the household in the delinquent state is given by

$$V_k^d(\tilde{b}, y) = u(y) - \xi_k + \beta_k \mathbb{E} V_k((1 - \delta)\tilde{b} + o'(y')m', y'). \quad (5)$$

The timeline in every period is as follows. First, the household learns its current income and medical bill. Then, it decides whether or not to repay its outstanding debt obligations. If it decides to repay its debt obligations, it chooses the level of consumption and borrowing. If it decides to renege on its debt obligations, it consumes its income, suffers a utility loss, and at the end of the period may receive a haircut.

We denote the delinquency policy function of a household of type k with total debt obligations $\tilde{b}$ by

$$d_k(\tilde{b}, y) = \mathbb{1} \left[V_k^r(\tilde{b}, y) < V_k^d(\tilde{b}, y) \right], \quad (6)$$

where $\mathbb{1}$ is the indicator function. The function $d_k(\tilde{b}, y)$ equals 1 when the household defaults. When indifferent, we assume the household repays its debt obligations.

Credit Supply. Credit to households is supplied by risk-neutral credit card companies. Perfect competition among these companies ensures an expected zero-profit condition holds in equilibrium. We assume credit card companies have unlimited access to funds at the risk-free interest rate, which we denote by r^f .

Debt in the economy is a hybrid of short-term and long-term debt. Debt is of short maturity in nature, as households need to repay their debt obligations in the following period.³² However, when debt becomes delinquent, credit card companies do not receive payments within that period. Instead, they need to wait until the household decides to repay its debt—either because it received a sufficiently large haircut or because its income level changes so that it decides to repay.

We assume perfect information, so that credit card companies know the household type k as well as its current income level. The zero-profit condition that pins down the price of debt, $q_k(b', y)$, is

$$\begin{aligned} q_k(b', y)(1 + r_f) &= [1 - \mathbb{E}(d_k(b' + o(y')m', y'))] \\ &\quad + \mathbb{E}[d_k(b' + o(y')m', y')(1 - \delta')q_k((1 - \delta')(b' + o(y')m'), y')]. \end{aligned} \quad (7)$$

That is, the credit card companies equate the cost of their loan (LHS) with its return (RHS). The return from the loan is the sum of two parts. If the debt is repaid, the credit card company receives its full face value. If the debt is delinquent, which happens with probability $\mathbb{E}(d_k(b' + o(y')m', y'))$, the creditor is left with a claim on a delinquent debt. The worth of such a claim, after a haircut $(1 - \delta')$, is given by $(1 - \delta')q_k((1 - \delta')(b' + o(y')m'), y')$. The fixed point function $q_k(\cdot)$ that solves equation (7) is the debt pricing schedule. As the following proposition shows, this function exists and is unique.³³

Proposition 1. *Given a default policy function, $d_k(\tilde{b}, y)$, there exists a unique pricing schedule $q_k(b', y)$ that satisfies equation (7).*

of delinquency, as might arise from repeated negotiations with creditors or increasingly favorable settlement offers.

³²Note that households can potentially roll over their debt by borrowing to cover their current debt obligations.

³³The proof is in Appendix D.

3.2 Stationary Equilibrium

We denote the joint distribution of households of type k across total net asset position and income levels at the beginning of the period, after the expenditure shock realization, by $\Lambda_k(\tilde{b}, y)$. Four components characterize the law of motion for this joint distribution: (i) the borrowing decision of households, $b'_k(\tilde{b}, y)$, (ii) the debt pricing schedule, $q_k(b', y)$, (iii) the household's default decision, $d_k(\tilde{b}, y)$, and (iv) the exogenous processes of income, expenditure shocks, and haircuts. The law of motion for the joint distribution is defined as follows. For all Borel sets $\mathcal{B} \times \mathcal{Y} \subset \mathbb{R} \times \mathbb{R}^+$,

$$\begin{aligned} \Lambda'_k(\mathcal{B} \times \mathcal{Y}) = & \int_{y' \in \mathcal{Y}} \int_{m'} \int_{\mathcal{B}_k(\tilde{b}, y, m')} d\Lambda_k(\tilde{b}, y) dG(m'|y') dF(y'|y) \\ & + \int_{\delta} \int_{y' \in \mathcal{Y}} \int_{m'} \int_{y, ((1-\delta)\tilde{b} + o(y')m') \in \mathcal{B}} \mathbb{1}[d_k(\tilde{b}, y)] d\Lambda_k(\tilde{b}, y) dG(m'|y') dF(y'|y) dH(\delta), \end{aligned} \quad (8)$$

and

$$\mathcal{B}_k(\tilde{b}, y, m') = \{(\tilde{b}, y) \text{ s.t. } d_k(\tilde{b}, y) = 0 \text{ and } b'_k(\tilde{b}, y) + o(y')m' \in \mathcal{B}\},$$

where $F(\cdot)$, $G(\cdot)$, and $H(\cdot)$ are the CDFs of the different exogenous variables. In the stationary equilibrium, the distribution $\Lambda_k(\tilde{b}, y)$ is constant over time. The definition of the stationary Markov-perfect equilibrium is as follows.

Definition 1 (Equilibrium). *A stationary Markov-perfect equilibrium is given by a default policy function $d_k(\tilde{b}, y)$, a borrowing policy function $b'_k(\tilde{b}, y)$, a debt pricing schedule $q_k(b', y)$, and a joint distribution of households across total debt and income levels, $\Lambda_k(\tilde{b}, y)$, for each household type k , such that*

1. *The default and borrowing policy functions solve the household's problem given the debt pricing schedule.*
2. *The debt pricing schedule satisfies the zero-profit condition, (7).*
3. *The joint distribution of households across debt and income levels is stationary.*

3.3 Health Insurance and Household Borrowing Behavior

We model the expansion of health insurance as a change in the out-of-pocket share of medical expenditure that households of different income levels face, $o(y)$. A more generous health insurance policy corresponds to a reduction in the out-of-pocket share that some households pay for medical shocks.

Expanding health insurance affects households' accumulation of credit card debt in several ways, both directly and indirectly. There are two direct effects of the policy. First, more generous health insurance increases households' disposable income. As a result, household debt levels decline. Second, the insurance against medical expenditure shocks reduces delinquency, which raises credit card debt for two reasons. When a delinquent household receives a medical expenditure shock, its additional debt is not counted as credit card debt but rather as medical debt, whereas a household that is not in delinquency may borrow to cover that expenditure. In addition, a household that avoids delinquency continues to roll over its credit card debt, which would otherwise have been charged off and gradually written down through haircuts. Therefore, while the first direct effect reduces credit card debt, the second raises it, and the overall direct effect is ambiguous.

To study the indirect channels of the policy, consider the household's optimality condition with respect to debt accumulation. To ease notation, in this subsection we omit the preference type k . The borrowing optimality condition is given by

$$u'(c) \frac{\partial (q(b', y)b')}{\partial b'} = \beta \mathbb{E} \mathbb{1}_{V^r \geq V^d} u'(c(b' + o(y')m', y')) + \beta \mathbb{E} \mathbb{1}_{V^r < V^d} \left(-\partial_b V^d(b' + o(y')m', y') \right) \quad (9)$$

The household equates the benefits from borrowing (LHS) to the utility costs of borrowing (RHS). Increasing debt obligations, b' , raises the household's current funds by $\frac{\partial (q(b', y)b')}{\partial b'}$. Note that the household internalizes how its borrowing decision affects the interest rate it pays on debt. There are two potential costs of borrowing. If the household repays its debt in the following period ($V^r \geq V^d$), the marginal cost of debt obligations is simply the marginal utility of consumption. Alternatively, if the household goes delinquent ($V^d > V^r$), the household's cost of debt obligations is $-\frac{\partial V}{\partial b}$.

The first indirect channel through which social insurance affects household debt is a reduction in the *precautionary savings motive*. A reduction in $o(y')$ reduces the volatility of out-of-pocket medical expenditure, $o(y')m'$, lowering the volatility of future consumption. If the utility function features prudence ($u'''(\cdot) > 0$), as it does in our calibration, then this reduction in volatility results in a smaller expected utility cost of borrowing. The reduction in the marginal cost of borrowing induces households to take on more debt. That is, social insurance increases debt by weakening the motive to self-insure through precautionary savings.

The second indirect channel is the *debt aversion* channel. A more generous insurance policy raises the probability of repayment, $\mathbb{E} \mathbb{1}_{V^r \geq V^d}$, as medical expenditures that would have pushed households into delinquency are now partially insured. Holding the interest rate schedule fixed, this makes debt overpriced from the household's perspective: the spread it pays still reflects the old delinquency risk, while it now expects to repay in full more often. The expected marginal cost of borrowing rises, and households respond by borrowing less. Therefore, through the debt aversion channel, social insurance reduces household debt levels.

Neither the precautionary savings motive nor the debt aversion channels depend on lenders changing their behavior. That is, they do not depend on the supply side. We refer to the combined effect of these two channels as the *credit demand* channel.

The final indirect channel is the *credit supply* channel. The reduction in the probability of delinquency induces lenders to lower interest rate spreads, raising $q(b', y)$. This raises the benefits from taking on more debt obligations b' . For each unit of consumption promised to be repaid in the following period, households receive more units of consumption in the current period. This induces households to increase their debt obligations. So, through the credit supply channel, social insurance increases household debt levels.

Overall, the effect of social insurance on the aggregate level of household debt is ambiguous. In the next section, we take the model to the data and quantitatively assess the strength of the different channels.

3.4 Discussion

Before taking the model to the data, we discuss several of its abstractions and how they bear on our results. First, we do not explicitly model household health. Therefore, our welfare calculations reflect improvements in *financial* health rather than *physical* health. In reality, the welfare benefits from improved physical health are also likely large, as Medicaid expansions significantly reduced mortality ([Goodman-Bacon, 2018](#); [Miller et al., 2021b](#)).

Second, our model does not feature moral hazard in terms of either labor supply, risky health behaviors, or use of medical services. In principle, some households face an incentive to reduce labor supply in order to qualify for Medicaid. However, empirical estimates of the labor supply responses to the ACA’s Medicaid expansions are centered on zero and precise enough to rule out economically significant responses (e.g., [Leung and Mas, 2018](#); [Kaestner et al., 2017](#); [Baicker et al., 2014](#)). Regarding risky behaviors, research also estimates precise null effects of the ACA’s Medicaid expansions on risky behaviors (e.g., smoking and lack of exercise, [Simon, Soni and Cawley, 2017](#)). However, research generally estimates moderate moral hazard responses in terms of overall health expenditures ([Manning et al., 1987](#); [Finkelstein et al., 2012](#); [Garthwaite et al., 2019](#)), although a recent analysis of the Medicaid expansion under the ACA finds no impact on total (pre-insurance) medical expenditures ([Shupe, 2023](#)). Because households do not increase total medical expenditures in our model, the model likely understates the borrowing response. While the additional costs of the expansion that would arise from increased expenditure are not included in our welfare calculation, these costs should be weighed against the welfare gains from improved physical health.

Third, we assume that lenders behave competitively. While there is evidence that the credit card market in the US is not perfectly competitive ([Herkenhoff and Raveendranathan, 2025](#)), the degree of competition in credit markets need not shape the way the economy responds to the expansion of Medicaid. In Appendix E.2, we extend the model to incorporate oligopolistic competition in credit markets. In equilibrium, the price of debt equals the actuarially fair price minus a constant markdown that reflects the number of lenders and the degree of substitutability across lenders, but is independent of the borrower’s default risk. Social insurance operates through the actuarially fair component, so the improvement in debt prices generated by lower default risk passes through to households regardless of the markdown. Calibrating the markdown to match the average excess spread estimated by [Herkenhoff and Raveendranathan \(2025\)](#), we find that imperfect competition lowers the level of welfare in the stationary equilibrium, but leaves the welfare *gains* from the expansion of Medicaid essentially unchanged.

Fourth, households in our model are rational, so our welfare calculations follow the standard revealed-preference benchmark. If some credit card borrowing reflects behavioral biases rather than consumption smoothing (e.g., [Heidhues and Kőszegi, 2010](#); [Zinman, 2014](#)), this benchmark could overstate the welfare gains from the expansion of credit supply. We believe this is unlikely to change our conclusions for two reasons. First, if the additional borrowing were harmful, financial distress should rise alongside debt. We find the opposite in Section 2: the expansion reduced delinquency, debt in collections, and borrowing costs, while increasing credit scores. Second, the welfare gains from credit supply in the model arise from the lower price of credit rather than from additional

borrowing. Lower interest rates benefit households on the borrowing they already do, regardless of whether the additional borrowing they induce is optimal.

Finally, we also abstract away from explicitly modeling collateralized forms of debt such as auto loans and mortgages. As discussed earlier in Section 2.5, we find that auto debt does react to the Medicaid expansion but mortgage debt does not. This suggests excluding mortgage debt is likely innocuous for the two main applications of our model: studying its impact on household borrowing and welfare. Regarding auto loans, this abstraction makes our analysis conservative. If anything, it understates the increase in total household debt and welfare.

4 Quantitative Analysis

In this section, we evaluate the macroeconomic effects of the Medicaid expansion using our model. We proceed in three steps. First, we calibrate the model to the US economy, matching features of the economy prior to the expansion. Second, we simulate the expansion of Medicaid and show that the model broadly matches our empirical estimates from Section 2. Finally, we quantify the aggregate effects of the expansion on household debt and welfare.

4.1 Calibration

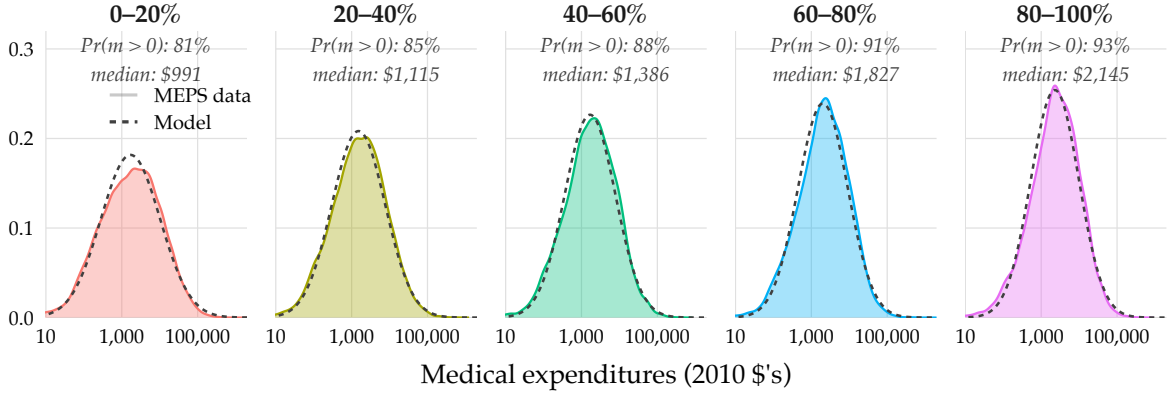
We set a period in the model to correspond to one year and calibrate the risk-free interest rate in the economy to 2%. The structural parameters fall into five categories: (i) the income process, (ii) the medical expenditure shock process, (iii) the health insurance system, (iv) household preferences, and (v) the haircut process. Our calibration strategy is to discipline the parameters in the first three categories directly using the data, and the remaining parameters by matching distributional features of households' saving and borrowing behavior. Importantly, our empirical estimates from Section 2 are not used in the calibration; we hold them out to validate the model.

Income Process Parameters. We calibrate the income process parameters to match key moments of household income dynamics from the Panel Study of Income Dynamics (PSID). Our sample includes households whose head is aged 25–55, over the period 1999–2019. The three moments we target are the two-year autocorrelation of log household income (0.826), the standard deviation of two-year log-income changes (0.478), and the kurtosis of two-year log-income changes (4.672), where we first residualize log household income on age and education fixed effects.³⁴ The calibration yields an arrival rate of income shocks $\lambda_y = 0.45$, so that income shocks arrive on average every 2.2 years. The calibrated persistence of income conditional on an income shock is $\rho = 0.80$, and the variance of income shocks is $\sigma_{\epsilon,y}^2 = 0.24$.³⁵

³⁴Following Aguiar et al. (2025), we exclude household-year observations with real family income below \$2,000 in 2019 dollars. To limit the influence of outliers, we trim the top and bottom 1% of two-year log-income changes. The data moments are in line with the income moments in Guvenen et al. (2021), who use Social Security Administration data on the annual income of males in the US.

³⁵In Appendix D, we show how the moments map to these three parameters. Note that while the persistence of income conditional on an income shock is 0.8, the unconditional income persistence is 0.91.

Figure 5: Distribution of Annual Medical Expenditure



Notes: Data come from the 2000–2019 MEPS. The sample consists of households with members between ages 25 and 55, sorted into within-year quintiles of income residualized on age and education. Solid lines plot weighted kernel densities of log medical expenditures; dashed lines plot the calibrated log-normal densities. Each curve is scaled to integrate to $Pr(m > 0)$, the quintile’s share of households with positive expenditures. Medians include households with zero spending.

Medical Expenditure Parameters. We divide households into five income quintiles and assume that the stochastic medical expenditure process is identical within each quintile. We use the Medical Expenditure Panel Survey (MEPS) between 2000–2019 to calibrate the parameters governing medical expenditure shocks, using total medical expenditures, including both out-of-pocket payments and expenditures covered by insurers and other payers.³⁶ For each income quintile, we assume that there is a probability λ_q^m that a medical expenditure shock realizes, where q indicates the income quintile. Conditional on a positive medical expenditure shock, we assume it follows a log-normal distribution with mean μ_q^m and standard deviation σ_q^m . The calibrated parameters are displayed in Table 7. Figure 5 compares the calibrated distribution with its empirical counterpart for each quintile. While the median medical expenditure shock is modest, about 4% of median household income, the calibrated process features substantial risk: a shock one standard deviation above the mean (in logs) amounts to 19% of median income, ranging from 16% to 22% across income quintiles. A two-standard-deviation shock corresponds to nearly a full year of income.

Health Insurance Parameters. The MEPS data also report insurance type and out-of-pocket (OOP) expenditures. To construct the OOP share as a function of income, we first split insurance types into three categories: Medicaid, non-Medicaid insurance, and uninsured. Non-Medicaid insurance includes households with private insurance (e.g., employer-sponsored) and Medicare. We compute the average OOP share of households aged 25–55 in each insurance type. The average OOP shares are 10.5% (Medicaid), 31.7% (other/non-Medicaid), and 69.5% (uninsured).³⁷

Second, using a shifted exponential function, i.e., $f(x) = ae^{bx} + c$, we approximate the share

³⁶Consistent with our PSID analysis, we restrict the sample to households aged 25–55, residualize income on age and education fixed effects, and then partition households into their income quintile within each year.

³⁷Uninsured patients can be required to pay less than 100% of their expenses when either they negotiate lower payments with hospitals or when other sources cover their payments. Other sources of payment include non-health sources of insurance (e.g., homeowners or auto insurance in relation to injuries), the Veterans Administration, and Workers’ Comp.

of households with each insurance type. We compute these functions both before and after the expansion of Medicaid. Figure 3 displays the empirical shares alongside our approximations. The stationary distribution of our benchmark model uses the pre-expansion approximations. We combine the relationship between insurance type and income along with the OOP share by insurance type to impute the average OOP share by income. Specifically, the OOP share in our model is

$$o(y) = \Pr(\text{Medicaid}|y) \times 10.5\% + \Pr(\text{Other}|y) \times 31.7\% + \Pr(\text{Uninsured}|y) \times 69.5\%. \quad (10)$$

A key advantage of this two-step approach, relative to directly estimating the OOP share along the income distribution, is that we can conduct counterfactuals in which we change the share of households under each insurance type.

Preference and Haircut Parameters. We assume that households have log utility in consumption, $u(c) = \ln(c)$, and that there are three types of households in the economy, $K = 3$.³⁸ This results in eight preference parameters: the discount factor β_k and the utility cost of delinquency ζ_k for each of the three types, as well as the weights of each type.³⁹ In addition, there are two haircut process parameters: the arrival rate of haircuts, λ_δ , and the shape parameter of the Power distribution, α . We estimate these ten parameters jointly to match twelve moments: ten moments of the asset distribution along the income distribution and two delinquency moments.

To construct the asset distribution moments, we use the 2011 and 2013 waves of the PSID—the first two waves that separately record credit-card balances, student loans, and unpaid medical bills, and the last two waves that precede the expansion of Medicaid. We measure net liquid wealth following the classification of Kaplan and Violante (2014), as implemented in the PSID by Aguiar et al. (2025): liquid assets comprise transaction accounts and directly held equity, while liquid debt comprises credit- and store-card balances and unpaid medical bills. We depart from Aguiar et al. (2025) in one respect: we exclude student loans from liquid debt, as households cannot draw on them to finance current consumption. When a household’s net liquid position is negative, we attribute the position first to credit-card debt.⁴⁰ We sort households into within-year quintiles of income residualized on age and education fixed effects, consistent with our income-process calibration. For each income quintile, we compute two moments: average liquid savings and average credit-card debt. Both are normalized by median household income.

The two delinquency moments come from the Experian data: the share of consumers entering credit-card delinquency each year (5.6%) and the share of newly delinquent consumers still delin-

³⁸Three is the smallest number of types that allows the model to match the joint distribution of liquid wealth and credit card debt. With fewer types, the model generates a wealth profile that is too steep along the income distribution: households in the bottom quintiles hold essentially no liquid wealth, while the top quintile holds too much.

³⁹Since the weights sum to one, there are only two degrees of freedom in choosing them.

⁴⁰We cap the attributed credit-card debt at the household’s gross credit-card balance and assign any remainder to unpaid medical bills. The moments are essentially unchanged if we instead attribute negative positions first to unpaid medical bills, as households carrying both credit-card debt and unpaid medical bills typically hold few liquid assets. Our measure omits the PSID’s remaining non-collateralized debt categories—legal bills, loans from relatives, and other unsecured debts—which together account for only 7% of unsecured household debt in our sample.

Table 6: Estimation Targets and Model Fit

	Q1	Q2	Q3	Q4	Q5
<i>Liquid savings (% of median income)</i>					
Data	6.6	19.9	34.4	36.6	87.8
Model	7.6	13.2	22.7	40.2	101.7
<i>Credit card debt (% of median income)</i>					
Data	2.0	2.7	2.9	3.2	3.0
Model	2.2	2.5	2.8	3.4	3.1
		Data		Model	
Delinquency flow		5.6%		5.5%	
Delinquency hazard		55.0%		53.8%	

Notes: The top panel reports average liquid savings and average credit card debt by income quintile, as a share of median household income (PSID). The bottom panel reports the share of consumers entering credit-card delinquency each year and the share of newly delinquent consumers still delinquent one year later (Experian).

quent one year later (55%).⁴¹ The first moment disciplines the level of delinquency risk that lenders price, the second the persistence of delinquency spells, which the haircut process governs.

Note that not all debt in the model corresponds to credit card debt in the data. Consider a low-income household that faces high interest rate spreads and therefore chooses not to borrow. Suppose this household suffers a large medical expenditure shock and does not pay the resulting bill. This household carries delinquent debt, yet no lender ever extended it a loan. Its debt is medical debt, not credit card debt. Accordingly, credit card debt in the model comprises all non-delinquent debt, together with delinquent debt originally extended by a lender. We apply the same distinction when computing the model's delinquency moments.

We estimate the ten parameters jointly by minimizing the weighted sum of squared percentage deviations between the twelve model moments and their empirical counterparts. The two delinquency moments each receive a weight of 2.5, so that the delinquency block carries the same total weight as each set of five asset distribution moments. All other moments receive a weight of one. Table 6 reports the twelve targeted moments alongside their model counterparts. With ten parameters and twelve targets, the model matches the data well: it reproduces the level and shape of both the savings and the credit card debt profiles along the income distribution, and it closely matches the two delinquency moments.

Table 7 reports the parameter estimates. The three preference types can be roughly labeled as: (i) the impatient strategic defaulters, (ii) the credit-excluded households, and (iii) the patient liquidity-driven defaulters. We now discuss each type's behavior in equilibrium and its role in matching the targeted moments.

The impatient strategic defaulters make up 48% of the population. Their discount factor of 0.82 leads them to borrow despite sizeable interest rate spreads, and they hold most of the credit card debt

⁴¹ A consumer enters delinquency in year t if their most recent 30–180-day delinquency on a credit-card account occurred within the past twelve months, having had no recent delinquency as of year $t - 1$. Persistence is measured as having any past-due or derogatory credit-card status one year after entry. We observe each consumer's credit record once a year, in June, over 2010–2021.

Table 7: Model Parameters

Parameter description	Parameter	Value	Calibration Source
Risk-free rate	r_f	2%	Standard value
Arrival rate of income shocks	λ_y	0.450	PSID income moments
Persistence of income shocks	ρ	0.798	PSID income moments
Variance of income shocks	$\sigma_{\epsilon,y}^2$	0.239	PSID income moments
Arrival rate of medical shocks	λ_q^m	[0.82, 0.93]	MEPS expenditure moments
Mean log medical expenditure	μ_q^m	[−3.44, −3.02]	MEPS expenditure moments
Std. dev. of log medical expenditure	σ_q^m	[1.46, 1.80]	MEPS expenditure moments
Discount factors	$\{\beta_k\}$	{0.82, 0.82, 0.95}	SMM (Table 6)
Disutility of delinquency	$\{\xi_k\}$	{0, 0.08, 1.4}	SMM (Table 6)
Type weights	$\{\omega_k\}$	{0.42, 0.48, 0.10}	SMM (Table 6)
Arrival rate of haircuts	λ_δ	0.98	SMM (Table 6)
Haircut distribution shape	α	0.95	SMM (Table 6)

Notes: Medical expenditure parameters vary by income quintile; the table reports their ranges, and Appendix Table E.1 reports the values by quintile, with expenditure measured in units of median household income. SMM parameters are estimated jointly to match the moments in Table 6. Type-specific preference parameters are ordered so that the k -th entries of $\{\beta_k\}$, $\{\xi_k\}$, and $\{\omega_k\}$ describe the same type: the first two entries correspond to the credit-excluded households and the strategic defaulters, respectively, and the third to the patient households.

in the economy. Their disutility of delinquency, $\xi = 0.08$, is equivalent to an 8% drop in consumption. This cost is large enough to sustain repayment in equilibrium, yet small enough that these households go delinquent when their debt burden grows large relative to income. Their delinquency behavior generates the annual flow into delinquency, and their borrowing helps generate the level and shape of the credit card debt profile along the income distribution.

The lower discount factor of the impatient strategic defaulters (0.82) should not be interpreted as a strong desire to borrow. The utility cost of delinquency (ξ) amplifies the precautionary savings motive: households restrain their borrowing to avoid states in which they must either repay when their consumption is low or incur the delinquency cost. To gauge the strength of this force, we compute the risk-free rate that would clear the asset market in an economy populated only by strategic defaulters, with zero net supply of assets. The resulting rate is 2.18%.⁴² Under complete markets, this rate would correspond to a discount factor of 0.98. That is, the strategic defaulters' effective patience is far higher than their discount factor alone suggests.

The credit-excluded households make up 42% of the population. These households bear no utility cost of delinquency. Lenders anticipate that they would not repay and thus refuse to extend them credit, so in equilibrium these households neither borrow nor save. Their mass resembles the 41% of households in the PSID classified as hand-to-mouth in Aguiar et al. (2025), who also document that this status is highly persistent and driven by preferences rather than by adverse shocks, consistent with our modeling of this group as a permanent type. Because these households do not participate in

⁴²Income risk alone accounts for about half of this effect. In an incomplete-markets economy with the same income process but without medical expenditure shocks or the option to go delinquent, the market-clearing rate for the strategic defaulters is 11.8%, compared to 22% under complete markets. Medical expenditure risk and delinquency drive the rate further down to 2.18%.

the credit market, their discount factor is only weakly identified; we set it equal to that of the strategic defaulters.⁴³ Note that the presence of this type works against the mechanism we emphasize in the paper: since these households do not borrow, a larger credit-excluded share attenuates, rather than amplifies, the model’s debt response to the expansion of Medicaid.

The patient liquidity-driven defaulters make up the remaining 10% of the population. With a discount factor of 0.95, these households hold the bulk of liquid wealth in the economy, and their weight and discount factor are pinned down by the level of liquid savings in the top income quintiles. They also account for part of the credit card debt at the bottom of the income distribution: when adverse income shocks deplete their savings, they borrow, and they do so cheaply because lenders expect them to repay. Their disutility of delinquency, $\xi = 1.4$, equivalent to a 75% consumption drop, makes strategic default very costly. These households default on their debt only when large medical expenditure shocks exhaust their ability to repay.

Turning to the haircut process, the estimated arrival rate is $\lambda_\delta = 0.98$ and the estimated shape parameter is $\alpha = 0.95$. These values imply that a delinquent household receives a settlement offer nearly every year and that the average offer forgives 49% of the outstanding debt. In expectation, 48% of a delinquent household’s debt is therefore written down within a year. This expected relief rate is primarily identified by the delinquency hazard: more generous relief would get households out of delinquency faster than the 55% one-year persistence observed in the data, while less generous relief would keep them in delinquency for longer. The large haircut magnitude is consistent with the deep discounts at which lenders sell delinquent credit card debt to collection agencies. In the model, the average delinquent debt is valued at 13.4 cents per dollar of face value. Strikingly, this is nearly identical to the average price at which credit card debt was sold to debt buyers over the period 2021–2024, which equals 13.5 cents.⁴⁴ The average haircut is also in line with the terms of negotiated settlements: among large credit card issuers, delinquent accounts that reach a settlement pay roughly half of the balance owed ([Consumer Financial Protection Bureau, 2023](#)).

4.2 Equilibrium Properties

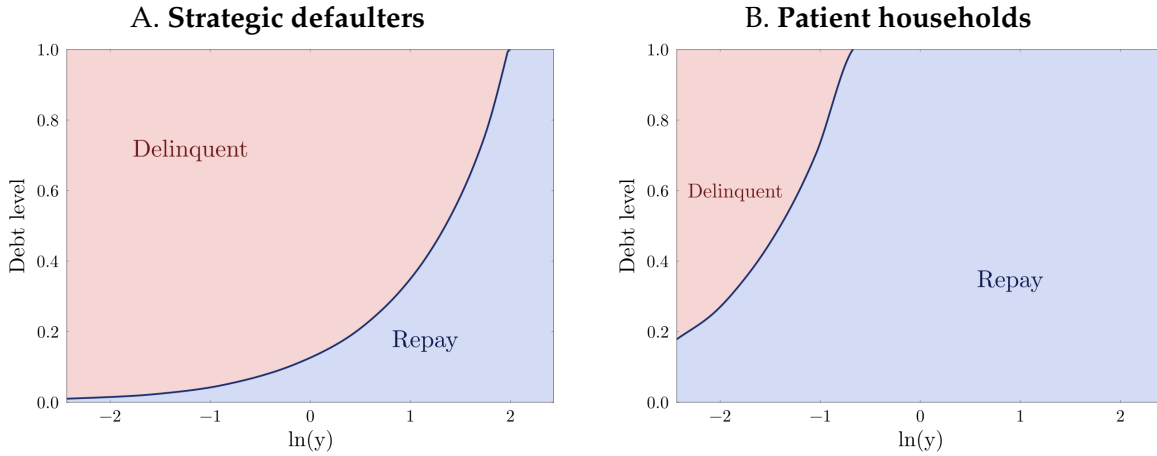
We now examine the equilibrium properties of the estimated model. Figure 6 presents the regions of the state space in which households repay their debt obligations and those in which they go delinquent, separately for the strategic defaulters (Panel A) and the patient households (Panel B).⁴⁵ For both types, the debt level above which a household reneges rises with income. Because the utility function is concave, low-income households are more tempted to renege on their debt and increase their contemporaneous consumption. The two panels differ in the location of the delinquency region. Strategic defaulters renege at moderate debt levels across most of the income distribution, and can therefore sustain meaningful debt only when their income is high. Patient households repay in

⁴³Setting the discount factor of the credit-excluded households to any value below 0.82 has effectively no effect on model fit and the behavior of the economy.

⁴⁴The Consumer Financial Protection Bureau’s reports on the consumer credit card market ([Consumer Financial Protection Bureau, 2023, 2025](#)) put the average purchase price of sold credit card debt at 14, 16, 13, and 11 cents per dollar of account balance in 2021 through 2024, respectively.

⁴⁵While the upper limit of debt in Figure 6 is 1, we solve the model with the upper limit of debt being 2. The share of households with debt level above 1 is less than 1%.

Figure 6: Delinquency Regions by Household Type



Notes: Each panel presents the regions of the state space in which households of the indicated type repay their debt obligations or go delinquent. Income and debt are in units of median income. The debt level is capped at 1 for illustration purposes.

nearly every state; they go delinquent only when their income is very low and their debt obligations exceed roughly 20% of median income.

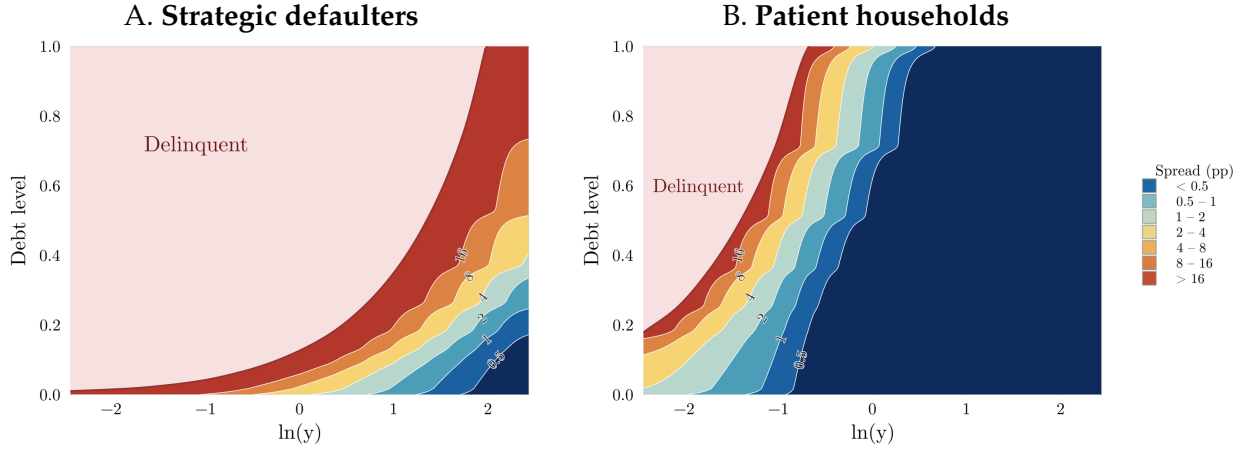
The delinquency regions shape the interest rates households face. Figure 7 presents the equilibrium interest rate spread as a function of the household's current income and the debt obligations it promises to repay in the following period. Because debt is priced competitively, the spread reflects the probability that the household goes delinquent next period. For strategic defaulters (Panel A), spreads rise steeply with promised repayment. The pricing schedule acts as an effective credit limit: while no explicit borrowing constraint exists in the model, high spreads bound the amount of funds households can borrow, and the bound tightens as income falls. For patient households (Panel B), the delinquency probability is low at moderate and high income levels, and their spreads are close to zero. Spreads start rising only when income is moderately low.

The distribution of equilibrium interest rates in the model resembles its empirical counterpart. Table 8 reports the model's distribution of APRs and the empirical distribution of effective APRs—interest and borrowing-related fees relative to balances—reported in Nelson (2025) for the near-universe of US credit cards. In the model, the median interest rate among borrowing households is 16.1%, with an interquartile range of 10.7% to 26.1%. Nelson (2025) reports a median of 15.8%, with an interquartile range of 10.1% to 24.7%. Even the tails of the two distributions align: the 10th and 90th percentiles are 5.0% and 35.1% in the model, against 4.5% and 35.8% in Nelson (2025). We emphasize that these rates are untargeted.⁴⁶ Additionally, both distributions are also similar to the distribution of effective APRs that we impute in the Experian data.

The interest rate schedules faced by low-income households effectively ration their access to credit. Part of the delinquency risk that lenders price stems from uninsured medical expenditure

⁴⁶The empirical distribution refers to accounts with no annual fee (the majority of accounts), from Appendix Table 2 of Nelson (2025), whose sample covers the period 2008–2014.

Figure 7: Interest Rate Spreads by Household Type



Notes: Each panel presents the equilibrium interest rate spread as a function of current income and promised debt repayment for households of the indicated type. Income and debt are in units of median income.

shocks. Increasing insurance against these shocks can expand credit access by lowering equilibrium interest rate spreads. In the next section, we study one such policy: the expansion of Medicaid.

4.3 Medicaid Expansion

We now use the model to study the expansion of Medicaid. To mimic the expansion, we replace the pre-expansion Medicaid coverage function with its post-expansion counterpart, estimated from the same ACS data (left panel of Figure 3). We keep the share of households covered by non-Medicaid insurance unchanged. This assumption is consistent with the data: the share of households with non-Medicaid insurance is virtually unchanged following the expansion (right panel of Figure 3), indicating that the expansion raised insurance coverage rather than crowding out private insurance. Overall, the expansion raises the share of households covered by Medicaid by 8.4 percentage points. The out-of-pocket share of households covered by Medicaid remains at its baseline value of 10.5%.

Table 8: Effective Credit Card APRs: Model, Data, and Nelson (2025)

Source	Percentile of the effective APR distribution				
	10th %	25th %	50th %	75th %	90th %
Model	5.0	10.7	16.1	26.1	35.1
Nelson (2025)	4.5	10.1	15.8	24.7	35.8
Data (imputed)	5.8	9.1	14.3	22.1	42.0

Notes: Each cell reports a percentile of the effective annual percentage rate (APR) distribution in percent. The first row reports these moments for our model. The second row reports the distribution reported in Appendix Table 2 of Nelson (2025), which reports fee-inclusive (i.e., effective) APRs for credit cards without annual fees using administrative card-level data. The last row reports the distribution of effective APRs we imputed using the Experian data following the procedures in Guttman-Kenney and Shahidinejad (2023) and Koşar et al. (2025).

The expansion is unanticipated and permanent.⁴⁷ We solve for the transition dynamics of the economy from the pre-expansion stationary distribution to the post-expansion one. We next provide further model validation by comparing model-implied responses of credit card borrowing, credit limits, and delinquency to their empirical counterparts. We then quantify the aggregate effects of the expansion on household debt and welfare, and decompose them into the direct, credit demand, and credit supply channels.

4.3.1 Replicating the Empirical Analysis in the Model

We first ask whether the model can reproduce the empirical estimates of Section 2. This exercise serves as an out-of-sample validation of the model’s key elasticities. To replicate the empirical design, we allocate households into ZIP codes. Using data on the income distribution of every US ZIP code from the IRS Statistics of Income (SOI), we assign households to ZIP codes in every period based on their current income, so that each ZIP code matches its observed income distribution at every date.⁴⁸ Because Medicaid coverage in the model varies with income, ZIP codes differ in their exposure to the expansion: enrollment rises more in ZIP codes with more low-income households. This variation in treatment intensity across ZIP codes mirrors the variation we exploit in the data.

Within the model, we compute the model counterpart of the DiD coefficient from Section 2 over the first five years following the expansion. For each ZIP code and year, we compute the change in each outcome relative to its level absent the expansion, and then regress these changes on the increase in the ZIP code’s Medicaid enrollment. One difference between the model and the data is that treatment intensity in the data is the share of a ZIP code’s population that becomes newly *eligible* for Medicaid, while in the model it is the share that becomes newly *enrolled*. To express the model-implied coefficients in eligibility units, we scale them by the ratio of the two: the average rise in Medicaid enrollment across ZIP codes in the model is 11.3 percentage points, while the average rise in eligibility across ZIP codes in the data is 19.45 percentage points (Table 2), implying a take-up rate of 58%.⁴⁹

Table 9 compares the model-implied responses to our empirical estimates. The model closely reproduces the response of revolving credit card debt—the coefficient of 0.42 lies within one standard error of the empirical estimate of 0.38. Credit limits, defined as the maximal amount of borrowing a household can obtain, expand in both the model and the data. The model’s response is somewhat more modest than the empirical point estimate—0.25 against 0.31—though the two are within one standard error of one another. In both the model and the data, utilization barely moves: the additional credit extended by lenders keeps pace with the additional debt households take on. Delinquency falls

⁴⁷We note that in the model, Medicaid expands everywhere simultaneously, whereas in the data only a subset of states expanded, and they did so in a staggered fashion.

⁴⁸Specifically, we use the SOI data to obtain each ZIP code’s population shares across twenty bins of the national income distribution, which we map into the model’s income states. Conditional on income, every ZIP code inherits the national composition of wealth, delinquency status, and household types at each date, so ZIP codes differ only in their income mix. Appendix E.3 reports results under the alternative of assigning households to ZIP codes once, at the policy date, and following them over time.

⁴⁹This implied take-up rate is consistent with direct estimates in the literature. The estimates in Abdus (2025) imply that take-up among Medicaid-eligible adults averaged 60% over 2014–2019.

Table 9: The Response to the Medicaid Expansion: Model versus Data

	log(CC Rev. Bal.)	log(CC Lim.)	CC Util. (pp)	Delinq. (pp)	Collections (pp)
Data	0.383 (0.063)	0.313 (0.090)	0.011 (0.015)	−0.022 (0.010)	−0.119 (0.022)
Model	0.422	0.251	0.041		−0.074

Notes: Each column reports the effect of a one percentage point increase in the share of a ZIP code’s population that is newly eligible for Medicaid, in the data (standard errors, clustered by state, in parentheses) and in the model. The data coefficients are from Section 2; the last two columns are the indicators for having debt 90 or more days past due and for having debt in collections. The model does not distinguish between delinquency and collections: its single counterpart is the share of households in credit-card delinquency, which spans both empirical measures. Model-implied coefficients are converted from enrollment to eligibility units using the implied take-up rate of 58%.

in both the model and the data, and the model’s response lies between the empirical estimates for 90-day delinquency and for debt in collections, consistent with the model’s delinquent state pooling the two. Overall, the model’s ZIP-level DiD responses of borrowing, credit limits, utilization, and default are consistent with those we estimate in the data.

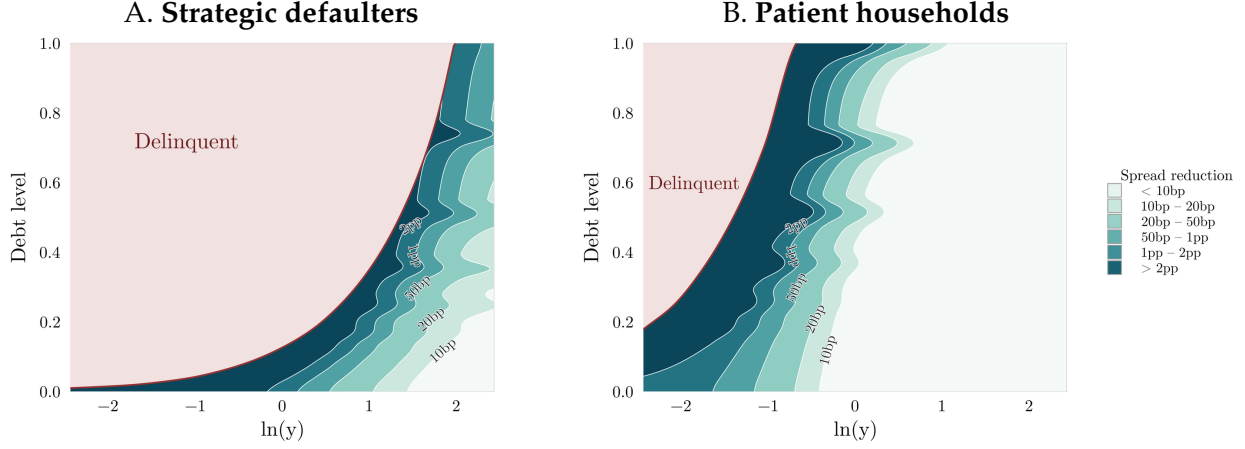
4.3.2 Aggregate Implications and Decomposition

We now turn to the aggregate effects of the expansion of Medicaid. By reducing the out-of-pocket share of expenditure shocks, the expansion lowers delinquency risk. The reduction in risk is passed through to interest rates, and households’ borrowing terms improve. Figure 8 presents the reduction in interest rate spreads across the state space. The reduction is largest close to the delinquency frontier, because the absolute reduction in delinquency risk is largest in this region of the state space. The lower interest rates households face raise their effective credit limits. The credit limit faced by strategic defaulters at the bottom of the income distribution expands by 17%, with the expansion declining monotonically with income. The long-run reduction in the median interest rate on credit card borrowing in the economy is 1.5 pp, falling from 16.1% to 14.6%. This decline is close to what is implied by our estimates in Appendix Table A.10: multiplying the impact of eligibility on effective APRs (−4.9 pp) by the 19.45 pp increase in the eligible population (Table 2) implies a 0.95 pp decline.

Cheaper credit induces households to borrow more. In the long run, the expansion of Medicaid raises aggregate credit card debt by 6.7%. The increase is concentrated at the bottom of the income distribution, where the reduction in spreads is largest. Credit card debt rises by 22% in the bottom income quintile and by 17% in the second, compared with 3% in the fourth, while the top quintile marginally reduces its borrowing.

The aggregate response builds gradually. Figure 9 plots the path of aggregate credit card debt following the expansion. In the first year following the expansion, credit card debt has closed 36% of the gap to its new long-run level, and by year five, 78%. In the baseline, lenders reprice credit immediately upon the expansion. In practice, lenders might learn about improved repayment behavior gradually as credit scores take time to adjust. To capture this, we also consider a version of the model where the debt pricing schedule sluggishly responds. Specifically, we assume the interest

Figure 8: Reduction in Interest Rate Spreads following the Medicaid Expansion



Notes: Each panel presents the reduction in the equilibrium interest rate spread following the expansion of Medicaid, as a function of current income and promised debt repayment, for households of the indicated type. The shaded bands indicate the size of the reduction, from 10 basis points to over 2 percentage points. The red region marks the states in which households go delinquent. Debt is in units of median income.

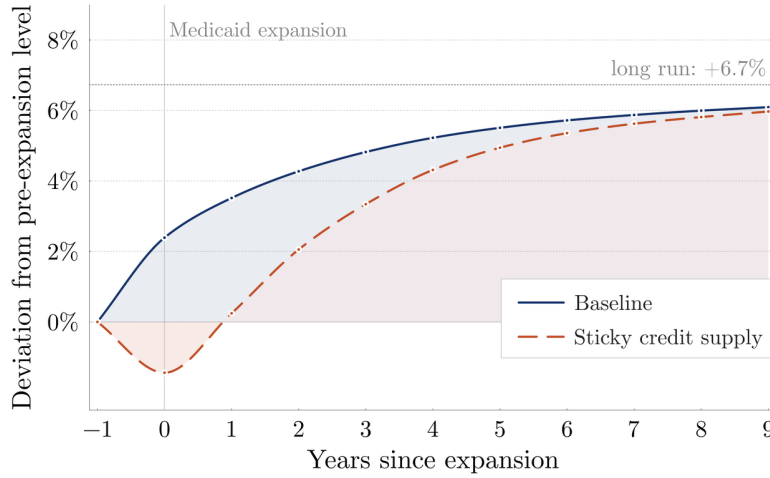
rate schedule in year t of the transition is the average between last year's schedule and the *fair* schedule.⁵⁰ Sticky credit supply slows the transition down: debt closes 64% of the long-run gap by year five, compared with 78% in the baseline. More strikingly, credit card debt initially *falls* below its pre-expansion level. This is because the direct and credit demand effects of coverage arrive immediately, reducing debt levels, while the credit supply channel arrives gradually.

Our model allows us to decompose the rise in credit card debt into the three channels laid out in Section 3.3: the direct effect, the credit demand channel, and the credit supply channel. We compute the direct effect by changing the out-of-pocket share, $o(y)$, while holding both households' consumption and delinquency policy functions and the debt pricing schedule fixed at their pre-expansion levels. The credit demand channel additionally allows households to re-optimize their consumption, borrowing, and default, still facing the pre-expansion debt pricing schedule. Finally, the credit supply channel also updates the debt pricing schedule as illustrated in Figure 8. Table 10 reports the decomposition, for both the baseline expansion and a budget-neutral version we introduce below.

The decomposition shows that the rise in credit card debt is entirely driven by the credit supply channel. The direct effect is small: holding behavior and prices fixed, better coverage reduces the medical expenditures households finance with debt, lowering credit card debt by 0.5%. The credit demand channel further reduces debt by a 5.4%. Recall that the demand channel nets two opposing forces. Its decline indicates that the debt aversion motive dominates the precautionary savings motive, leading to a reduction in credit card debt. Finally, the credit supply channel raises credit card debt by 12.6%, more than offsetting both. Absent the response of lenders, the expansion of Medicaid

⁵⁰That is, $q_t(b', y) = 0.5q_{t-1}(b', y) + 0.5q_t^f(b', y)$, where q_t^f is the actuarially fair schedule at date t which satisfies equation (7). By taking a simple average, we implicitly assume a repricing half-life of one year, consistent with the weight credit scoring models place on the most recent two years of payment history.

Figure 9: Transition Dynamics of Aggregate Credit Card Debt



Notes: The figure plots aggregate credit card debt following the expansion of Medicaid, as a percentage deviation from its pre-expansion level. The solid line is the baseline, in which the debt pricing schedule adjusts immediately. The dashed line is the sticky credit supply variant, in which the pricing schedule converges gradually to its post-expansion level. The dotted horizontal line marks the long-run increase of 6.7%.

would have *reduced* household debt.

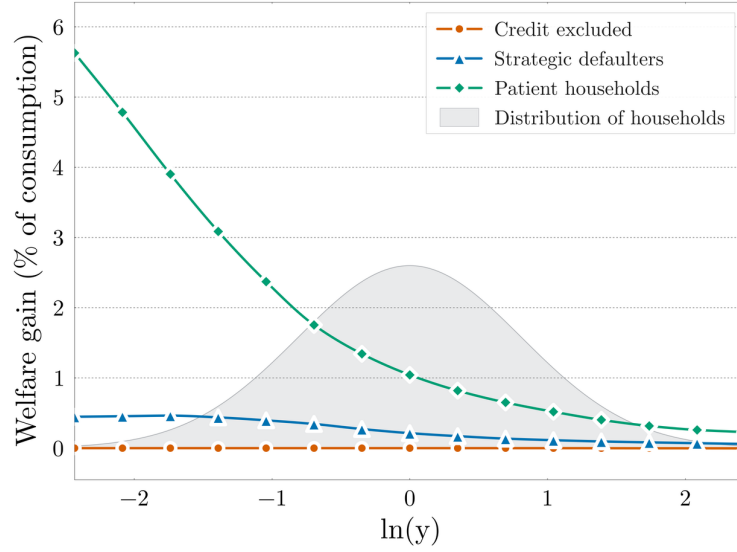
We now turn to the welfare implications of the expansion. Following [Chatterjee et al. \(2007\)](#), we compute welfare as the permanent percentage change in consumption that would make households indifferent between the pre-expansion economy and the economy with the expansion, taking the transition dynamics into account. The unfunded expansion generates an average welfare gain across all households of 0.24% in consumption-equivalent terms. The direct effect accounts for the majority of the gain, 0.18%, as households' out-of-pocket medical expenditure declines. The credit demand channel contributes essentially nothing. This is the envelope theorem at work. Households were optimizing their borrowing decision prior to the policy. So adjusting their borrowing decision following

Table 10: Decomposing the Effect of the Medicaid Expansion

	Total	Direct	Credit demand	Credit supply
<i>Not budget neutral</i>				
Credit card debt	+6.73%	−0.49%	−5.41%	+12.63%
Welfare	+0.24%	+0.18%	+0.00%	+0.07%
<i>Budget neutral</i>				
Credit card debt	+6.81%	−0.49%	−5.40%	+12.70%
Welfare	+0.22%	+0.16%	+0.00%	+0.07%

Notes: The table decomposes the effect of the Medicaid expansion on aggregate credit card debt and on welfare into the three channels described in the text. The credit card debt effect is the long-run change, in percent of the pre-expansion level. The welfare effect is in consumption-equivalent percent and takes into account the transition dynamics following the expansion. The top panel reports the unfunded expansion and the bottom panel reports the budget-neutral expansion, financed by a uniform income tax.

Figure 10: Welfare Gains from the Medicaid Expansion across Households



Notes: The figure presents the welfare gain from the unfunded expansion of Medicaid, in consumption-equivalent terms, as a function of household income, for each of the three household types. Welfare gains take into account the transition dynamics following the expansion. Income is expressed in logs, relative to median income. The shaded region plots the distribution of households across income levels.

the policy can only lead to second-order welfare gains. The credit supply channel, in contrast, contributes 0.07%, more than a quarter of the total welfare gain. Unlike their borrowing decisions, the interest rate schedule is not an object households choose, so its improvement generates first-order welfare gains.

We next consider a budget-neutral version of the expansion, financed by a uniform tax levied on all households. The expansion of Medicaid reduces uncompensated medical care, and we set the tax so that revenues equal the cost of the expansion net of this reduction.⁵¹ The resulting uniform tax is 0.02% of income. The bottom panel of Table 10 reports the results. The response of credit card debt is essentially unchanged, and the welfare gain falls modestly to 0.22%, with the entire reduction coming through the direct channel. The credit supply channel amplifies the direct welfare benefit by more than 40%, accounting for 30% of the total net welfare gains. Neglecting the effect of Medicaid on the supply of credit would therefore understate its welfare benefits by nearly a third.

The welfare gains from the expansion vary starkly across households. Figure 10 plots the average welfare gains by income for each of the three household types under the unfunded expansion. Two patterns stand out. First, within each type, gains decline with income, reflecting the concentration of the expanded coverage among low-income households. Second, across types, gains line up with the extent to which households actually pay their medical bills. The patient households gain the most: 1.3% of consumption on average, and nearly 6% at the bottom of the income distribution. Because

⁵¹Uncompensated medical care is defined as the sum of unpaid out-of-pocket medical bills net of the market value of medical debt due to unpaid medical bills.

these households repay their debts, they gain the most from the reduction in their out-of-pocket medical expenditures. The strategic defaulters gain a more modest 0.2%: they pay some of their medical bills, and they benefit from the improvement in their borrowing terms. The credit-excluded households do not benefit from the policy. Bearing no cost of delinquency, they do not pay their medical bills, so the bills that Medicaid now covers are bills they would not have paid.

5 Conclusion

This paper investigates how social insurance affects household debt. We exploit heterogeneity in the impact of the staggered expansions of Medicaid as a source of quasi-experimental variation in households' access to health insurance. Using a difference-in-differences design, we estimate that a 1 percentage point increase in the share of a ZIP code's population eligible for Medicaid increases revolving credit card balances by 0.38%. Consistent with credit supply increasing in response to expanded Medicaid, we find that credit limits rise, and that credit card inquiries become more successful in below-median-income ZIP codes. In support of improved financial resilience driving this rise in credit supply, we show that debt in collections falls, and delinquencies also fall despite the increase in debt. Credit scores increase as well.

Our paper builds on prior empirical work by quantifying the role of credit markets and by studying both aggregate and distributional outcomes. We develop an incomplete-markets heterogeneous-agent model in which households face income and medical expenditure shocks. The model reproduces our estimated response of credit card debt without targeting it, and we use it to recover the aggregate effect of expanding health insurance and to decompose it into the direct effect and the responses of credit demand and credit supply. The credit supply channel more than accounts for the rise in credit card debt, and it contributes 30% of the net welfare gains of the expansion. We conclude that social insurance can have a large impact on the quantity and distribution of household debt, and that the response of credit markets shapes much of this impact. Neglecting the credit supply channel could significantly understate the welfare gains of social insurance programs.

References

- Aaronson, Daniel, Sumit Agarwal, and Eric French, "The Spending and Debt Response to Minimum Wage Hikes," *American Economic Review*, 2012, 102 (7), 3111–3139.
- Abdus, Salam, "Trends in Medicaid Take-Up Among Eligible Adults After the Affordable Care Act Medicaid Expansions: 2014–2019," *Medical Care Research and Review*, 2025, 82 (1), 100–106.
- Adams, Robert, Vialty Bord, and Bradley Katcher, "Credit Card Profitability," *FEDS Notes*, 2022.
- Aguiar, Mark A, Mark Bils, and Corina Boar, "Who Are the Hand-to-Mouth?," *Review of Economic Studies*, 2025, 92 (3), 1293–1340.
- Aiyagari, Rao S., "Uninsured Idiosyncratic Risk and Aggregate Saving," *The Quarterly Journal of Economics*, 1994, 109 (3), 659–684.
- Allen, Heidi, Ashley Swanson, Jialan Wang, and Tal Gross, "Early Medicaid Expansion Associated with Reduced Payday Borrowing in California," *Health Affairs*, 2017, 36 (10), 1769–1776.
- Arellano, Cristina, "Default risk and income fluctuations in emerging economies," *American economic review*, 2008, 98 (3), 690–712.
- Attanasio, Orazio, Sagiri Kitao, and Giovanni L Violante, "Financing Medicare: A general equilibrium analysis," in "Demography and the Economy," University of Chicago Press, 2010, pp. 333–366.
- Baicker, Katherine, Amy Finkelstein, Jae Song, and Sarah Taubman, "The Impact of Medicaid on Labor Market Activity and Program Participation: Evidence from the Oregon Health Insurance Experiment," *American Economic Review*, 2014, 104 (5), 322–328.
- Bankrate, "Mortgage Rates by Credit Score," <https://www.bankrate.com/real-estate/what-credit-score-do-you-need-to-buy-a-house/> 2025.
- Bewley, Truman, "Stationary Monetary Equilibrium with a Continuum of Independently Fluctuating Consumers," *Contributions to Mathematical Economics in Honor of Gérard Debreu*, 1986, 79.
- Blanchet, Thomas, Juliette Fournier, and Thomas Piketty, "Generalized Pareto Curves: Theory and Applications," *Review of Income and Wealth*, 2022, 68 (1), 263–288.
- Braxton, J. Carter, Kyle Herkenhoff, and Gordon M. Phillips, "Can the Unemployed Borrow? Implications for Public Insurance," *Journal of Political Economy*, 2024, 132 (9), 3025–3076.
- Brevoort, Kenneth, Daniel Grodzicki, and Martin B. Hackmann, "The credit consequences of unpaid medical bills," *Journal of Public Economics*, 2020, 187, 104203.
- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer, "The effect of minimum wages on low-wage jobs," *The Quarterly Journal of Economics*, 2019, 134 (3), 1405–1454.
- Chatterjee, Satyajit and Burcu Eyigungor, "Maturity, indebtedness, and default risk," *American Economic Review*, 2012, 102 (6), 2674–2699.
- , Dean Corbae, Makoto Nakajima, and José-Víctor Ríos-Rull, "A Quantitative Theory of Unsecured Consumer Credit with Risk of Default," *Econometrica*, 2007, 75 (6), 1525–1589.
- Congressional Budget Office, "Letter to John Boehner regarding H.R. 6079, the Repeal of Obamacare Act," 2012, <https://perma.cc/Q9VQ-C63D>.

- Consumer Financial Protection Bureau**, “Consumer Credit Card Market Report,” Report to Congress, Consumer Financial Protection Bureau 2023.
- , “The Consumer Credit Card Market,” Technical Report, Consumer Financial Protection Bureau 2025.
- Deaton, Angus**, “Saving and Liquidity Constraints,” *Econometrica*, 1991, 59 (5), 1221–48.
- Dranove, David, Craig Garthwaite, and Christopher Ody**, “Uncompensated care decreased at hospitals in Medicaid expansion states but not at hospitals in nonexpansion states,” *Health Affairs*, 2016, 35 (8), 1471–1479.
- Drozd, Lukasz A. and Ricardo Serrano-Padial**, “Modeling the Revolving Revolution: The Debt Collection Channel,” *American Economic Review*, 2017, 107 (3), 897–930.
- Druehl, Jeppe and Casper Nardal Jørgensen**, “Precautionary Borrowing and the Credit Card Debt Puzzle,” *Quantitative Economics*, 2018, 9 (2), 785–823.
- Experian**, “State of the Automotive Finance Market: Q3 2025,” <https://www.experian.com/content/dam/noindex/na/us/automotive/finance-trends/2025/Experian-SAFM-Q3-2025.pdf> September 2025.
- , “2026 State of the U.S. Housing Market Report,” <https://www.experian.com/content/dam/marketing/na/thought-leadership/business/documents/2026-state-of-housing-report.pdf> January 2026.
- Finkelstein, Amy, Sarah Taubman, Bill Wright, Mira Bernstein, Jonathan Gruber, Joseph P Newhouse, Heidi Allen, Katherine Baicker, and Oregon Health Study Group**, “The Oregon health insurance experiment: evidence from the first year,” *The Quarterly journal of economics*, 2012, 127 (3), 1057–1106.
- Fulford, Scott and David Low**, “Expense Shocks Matter,” CFPB Office of Research Working Paper 24-08, Consumer Financial Protection Bureau 2024.
- Gallagher, Emily A., Radhakrishnan Gopalan, and Michael Grinstein-Weiss**, “The Effect of Health Insurance on Home Payment Delinquency: Evidence from ACA Marketplace Subsidies,” *Journal of Public Economics*, 2019, 172, 67–83.
- , —, **Michal Grinstein-Weiss, and Jorge Sabat**, “Medicaid and Household Savings Behavior: New Evidence from Tax Refunds,” *Journal of Financial Economics*, 2020, 136 (2), 523–546.
- Garthwaite, Craig, John A Graves, Tal Gross, Zeynal Karaca, Victoria R Marone, and Matthew J Notowidigdo**, “All Medicaid Expansions Are Not Created Equal: The Geography and Targeting of the Affordable Care Act,” *Brookings Papers on Economic Activity*, 2019, 2019 (2), 1–92.
- Goldsmith-Pinkham, Paul, Maxim Pinkovskiy, and Jacob Wallace**, “The Great Equalizer: Medicare and the Geography of Consumer Financial Strain,” 2025.
- Goodman-Bacon, Andrew**, “Public insurance and mortality: evidence from Medicaid implementation,” *Journal of Political Economy*, 2018, 126 (1), 216–262.
- , “Difference-in-differences with variation in treatment timing,” *Journal of Econometrics*, 2021, 225 (2), 254–277.

- , “The long-run effects of childhood insurance coverage: Medicaid implementation, adult health, and labor market outcomes,” *American Economic Review*, 2021, 111 (8), 2550–93.
- Gordon, Grey and Shi Qiu**, “A Divide and Conquer Algorithm for Exploiting Policy Function Monotonicity,” *Quantitative Economics*, 2018, 9 (2), 521–540.
- Gross, David B. and Nicholas S. Souleles**, “Do Liquidity Constraints and Interest Rates Matter for Consumer Behavior? Evidence from Credit Card Data,” *Quarterly Journal of Economics*, 2002, 117 (1), 149–185.
- Gross, Tal and Matthew J. Notowidigdo**, “Health Insurance and the Consumer Bankruptcy Decision: Evidence from Expansion Medicaid,” *Journal of Public Economics*, 2011, 95 (7), 767–778.
- Guttman-Kenney, Benedict and Andres Shahidinejad**, “Unraveling Information Sharing in Consumer Credit Markets,” SSRN Working Paper 4629496, Social Science Research Network 2023.
- Guvenen, Fatih, Fatih Karahan, Serdar Ozkan, and Jae Song**, “What Do Data on Millions of U.S. Workers Reveal About Lifecycle Earnings Dynamics?,” *Econometrica*, 2021, 89 (5), 2303–2339.
- Hatchondo, Juan Carlos and Leonardo Martinez**, “Long-duration bonds and sovereign defaults,” *Journal of international Economics*, 2009, 79 (1), 117–125.
- Heidhues, Paul and Botond Köszegi**, “Exploiting Naïveté about Self-Control in the Credit Market,” *American Economic Review*, 2010, 100 (5), 2279–2303.
- Herkenhoff, Kyle F and Gajendran Raveendranathan**, “Who bears the welfare costs of monopoly? the case of the credit card industry,” *Review of Economic Studies*, 2025, 92 (5), 3067–3111.
- Himmelstein, David U, Deborah Thorne, Elizabeth Warren, and Steffie Woolhandler**, “Medical bankruptcy in the United States, 2007: results of a national study,” *The American journal of medicine*, 2009, 122 (8), 741–746.
- Hsu, Joanne W., David A. Matsa, and Brian T. Melzer**, “Unemployment Insurance as a Housing Market Stabilizer,” *American Economic Review*, 2018, 108 (1), 49–81.
- Hu, LuoJia, Robert Kaestner, Bhashkar Mazumder, Sarah Miller, and Ashley Wong**, “The Effect of the Affordable Care Act Medicaid Expansions on Financial Wellbeing,” *Journal of Public Economics*, 2018, 163, 99–112.
- Hubbard, R Glenn, Jonathan Skinner, and Stephen P Zeldes**, “Precautionary saving and social insurance,” *Journal of political Economy*, 1995, 103 (2), 360–399.
- Huggett, Mark**, “The Risk-Free Rate in Heterogeneous-Agent Incomplete-Insurance Economies,” *Journal of Economic Dynamics and Control*, 1993, 17 (5–6), 953–969.
- Imrohoroglu, Ayşe**, “Cost of business cycles with indivisibilities and liquidity constraints,” *Journal of Political economy*, 1989, 97 (6), 1364–1383.
- Kaestner, Robert, Bowen Garrett, Jiajia Chen, Anuj Gangopadhyaya, and Caitlyn Fleming**, “Effects of ACA Medicaid expansions on health insurance coverage and labor supply,” *Journal of Policy Analysis and Management*, 2017, 36 (3), 608–642.
- Kaplan, Greg and Giovanni L Violante**, “A model of the consumption response to fiscal stimulus payments,” *Econometrica*, 2014, 82 (4), 1199–1239.

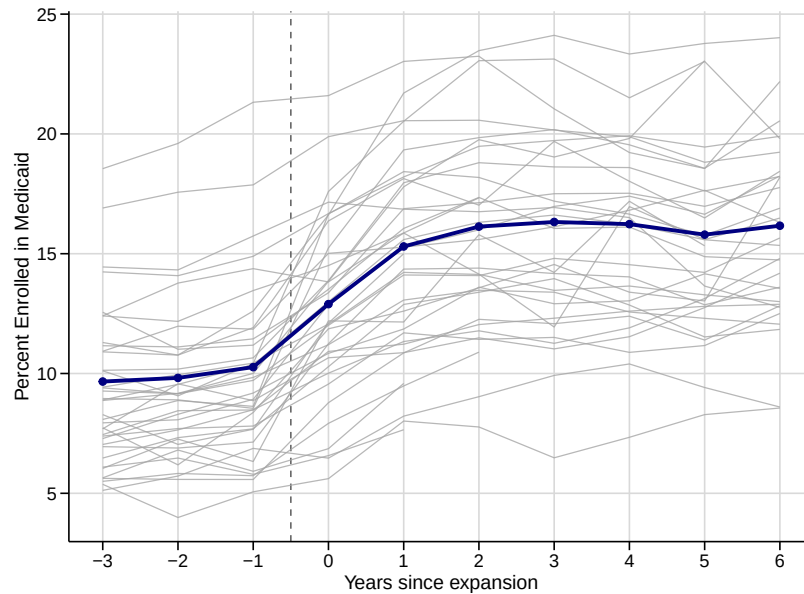
- Kluender, Raymond, Neale Mahoney, Francis Wong, and Wesley Yin**, “The Effects of Medical Debt Relief: Evidence from Two Randomized Experiments,” *Quarterly Journal of Economics*, 2025, 140 (2), 1187–1241.
- Koşar, Gizem, Davide Melcangi, Laura Pilossoph, and David G. Wiczer**, “Stimulus through Insurance: The Marginal Propensity to Repay Debt,” Working Paper 34399, National Bureau of Economic Research October 2025.
- Krueger, Dirk and Fabrizio Perri**, “Public versus private risk sharing,” *Journal of Economic Theory*, 2011, 146 (3), 920–956.
- Leung, Pauline and Alexandre Mas**, “Employment Effects of the ACA Medicaid Expansions,” *Industrial Relations*, 2018, 57, 206–234.
- Livshits, Igor, James MacGee, and Michele Tertilt**, “Consumer Bankruptcy: A Fresh Start,” *American Economic Review*, 2007, 97 (1), 402–418.
- Manning, Willard G, Joseph P Newhouse, Naihua Duan, Emmett B Keeler, and Arleen Leibowitz**, “Health insurance and the demand for medical care: evidence from a randomized experiment,” *The American economic review*, 1987, pp. 251–277.
- Matsa, David A., Brian T. Melzer, and Michal Zator**, “Dual Credit Markets: Income Risk, Household Debt, and Consumption,” *Working Paper*, 2026.
- Mazumder, Bhashkar and Sarah Miller**, “The effects of the Massachusetts health reform on household financial distress,” *American Economic Journal: Economic Policy*, 2016, 8 (3), 284–313.
- Miller, Sarah, Luojia Hu, Robert Kaestner, Bhashkar Mazumder, and Ashley Wong**, “The ACA Medicaid Expansion in Michigan and Financial Health,” *Journal of Policy Analysis and Management*, 2021, 40 (2), 348–375.
- , **Norman Johnson, and Laura R Wherry**, “Medicaid and mortality: new evidence from linked survey and administrative data,” *The Quarterly Journal of Economics*, 2021, 136 (3), 1783–1829.
- Mitman, Kurt**, “Macroeconomic Effects of Bankruptcy and Foreclosure Policies,” *American Economic Review*, 2016, 106 (8), 2219–2255.
- Nardi, Mariacristina De, Eric French, and John B Jones**, “Why do the elderly save? The role of medical expenses,” *Journal of political economy*, 2010, 118 (1), 39–75.
- Nelson, Scott T**, “Private information and price regulation in the us credit card market,” *Econometrica*, 2025, 93 (4), 1371–1410.
- Rambachan, Ashesh and Jonathan Roth**, “A more credible approach to parallel trends,” *Review of Economic Studies*, 2023, 90 (5), 2555–2591.
- Roth, Jonathan, Pedro H. C. Sant’Anna, Alyssa Bilinski, and John Poe**, “What’s Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature,” *Journal of Econometrics*, 2023, 235 (2), 2218–2244.
- Shupe, Cortnie**, “Public Health Insurance and Medical Spending: The Incidence of the ACA Medicaid Expansion,” *Journal of Policy Analysis and Management*, 2023, 42 (1), 137–165.

- Simon, Kosali, Aparna Soni, and John Cawley**, “The impact of health insurance on preventive care and health behaviors: evidence from the first two years of the ACA Medicaid expansions,” *Journal of Policy Analysis and Management*, 2017, 36 (2), 390–417.
- Sullivan, James X.**, “Borrowing during Unemployment: Unsecured Debt as a Safety Net,” *Journal of Human Resources*, 2008, 43 (2), 383–412.
- Tauchen, George**, “Finite State Markov-Chain Approximations to Univariate and Vector Autoregressions,” *Economics Letters*, 1986, 20 (2), 177–181.
- Telyukova, Irina A.**, “Household Need for Liquidity and the Credit Card Debt Puzzle,” *Review of Economic Studies*, 2013, 80 (3), 1148–1177.
- Zeldes, Stephen P.**, “Optimal consumption with stochastic income: Deviations from certainty equivalence,” *The Quarterly Journal of Economics*, 1989, 104 (2), 275–298.
- Zinman, Jonathan**, “Where is the missing credit card debt? Clues and implications,” *Review of income and wealth*, 2009, 55 (2), 249–265.
- , “Consumer Credit: Too Much or Too Little (or Just Right)?,” *Journal of Legal Studies*, 2014, 43 (S2), S209–S237.

Online Appendix

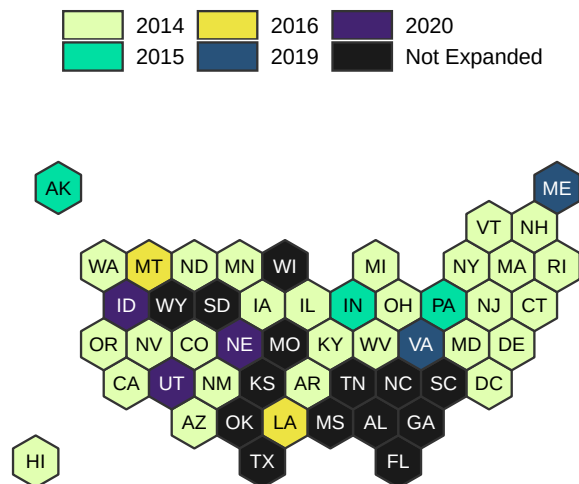
A Additional Figures and Tables

Figure A.1: Medicaid Enrollment Around the Expansions



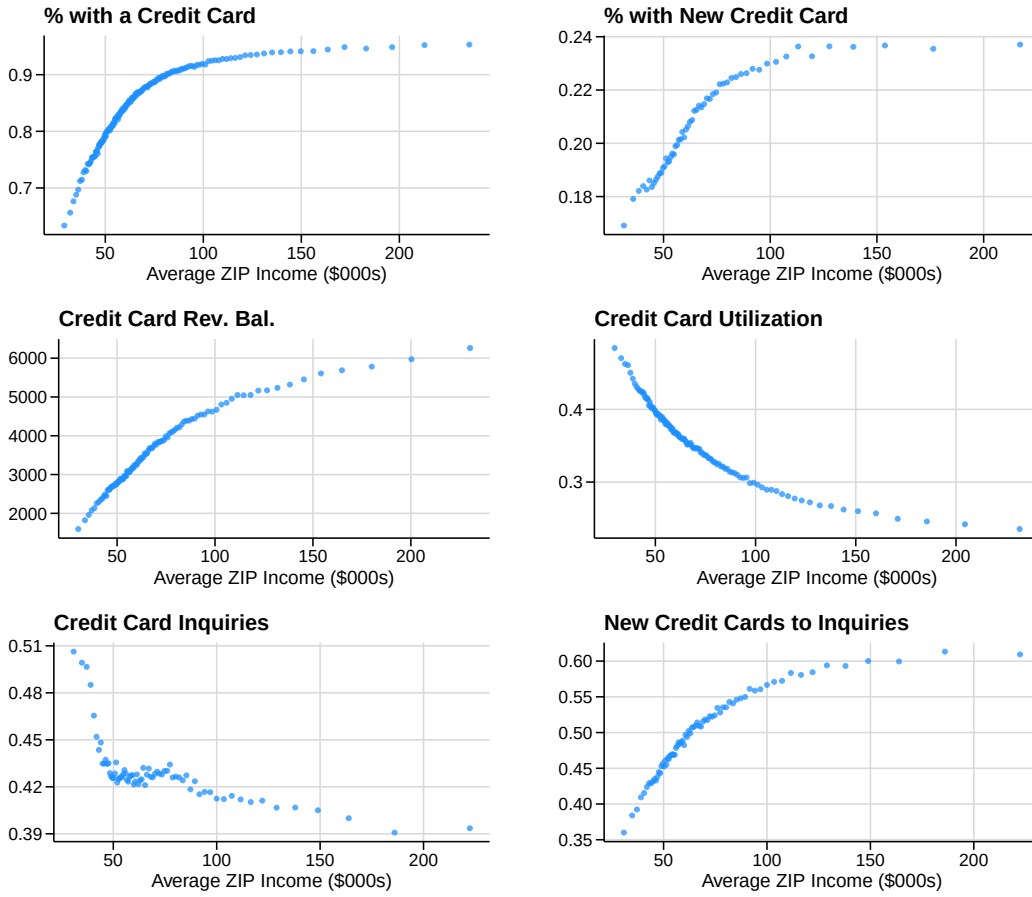
Notes: This figure plots Medicaid enrollment of prime-age adults (25–55) for states that expanded Medicaid, where time zero is the year a state’s expansion took effect. Each gray line is a state, and the navy line is the average across states. Source: American Community Survey (ACS), 2010 to 2021.

Figure A.2: Medicaid Expansion Dates



Notes: Expansion dates come from the Kaiser Family Foundation.

Figure A.3: ZIP-Level Credit Access and Income Facts



Notes: This figure plots binscatters of credit outcomes against average income, with all variables measured as the average within a ZIP code for a given year. All nominal variables deflated to 2020 dollars.

Table A.1: Adding Flexible Income Controls (Revolving Credit Card Balances)

	(1)	(2)	(3)	(4)
$\text{NewElig}_{zs} \times \text{Post}_{st}$	0.478*** (0.098)	0.502*** (0.086)	0.383*** (0.063)	0.365*** (0.061)
Obs	339,785	339,785	339,785	339,785
R2	0.952	0.952	0.952	0.953
Percentile Controls	None	Quartiles	Deciles	5th, 10th, ... 95th

Notes: This table reports results from regressions that control for additional moments from the ZIP code's income distribution for our main outcome of interest: log revolving credit card balances. All columns control for log average AGI (coefficients omitted). Column (2) adds controls for logged quartiles (i.e., the 25th, 50th, and 75th percentiles of the ZIP code's income distribution). Column (3) instead uses deciles (the 10th, 20th, ..., 90th percentiles). Column (4) uses 19 percentiles ranging from the 5th to 95th percentile. Standard errors are clustered by state. Nominal variables are CPI-adjusted to be in terms of 2020 dollars. Statistical significance: 10%*, 5%** , and 1%***.

Table A.2: Placebo Test: Income Interactions

	(1)	(2)	(3)
$\text{NewElig}_{zs} \times \text{Post}_{st}$	0.383*** (0.063)	0.376*** (0.056)	0.452*** (0.100)
$\text{HighInc}_{zst} \times \text{Post}_{st}$		-0.001 (0.004)	
$\log(\text{AGI}_{zst}^{\text{avg}}) \times \text{Post}_{st}$			0.010 (0.011)
Obs	339,785	339,785	339,785
R2	0.952	0.952	0.952

Notes: The dependent variable in all columns is log revolving credit card balances. Column (1) here reproduces our baseline estimate from Column (4) of Table 3 to facilitate comparison. Each specification uses county-time and ZIP-code fixed effects and controls for logged ZIP-level average AGI and logged deciles of the ZIP code's income distribution (coefficients omitted). Column (2) adds an interaction of the post-expansion indicator with a binary indicator HighInc_{zst} which equals one if ZIP code z had average AGI above the median of all ZIP codes' average AGI in year t . An un-interacted HighInc_{zst} is included too but its coefficient is omitted for brevity. Column (3) adds an interaction of the post-expansion indicator with ZIP z 's logged average AGI in year t (recall that un-interacted log AGI is already included as a control). Standard errors are clustered by state. Nominal variables are CPI-adjusted to 2020 dollars. Statistical significance: 10%*, 5%** , 1%***.

A.1 Treatment Effect Heterogeneity Robustness

We implement the stacked DiD as follows. We first split our sample into several groups based on their expansion year (2014, 2015, 2016, 2019, or 2020). For each group of expanders, we append rows for the non-expanders. We create a group identifier for each group and then "stack" observations for each group into a single data table. We then estimate a modified DiD regression, specified below:

$$Y_{zctg} = \beta (\text{Post}_{stg} \times \text{NewElig}_{zs}) + \phi_{zg} + \phi_{ctg} + \gamma X_{zct} + \varepsilon_{zct}.$$

The regression above modifies our baseline, Equation (1), by allowing the ZIP and county-time fixed effects (ϕ_{zg} and ϕ_{ctg} , respectively) to vary by group g . Estimating the modified equation results in effectively estimating the treatment effect for each group separately and then taking an average of these estimated treatment effects.

We create "balanced panels" by restricting the stacked subsamples to include the same number of pre- and post-expansion years. Since only two and one years of post-expansion data are available for the 2019 and 2020 expansions (respectively), in this robustness analysis we use only data for 2014, 2015, and 2016 expanders (constituting 93% of our original sample). Because including later expansions comes at the cost of including fewer years of data, and treatment effects appear to grow over time (Figure 4), we present three sets of results that vary in whether they include the 2015 and 2016 expanders. To gauge the bias induced by "bad comparisons," the tables also display TWFE estimates for the same subsamples.

Table A.3: Comparison of Non-Stacked and Stacked DiD Estimates (1/3)

	Expand = 2014		Expand $\leq$ 2015		Expand $\leq$ 2016	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Outcome = 1[Has CC]						
NewElig _{zs} $\times$ Post _{st}	0.078*** (0.017)	0.084*** (0.018)	0.067*** (0.016)	0.065*** (0.017)	0.050*** (0.016)	0.043** (0.018)
Obs	57,235	92,797	58,764	122,767	53,543	138,822
R2	0.970	0.976	0.971	0.978	0.974	0.980
Panel B: Outcome = 1[New CC]						
NewElig _{zs} $\times$ Post _{st}	0.150*** (0.015)	0.149*** (0.016)	0.142*** (0.015)	0.134*** (0.017)	0.131*** (0.014)	0.121*** (0.017)
Obs	57,235	92,797	58,764	122,767	53,543	138,822
R2	0.825	0.851	0.833	0.861	0.848	0.870
Panel C: Outcome = log(CC Bal.)						
NewElig _{zs} $\times$ Post _{st}	0.703*** (0.069)	0.742*** (0.077)	0.629*** (0.055)	0.644*** (0.067)	0.564*** (0.047)	0.567*** (0.058)
Obs	181,119	291,879	195,058	409,100	176,716	473,375
R2	0.968	0.968	0.968	0.968	0.970	0.969
Panel D: Outcome = log(CC Bal. Rev.)						
NewElig _{zs} $\times$ Post _{st}	0.362*** (0.065)	0.401*** (0.073)	0.309*** (0.049)	0.321*** (0.060)	0.254*** (0.043)	0.248*** (0.055)
Obs	181,117	291,877	195,056	409,098	176,714	473,373
R2	0.953	0.954	0.952	0.954	0.954	0.956
Window	-3 to +6 years		-3 to +5 years		-3 to +4 years	
Stacked	No	Yes	No	Yes	No	Yes

Notes: This table reports results from estimating the "stacked" version of the DiD in Equation (1). Columns (1) and (2) use the subsample of observations whose states expanded in 2014 using data from 3 years before to 6 years after expansion (the window). Columns (3) and (4) expand the sample to observations whose states expanded in 2015 or earlier but use a narrower window of 3 years before to 5 years after. Columns (5) and (6) expand the sample to include observations whose states expanded in 2016 or earlier, but restrict the window to 3 years before to 4 years after expanding. Each panel reports these six regressions for different outcome variables. To facilitate comparisons, the odd-numbered columns display non-stacked regression results for the subsamples used in the stacked regressions. Each specification uses county-time and ZIP-code fixed effects and controls for logged ZIP-level average adjusted gross income (AGI) and logged deciles of the ZIP code's income distribution (coefficients omitted). Standard errors are clustered by state. Nominal variables are CPI-adjusted to be in terms of 2020 dollars. Statistical significance: 10%*, 5%***, and 1%***.

Table A.4: Comparison of Non-Stacked and Stacked DiD Estimates (2/3)

	Expand = 2014		Expand $\leq$ 2015		Expand $\leq$ 2016	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Outcome = CC Inquiries (#)						
NewElig _{zs} $\times$ Post _{st}	0.359*** (0.034)	0.363*** (0.034)	0.338*** (0.032)	0.336*** (0.033)	0.308*** (0.030)	0.297*** (0.033)
Obs	57,235	92,797	58,764	122,767	53,543	138,822
R2	0.924	0.932	0.927	0.937	0.932	0.939
Panel B: Outcome = log(CC Limits)						
NewElig _{zs} $\times$ Post _{st}	0.387*** (0.085)	0.412*** (0.093)	0.313*** (0.065)	0.294*** (0.080)	0.263*** (0.059)	0.221*** (0.077)
Obs	57,235	92,797	58,764	122,767	53,543	138,822
R2	0.985	0.986	0.986	0.986	0.988	0.987
Panel C: Outcome = CC Utilization (%)						
NewElig _{zs} $\times$ Post _{st}	0.015 (0.015)	0.011 (0.015)	0.032* (0.017)	0.029 (0.018)	0.031* (0.017)	0.027 (0.019)
Obs	57,235	92,797	58,764	122,767	53,543	138,822
R2	0.957	0.956	0.954	0.953	0.955	0.954
Panel D: Outcome = New CC to Inquiries Ratio						
NewElig _{zs} $\times$ Post _{st}	0.054* (0.031)	0.035 (0.034)	0.066** (0.029)	0.037 (0.032)	0.073** (0.029)	0.044 (0.031)
Obs	57,235	92,797	58,764	122,767	53,543	138,822
R2	0.830	0.831	0.828	0.828	0.836	0.833
Window	-3 to +6 years		-3 to +5 years		-3 to +4 years	
Stacked	No	Yes	No	Yes	No	Yes

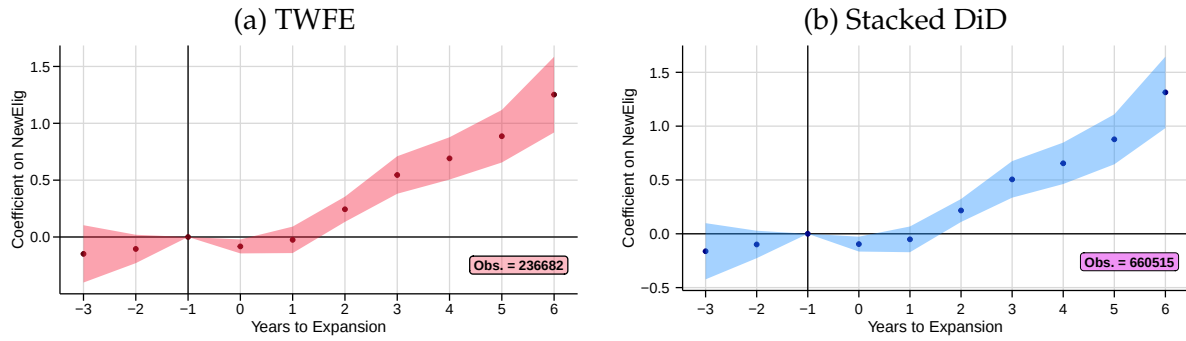
Notes: This table reports results from estimating the "stacked" version of the DiD in Equation (1). Columns (1) and (2) use the subsample of observations whose states expanded in 2014 using data from 3 years before to 6 years after expansion (the window). Columns (3) and (4) expand the sample to observations whose states expanded in 2015 or earlier but use a narrower window of 3 years before to 5 years after. Columns (5) and (6) expand the sample to include observations whose states expanded in 2016 or earlier, but restrict the window to 3 years before to 4 years after expanding. Each panel reports these six regressions for different outcome variables. To facilitate comparisons, the odd-numbered columns display non-stacked regression results for the subsamples used in the stacked regressions. Each specification uses county-time and ZIP-code fixed effects and controls for logged ZIP-level average adjusted gross income (AGI) and logged deciles of the ZIP code's income distribution (coefficients omitted). Standard errors are clustered by state. Nominal variables are CPI-adjusted to be in terms of 2020 dollars. Statistical significance: 10%*, 5%***, and 1%***.

Table A.5: Comparison of Non-Stacked and Stacked DiD Estimates (3/3)

	Expand = 2014		Expand $\leq$ 2015		Expand $\leq$ 2016	
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Outcome = 1[90+ DPD]						
NewElig _{zst} $\times$ Post _{st}	-1.00e-3 (1.16e-2)	-5.29e-3 (1.13e-2)	3.85e-3 (9.22e-3)	-4.15e-4 (9.06e-3)	6.52e-3 (8.35e-3)	1.72e-3 (8.59e-3)
Obs	57,235	92,797	58,764	122,767	53,543	138,822
R2	0.888	0.895	0.889	0.896	0.894	0.899
Panel B: Outcome = 1[Utilities 90+ DPD]						
NewElig _{zst} $\times$ Post _{st}	-0.115** (0.053)	-0.118** (0.053)	-0.109** (0.042)	-0.109** (0.043)	-0.099** (0.039)	-0.098** (0.040)
Obs	57,235	92,797	58,764	122,767	53,543	138,822
R2	0.910	0.916	0.912	0.923	0.921	0.932
Panel C: Outcome = 1[Any Col.]						
NewElig _{zst} $\times$ Post _{st}	-0.123*** (0.024)	-0.133*** (0.025)	-0.112*** (0.021)	-0.119*** (0.023)	-0.114*** (0.020)	-0.124*** (0.023)
Obs	57,235	92,797	58,764	122,767	53,543	138,822
R2	0.976	0.981	0.976	0.982	0.978	0.983
Panel D: Outcome = Collections/Income (%)						
NewElig _{zst} $\times$ Post _{st}	-0.037*** (0.008)	-0.038*** (0.009)	-0.033*** (0.007)	-0.034*** (0.008)	-0.034*** (0.007)	-0.035*** (0.008)
Obs	57,235	92,797	58,764	122,767	53,543	138,822
R2	0.926	0.937	0.925	0.937	0.928	0.938
Panel E: Outcome = Vantage Score						
NewElig _{zst} $\times$ Post _{st}	0.189* (0.096)	0.219** (0.103)	0.137* (0.075)	0.154* (0.089)	0.119* (0.067)	0.133 (0.085)
Obs	57,235	92,797	58,764	122,767	53,543	138,822
R2	0.986	0.987	0.986	0.988	0.987	0.988
Window	-3 to +6 years		-3 to +5 years		-3 to +4 years	
Stacked	No	Yes	No	Yes	No	Yes

Notes: This table reports results from estimating the "stacked" version of the DiD in Equation (1). Columns (1) and (2) use the subsample of observations whose states expanded in 2014 using data from 3 years before to 6 years after expansion (the window). Columns (3) and (4) expand the sample to observations whose states expanded in 2015 or earlier but use a narrower window of 3 years before to 5 years after. Columns (5) and (6) expand the sample to include observations whose states expanded in 2016 or earlier, but restrict the window to 3 years before to 4 years after expanding. Each panel reports these six regressions for different outcome variables. To facilitate comparisons, the odd-numbered columns display non-stacked regression results for the subsamples used in the stacked regressions. Each specification uses county-time and ZIP-code fixed effects and controls for logged ZIP-level average adjusted gross income (AGI) and logged deciles of the ZIP code's income distribution (coefficients omitted). Standard errors are clustered by state. Nominal variables are CPI-adjusted to be in terms of 2020 dollars. Statistical significance: 10%*, 5%***, and 1%***.

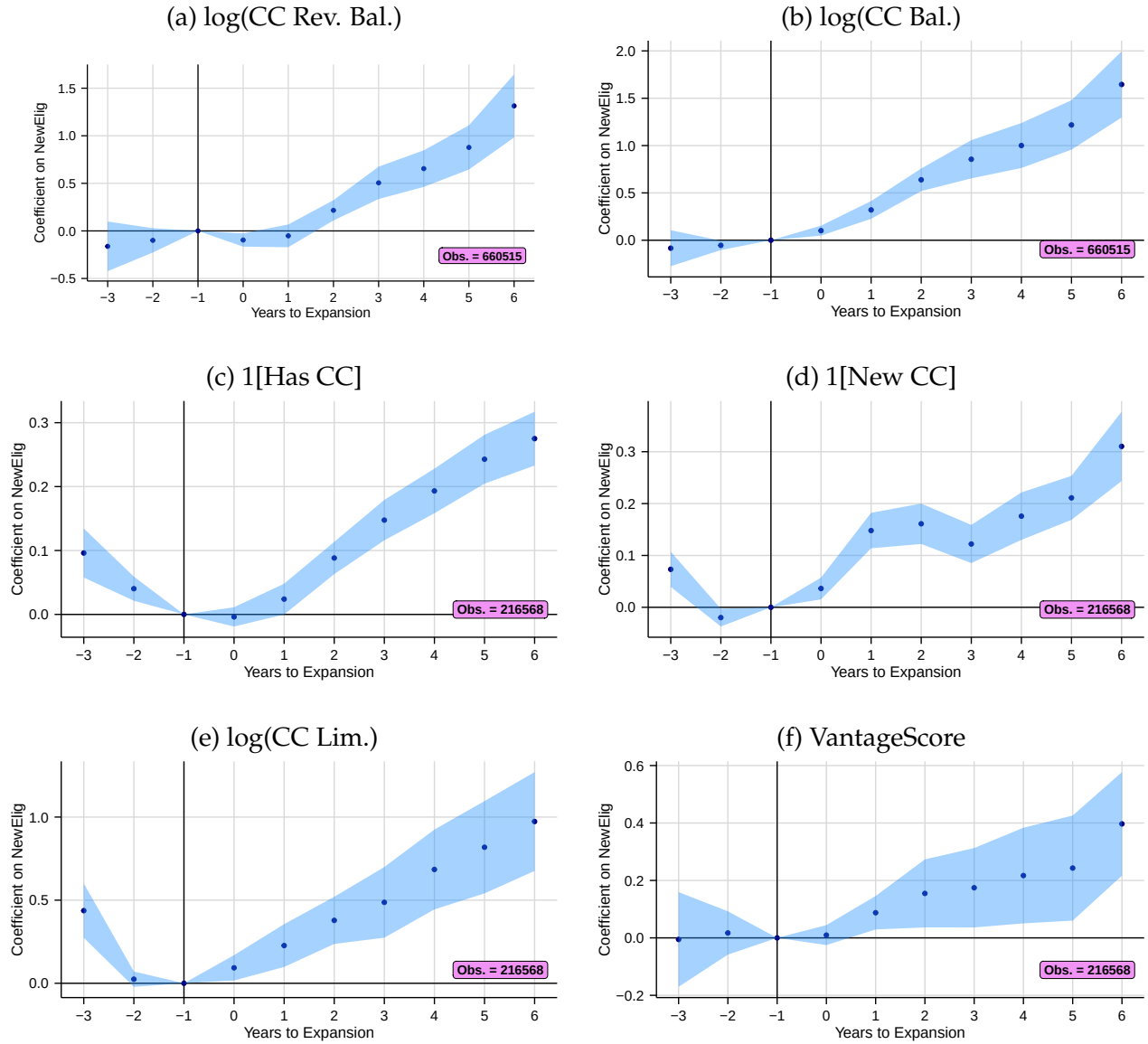
Figure A.4: Comparison of TWFE and Stacked DiD Estimates: log(CC Rev. Bal.)



Notes: Panel (a) reports dynamic difference-in-differences estimates from a standard two-way fixed effects specification. Panel (b) reports stacked difference-in-differences estimates following [Cengiz et al. \(2019\)](#), which addresses potential bias from heterogeneous treatment effects across cohorts. Shaded areas represent 95% confidence intervals for standard errors clustered at the state level. The sample size discrepancy arises because ZIPs from "never treated" states appear multiple times when stacking data by expansion year cohort.

A.2 Dynamic DiD (Additional Outcomes)

Figure A.5: Credit Card Outcomes and Credit Score



Notes: Each panel reports stacked difference-in-differences estimates following [Cengiz et al. \(2019\)](#). The shaded area represents the 95% confidence interval for standard errors clustered at the state level. The sample size discrepancy across panels is due to quarterly availability for credit card balances (total and revolving), whereas the remaining outcomes are measured annually. Note that the rise in total balances in early periods when revolving balances have not yet risen indicates that borrowers are making larger payments (such that their revolving balances do not increase despite increased card usage).

A.3 Heterogeneity by ZIP-Code Income

Table A.6: Borrowing Outcomes (Low versus High Income)

	1[Has CC] (1)	1[New CC] (2)	log(CC Bal.) (3)	log(CC Rev. Bal.) (4)
<i>Below-Median Income</i>				
NewElig _{zs} × Post _{st}	0.039* (0.020)	0.091*** (0.022)	0.478*** (0.059)	0.438*** (0.062)
Obs	63,518	63,518	166,280	166,280
R2	0.968	0.888	0.944	0.931
Mean	80%	20%	\$3,250	\$2,870
<i>Above-Median Income</i>				
NewElig _{zs} × Post _{st}	0.103*** (0.014)	0.076*** (0.017)	0.452*** (0.057)	0.086 (0.082)
Obs	65,868	65,868	173,505	173,505
R2	0.947	0.846	0.953	0.934
Mean	91%	23%	\$5,534	\$4,645

Notes: This table reports results from estimating the DiD in Equation (1) separately within subsamples (below- and above-median-income ZIP codes). We classify a ZIP as above or below median based on whether its income in a given year is above or below the median of ZIP-level average income. Each specification uses county-time and ZIP-code fixed effects and controls for logged ZIP-level average adjusted gross income (AGI) and logged deciles of the ZIP code's income distribution (coefficients omitted). Nominal variables are CPI-adjusted to be in terms of 2020 dollars. The dependent variable is labeled above the column number, and its mean is reported below the R2. Statistical significance: 10%*, 5%** and 1%***.

Table A.7: Demand and Supply Proxies (Low versus High Income)

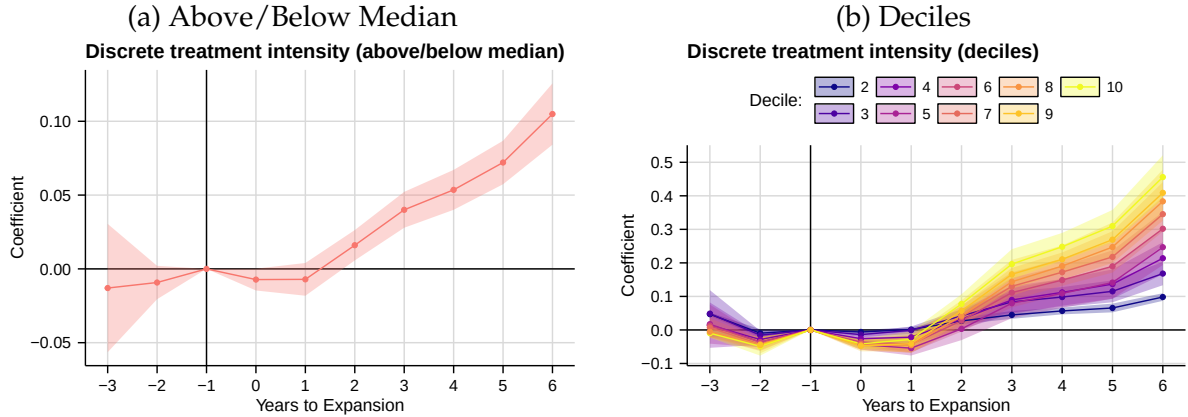
	Demand CC Inq. (#) (1)	log(CC Lim.) (2)	Supply CC Util. (%) (3)	New CC to Inq. (4)
<i>Below-Median Income</i>				
NewElig _{zs} × Post _{st}	0.280*** (0.044)	0.367*** (0.094)	0.022 (0.016)	0.061** (0.029)
Obs	63,518	63,518	63,518	63,518
R2	0.950	0.967	0.923	0.861
Mean	0.452	\$12,344	40%	0.460
<i>Above-Median Income</i>				
NewElig _{zs} × Post _{st}	0.147*** (0.038)	0.120 (0.137)	-0.009 (0.034)	-0.034 (0.028)
Obs	65,868	65,868	65,868	65,868
R2	0.932	0.976	0.946	0.841
Mean	0.416	\$24,474	31%	0.554

Notes: See notes to Appendix Table A.6.

Table A.8: Default and Credit Scores (Low versus High Income)

	1[Delinquency]		Debt in Collections		Credit Score
	1[90+ Days] (1)	1[90+ (Util.)] (2)	1[Any Col.] (3)	Col./Inc. (4)	Vantage Score (5)
<i>Below-Med. Inc.</i>					
NewElig _{zs} × Post _{st}	0.011 (0.019)	-0.151** (0.065)	-0.093*** (0.025)	-0.019** (0.008)	0.102 (0.066)
Obs	63,518	63,518	63,518	63,518	63,518
R2	0.879	0.932	0.974	0.931	0.979
Mean	8.9%	5.3%	31%	1.8%	660.32
<i>Above-Med. Inc.</i>					
NewElig _{zs} × Post _{st}	-0.035*** (0.009)	-0.049** (0.021)	-0.104*** (0.021)	-0.019*** (0.005)	0.389*** (0.109)
Obs	65,868	65,868	65,868	65,868	65,868
R2	0.861	0.913	0.965	0.884	0.975
Mean	5.2%	3.4%	16%	0.5%	711.53

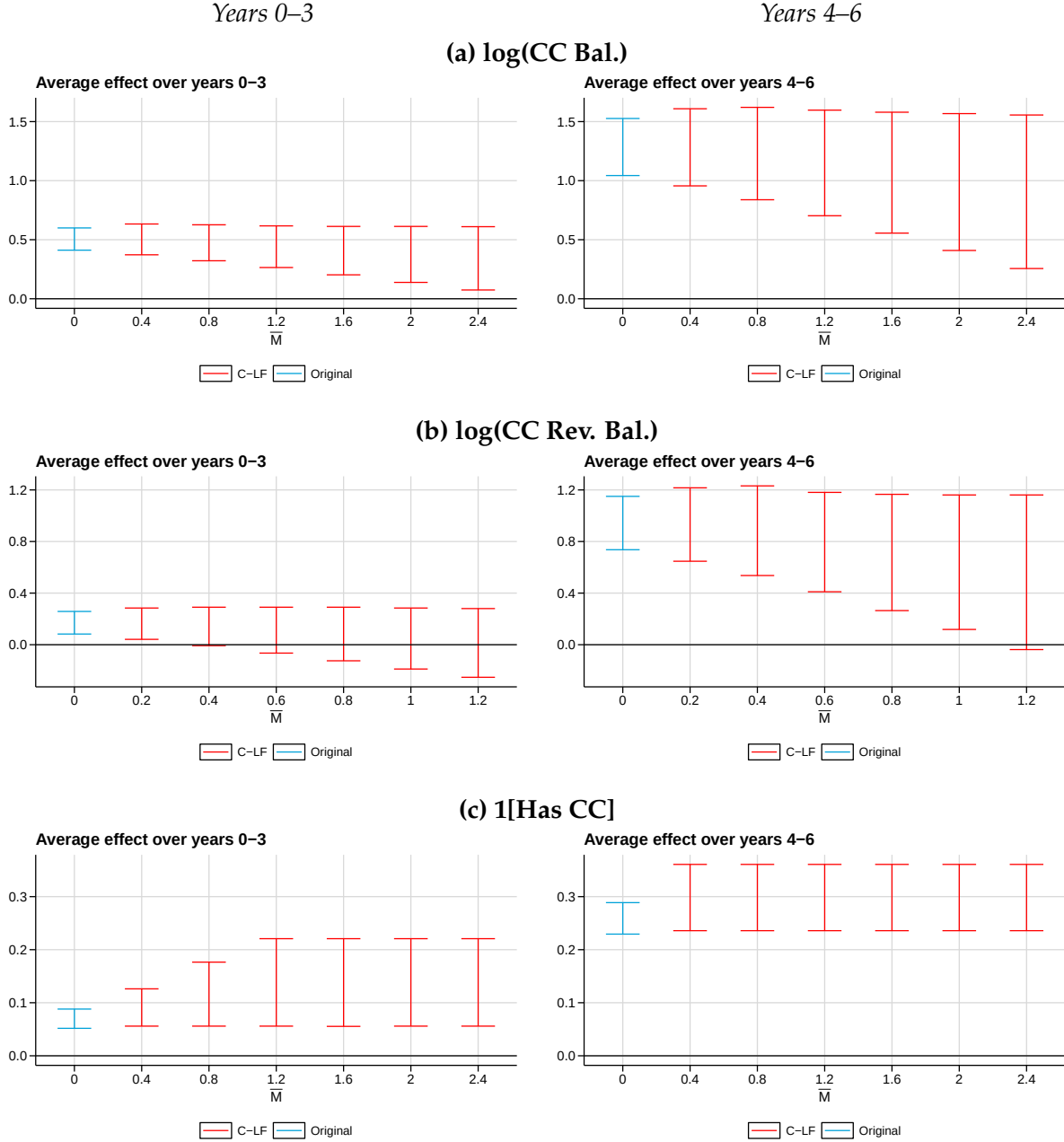
Notes: See notes to Appendix Table A.6.

Figure A.6: Dynamic DiD for Revolving Balances (Discrete Treatment Intensity)

Notes: This figure reports dynamic DiD estimates for logged revolving credit card balances, replacing the continuous treatment NewElig_{zs} with indicators for bins of it: above versus below the median in Panel (a) and deciles in Panel (b). We estimate $Y_{zcs\tau} = \sum_{k \geq 2} \sum_{\tau \neq -1} \beta_{k\tau} (D_{st}^{\tau} \times \text{Bin}_{zsk}) + \phi_z + \phi_{ct} + \gamma X_{zcs\tau} + \varepsilon_{zcs\tau}$, where D_{st}^{τ} indicates that state s is τ years from expansion. Each $\beta_{k\tau}$ is the change in log balances relative to $\tau = -1$ for ZIP codes in bin k relative to the bottom bin. Fixed effects and controls are as in Equation (1). Shaded areas denote 95% confidence intervals for standard errors clustered by state. These coefficients are not directly comparable to our continuous estimate because bins differ greatly in exposure and high-exposure ZIP codes have lower balances. Average NewElig_{zs} is 28.0% above the median versus 10.2% below, and 36.8% in the top decile versus 0.7% in the bottom; average revolving balances are \$3,342 versus \$4,492, and \$2,626 versus \$4,412. In dollars, the year-three estimate in Panel (a) (about 0.04) implies \$134 higher balances in above-median ZIP codes, versus \$257 implied by our continuous estimate ($0.383 \times 17.8 \times \$3,776$). The year-three top-decile estimate in Panel (b) (about 0.20) implies \$525, versus \$522 ($0.383 \times 36.1 \times \$3,776$).

A.4 Honest DiD Robustness

Figure A.7: Honest DiD Sensitivity Analysis (1 of 2)



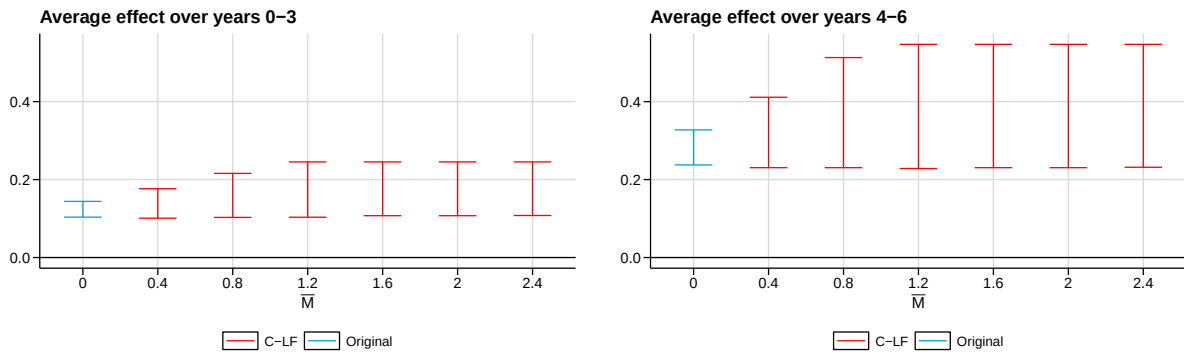
Notes: We report 90% confidence intervals for the estimated average treatment effect over two horizons: early (years 0–3, left column) and later (years 4–6, right column). Each row reports a different outcome. Red "least favorable" confidence intervals ("C-LF") are obtained via the "relative magnitudes" approach of [Rambachan and Roth \(2023\)](#), which allows various violations of the parallel trends assumption. Above, each $\bar{M}$ corresponds to a scenario allowing post-treatment violations of parallel trends to be up to $\bar{M}$ times the size of the average pre-trend. Estimates and confidence intervals resulting from our baseline regression are in blue ("original").

Figure A.8: Honest DiD Sensitivity Analysis (2 of 2)

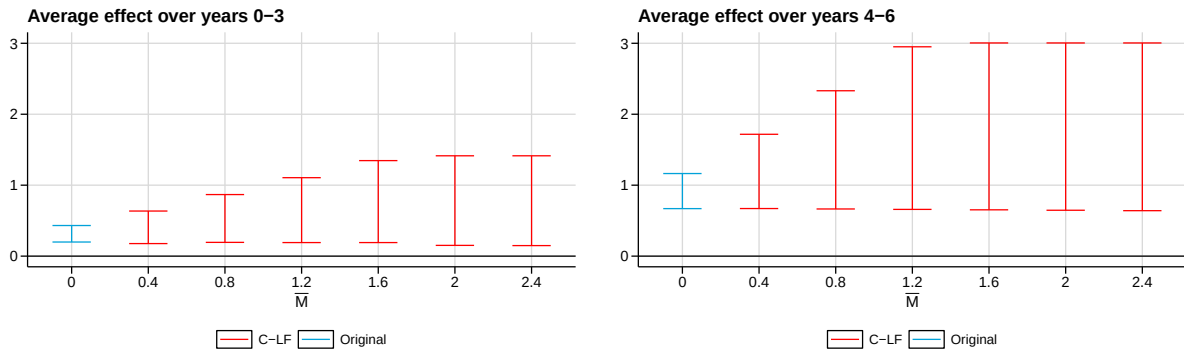
Years 0–3

Years 4–6

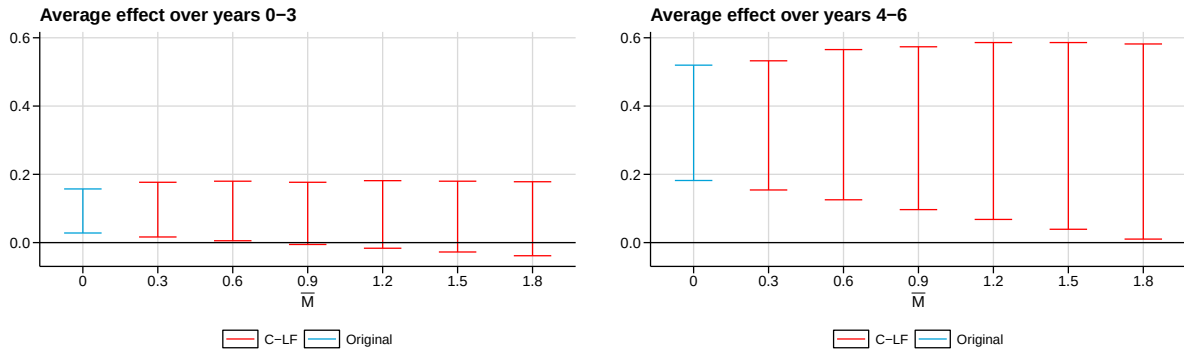
(d) 1[New CC]



(e) log(CC Lim.)



(f) VantageScore



Notes: See notes to Appendix Figure A.7.

A.5 Individual-Level DiD Results

Table A.9: Individual-Level DiD Results for Credit Card Debt

	1[Has CC] (1)	log(CC Bal.) (2)	log(CC Rev. Bal.) (3)
Panel A: County $\times$ Time FE			
$1[\text{Elig}_i] \times \text{Post}_{st}$	0.228*** (0.013)	0.547*** (0.020)	0.535*** (0.022)
Panel B: ZIP $\times$ Time FE			
$1[\text{Elig}_i] \times \text{Post}_{st}$	0.228*** (0.013)	0.534*** (0.016)	0.526*** (0.018)
Obs.	56,436,593	27,714,715	27,714,715
R2	0.648	0.708	0.659
Mean (all)	88%	\$11,697	\$10,629
Mean (newly elig.)	39%	\$2,550	\$2,307

Notes: This table reports results from estimating the individual-level DiD in Equation (11), where the treatment indicator $1[\text{Elig}_i]$ equals one if individual i became newly eligible for Medicaid due to a 2014 expansion. Both panels include individual and ZIP-code fixed effects; Panel A includes county-time fixed effects and ZIP-level income controls, while Panel B instead includes ZIP-time fixed effects (which absorb the ZIP-level controls). Note that individual and ZIP fixed effects are not generally collinear, as individuals may move across ZIP codes over time. Time fixed effects are at the year level in Column (1) and the quarter level in Columns (2)–(3). Columns (2)–(3) restrict to individuals with strictly positive total and revolving credit card balances in every observed quarter. Standard errors are clustered by state. The dependent variable is labeled above the column number; the bottom rows report its full-sample mean and its 2013 mean among the newly eligible. Statistical significance: 10%*, 5%***, and 1%***.

Table A.10: Individual-Level DiD Results for Limits, Scores, and APRs

	log(CC Lim.) (1)	Vantage Score (2)	APR (3)
Panel A: County $\times$ Time FE			
$1[\text{Elig}_i] \times \text{Post}_{st}$	0.739*** (0.031)	20.230*** (0.960)	−4.888*** (0.463)
Panel B: ZIP $\times$ Time FE			
$1[\text{Elig}_i] \times \text{Post}_{st}$	0.727*** (0.030)	20.158*** (0.976)	−4.857*** (0.453)
Obs.	36,555,441	56,436,593	14,097,490
R2	0.763	0.815	0.511
Mean (all)	\$28,873	693	14.2%
Mean (newly elig.)	\$3,677	581	23.8%

Notes: See notes to Appendix Table A.9.

A.6 Reactions of Other Forms of Debt

Table A.11: DiD Results for Auto and Mortgage Debt

	1[Has Auto] (1)	1[Has Mortgage] (2)	log(Auto Loan) (3)	log(Mortgage Bal.) (4)
$\text{NewElig}_{zs} \times \text{Post}_{st}$	0.147*** (0.014)	-0.045 (0.036)	0.918*** (0.071)	-0.060 (0.129)
Obs	129,386	129,386	129,386	129,386
R2	0.951	0.968	0.956	0.981
Mean	35%	30%	\$6,812	\$69,611

Notes: This table reports results from estimating the DiD in Equation (1) for auto and mortgage loans, using the same fixed effects and controls. Each specification includes county-time and ZIP-code fixed effects. Standard errors are clustered by state. The dependent variable is labeled above the column number, and its mean is reported in the bottom row. Statistical significance: 10%*, 5%** and 1%***.

B Measuring Medicaid Eligibility

This appendix describes how we measure of Medicaid eligibility at the $\text{ZIP} \times \text{year}$ level.

Data: We use the following datasets from 2009 to 2021: IRS SOI (ZIP-level individual income tax statistics), KFF (Medicaid eligibility limits), ACS (joint distribution of income and household size), and KFF (Medicaid expansion date data).

B.1 Target Quantity

We want to measure the fraction of adults eligible for Medicaid in year t and ZIP z . Let E denote the event a household is eligible for Medicaid. Denote the probability that a household is eligible in (t, z) by $P_{tz}(E) = P(E|t, z)$. The probability that a household of size n is eligible depends on the probability that its income is below the relevant cutoff c_{tz}^n . We can write the probability above as the sum of probabilities for these different cases using the law of iterated expectations:

$$P_{tz}(E) = P_{tz}(y \leq c) = \sum_{n=1}^N P_{tz}(y \leq c_{tz}^n | c = c_{tz}^n) P(c = c_{tz}^n)$$

where y is income and c_{tz}^n is the eligibility cutoff for a household of size n in $\{z, t\}$. We can rewrite each term using Bayes' rule to get

$$P_{tz}(y \leq c_{tz}^n | c = c_{tz}^n) P(c = c_{tz}^n) = P_{tz}(c = c_{tz}^n | y \leq c_{tz}^n) P(y \leq c_{tz}^n).$$

The first term is the probability that a household is of a given size:

$$P_{tz}(c = c_{tz}^n | y \leq c_{tz}^n) = P_{tz}(\tilde{N} = n | y \leq c_{tz}^n) = g_{tz}(n | y \leq c_{tz}^n)$$

where g_{tz} is the probability mass function of household size conditional on income being below the

relevant threshold. The second term $P(y \leq c_{tz}^n)$ is the probability that the household's income is below the threshold. Let $F_{tz}(\cdot)$ denote the cumulative distribution function of income in $\{t, z\}$. Then

$$P(y \leq c_{tz}^n) = F_{tz}(c_{tz}^n).$$

Household Size Notation: Households of size two may comprise either a head and a dependent or a head and a non-dependent (e.g., a parent and child versus a childless couple). The eligibility thresholds can differ across these situations. To capture both, we use 2 to indicate a household with a head and a dependent and 2* to indicate a two-person household with no dependents. Let n denote household size. We write $n \in \{1, 2, 2^*, 3, 4, \dots, N\}$, where N is the largest observed household size.

Note on Cutoffs: Income cutoffs vary over time because of changes in the threshold and federal poverty line. They also differ across states; specifically, Alaska and Hawaii generally have higher thresholds in each year.

B.2 Construction

We compute $P_{tz}(E)$ using the following steps:

1. We obtain the number of tax returns and total adjusted gross income (AGI) by income bracket for each ZIP code and year from the IRS SOI. SOI reports these tabulations for six AGI brackets with thresholds at \$25,000, \$50,000, \$75,000, \$100,000, and \$200,000. From these tabulations we recover, for each ZIP-year, the cumulative share of returns below each bracket threshold and the mean AGI within each bracket. This gives us eleven independent moments per ZIP-year (six within-bracket mean incomes and six bracket shares that sum to one) with which to characterize the local income distribution.
2. Medicaid thresholds generally fall between the SOI thresholds, so we interpolate the income CDF $\hat{F}_{tz}(\cdot)$ between the known points. We use the generalized Pareto interpolation method of [Blanchet et al. \(2022\)](#), implemented in the `gpinter` package. The method fits a smooth quantile function that reproduces the observed bracket shares and bracket means exactly and is Pareto-like within brackets; it therefore fills in the distribution *within* brackets while leaving the tabulated moments unchanged. We then evaluate $\hat{F}_{tz}(c_{tz}^n)$, where c_{tz}^n is the Medicaid eligibility limit for a household of size n . Note that this equals the second term of the probability, $P(y \leq c_{tz}^n)$. We compute this for each of the $n \in N$ household sizes defined above.
3. We keep one observation per ZIP and year.
4. We use the ACS data to calculate $P(\tilde{N} = n | y \leq c_{tz}^n)$ for each n . We do this using income and household size information available by ZIP code and year. Note that this equals the first term of the probability, $g_{tz}(n | y \leq c_{tz}^n)$.
5. We can calculate the probability that a household of size n is eligible for Medicaid by multiplying the first and second terms for each n .
6. Then, $P_{tz}(E)$ is equal to the sum of the probabilities calculated in the step above.

C Individual-Level Analysis

This appendix describes the data and methodology behind the individual-level analysis reported in Tables A.9 and A.10. The individual-level design complements our ZIP-level analysis: treatment is measured at the individual level using an independent source of income data, individual fixed effects absorb latent time-invariant heterogeneity in credit outcomes, and it permits a more demanding specification using ZIP-time fixed effects to absorb heterogeneity in ZIP-level economic trends, which our ZIP-level design cannot. The individual-level data also allow us to estimate the response of the effective interest rates consumers pay on credit card debt.

C.1 Data

We use individual-level credit records from Experian spanning 2010–2021. Annual person-year records provide the number of open credit cards (from which we define $\mathbf{1}[\text{Has CC}]$), the total credit limit on open cards, and the VantageScore 4.0 credit score. Quarterly credit card variables come from Experian’s Trended 3D archives: total credit card balances, payments, and scheduled minimum payments. These archives exist for 2014, 2016, 2018, and 2020, and each reports the eight quarters ending in the second quarter of the archive year, so together they cover every calendar quarter from 2012Q3 through 2020Q2 exactly once. Nominal dollar variables are CPI-adjusted to 2020 dollars.

We merge these records with Experian’s imputation of individuals’ 2014 income (their *Income Insight* module), which we purchased for 7.6 million consumers from our original sample who resided in either a state that expanded Medicaid in 2014 or a state that never expanded during the sample period. The imputation comes from a proprietary model that predicts W2 income using attributes from the consumer’s credit report, trained on a national sample of consumers for whom Experian has credit reports matched to W2 income. We purchased a single year of income data, choosing 2014 because it is the year with the most Medicaid expansions.

We apply the following sample restrictions. We keep the ZIP-time cells used in our ZIP-level analysis (those with at least 150 individuals) and require non-missing ZIP-level income controls. We drop individuals who move out of their 2014 state after 2014, fixing each person’s treatment assignment at their 2014 state of residence. Finally, for a subset of outcomes we impose additional restrictions; these are detailed in the notes to Tables A.9 and A.10.

C.2 Individual-Level Medicaid Eligibility

For each person i , we construct $\mathbf{1}[\text{Elig}_i]$, an indicator for becoming newly eligible for Medicaid due to a 2014 expansion. We first calculate the probability that i is eligible under their state’s actual 2014 Medicaid rules. We then calculate the probability that i would have been eligible in 2014 if, counterfactually, their state’s 2013 rules had remained in place, holding fixed their 2014 income. Formally,

$$\mathbf{1}[\text{Elig}_i] = \mathbf{1} [\text{Pr}_{s,2014}(E_i = 1 \mid y_i) - \text{Pr}_{s,2013}(E_i = 1 \mid y_i) \geq 0.5] ,$$

where $\text{Pr}_{s,t}$ denotes the probability evaluated under state s ’s Medicaid eligibility rules in year t , $E_i \in \{0, 1\}$ indicates Medicaid eligibility, and y_i is individual i ’s imputed 2014 income.

Medicaid eligibility depends on both income and household size, and household size is not observed in the credit bureau data. We therefore integrate over the distribution of household size conditional on income, estimated from the 2010 American Community Survey (ACS):

$$\Pr_{s,t}(E_i = 1 \mid y_i) = \sum_n \mathbf{1}[y_i \leq c_{s,t}^n] \Pr(n \mid y_i),$$

where $c_{s,t}^n$ is the Medicaid income threshold in state s and year t for a household of size n , constructed from the Kaiser Family Foundation limits and the federal poverty level (FPL) as in Appendix B, and $\Pr(n \mid y_i)$ is the ACS-implied probability that an individual with income y_i lives in a household of size n .⁵² We set $\mathbf{1}[\text{Elig}_i]$ to zero for individuals in never-expanded states.

C.3 Effective Credit Card APR

We estimate the effective annual percentage rate (APR) each consumer pays on revolving credit card debt, adapting the approaches of [Guttman-Kenney and Shahidinejad \(2023\)](#) and [Koşar et al. \(2025\)](#) to our person-level data. The effective APR is the realized, fee-inclusive price of borrowing: it reflects fee expenses, such as late fees, in addition to interest charges, and so better approximates the true marginal cost of borrowing than the posted interest rate alone.

For person i in quarter t , we observe the total statement balance across the person's credit cards (B_{it}), total payments (P_{it}), total scheduled minimum payments (M_{it}), and the number of open cards (N_{it}); the balance and payment variables are monthly averages within the quarter. The revolving (unpaid) balance is $D_{it} = \max\{0, B_{it} - P_{it}\}$.

To impute an effective APR, we follow [Guttman-Kenney and Shahidinejad \(2023\)](#) and assume that the minimum payment on an individual card equals $\max\{\mu, \theta b + (1 - \theta)f\}$, which is the typical structure of the formula used to determine minimum payments. Here, μ is the minimum payment floor, which we set equal to \$25, as estimated in [Guttman-Kenney and Shahidinejad \(2023\)](#). The second term in the max operator is the balance-based minimum $\theta b + (1 - \theta)f$, where b is the card's statement balance (i.e., including interest and fees) and f is its financing charges (interest and fees). We again use estimates in [Guttman-Kenney and Shahidinejad \(2023\)](#) to set $\theta = 0.01$. With these assumptions, and card-level data, we could back out f for each card with revolving balances. But because we observe only person-level (and not card-level) totals, we cannot tell how much of M_{it} reflects financing charges versus the card-level payment floors μ , as we do not observe the division of balances across cards. However, we can bound the total financing charge F_{it} :

$$F_{it}^U = \max\left\{0, \frac{M_{it} - \theta B_{it}}{1 - \theta}\right\}, \quad F_{it}^L = \max\left\{0, \frac{M_{it} - \theta B_{it} - N_{it}\mu}{1 - \theta}\right\}.$$

The upper bound F_{it}^U follows from assuming that no card is at its floor (and thus $m = \theta b + (1 - \theta)f$ for each card). The lower bound F_{it}^L is constructed by setting aside the full $\mu = 25$ for each card before counting anything as financing charges, giving the floors the maximum weight possible. We take the

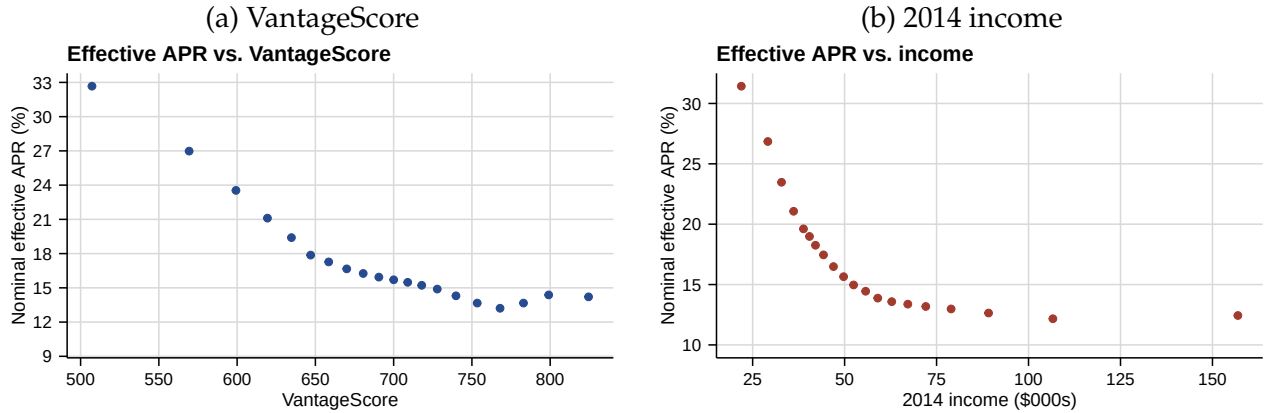
⁵²To best approximate the intended counterfactual, we use each state's 2013 Medicaid limits expressed as a percent of the FPL, but apply those limits to the 2014 dollar value of the FPL, so that the 2013 and 2014 thresholds differ only through the eligibility rules themselves.

midpoint between these two bounds. The implied effective monthly rate is $r_{it} = \frac{F_{it}}{B_{it}}$. To annualize this rate into an APR we multiply it by 12. Lastly, note that when $F_{it} = 0$ (i.e., consumers are not paying financing charges), the effective rate that they would pay is not identified. Our analyses using the imputed APR restrict to observations with non-zero imputed financing costs, thus focusing on observations whose effective APR is identified.

The effective APR measure is measured with error, as it is imputed rather than measured directly. Because it enters every regression as the dependent variable, classical measurement error does not bias our estimates for this outcome. We also exclude rates above 100% (0.8% of revolver-quarters); these cases appear to be driven by noise in the denominator resulting in small balances.

Two checks indicate that the imputation produces sensible borrowing costs. First, Table 8 in the main text compares percentiles of our imputed effective APR distribution to the fee-inclusive effective credit card interest rates that Nelson (2025) measures directly in card-level administrative data; the distributions are similar. Second, Figure C.1 plots the imputed effective APR against consumers' contemporaneous VantageScore and against their imputed 2014 income. The imputed rate declines steeply in both, consistent with risk-based pricing. The mean effective APR in our sample is 14.2% overall, and 23.8% in 2013 among consumers who became newly eligible in 2014.

Figure C.1: Effective APR versus Credit Scores and Income



Notes: Each panel plots the imputed effective APR (in percentage points) against 20 equal-count bins of the x-axis variable; each point is the within-bin mean of both variables. Panel (a) uses the contemporaneous (same-year) VantageScore. Panel (b) uses Experian's imputed 2014 income (in \$1,000s). The sample is all active-revolver person-quarters with a positive imputed rate, excluding rates above 100%. Panel (b) further restricts to APRs observed in 2014 to align with the timing of the measurement of income.

C.4 Regression Specification

With the individual-level treatment measure in hand, we estimate the following analog of our ZIP-level DiD:

$$Y_{it} = \beta (\text{Post}_{st} \times \mathbf{1}[\text{Elig}_i]) + \phi_i + \phi_z + \phi_{ct} + \gamma X_{zct} + \varepsilon_{it}, \quad (11)$$

where Y_{it} is a credit outcome of person i in period t . The period t is a quarter for the balance and effective APR outcomes and a year for the remaining outcomes. Post_{st} equals one in 2014-expansion states

from 2014 onward; it is always zero in never-expanded states. The specification mirrors Equation (1): it includes the same ZIP fixed effects (ϕ_z), county-time fixed effects (ϕ_{ct}), and ZIP-level controls X_{zcs} (log average income and log deciles of the ZIP income distribution), but adds an individual fixed effect ϕ_i . Because we observe income for a single year, ϕ_i implicitly controls for the individual's 2014 income. In our most demanding specification, we replace ϕ_z and ϕ_{ct} with ZIP-time fixed effects, which absorb the ZIP-level controls. Standard errors are clustered by state.

The coefficient β measures the effect of becoming newly eligible for Medicaid on an individual's credit outcomes. Tables A.9 and A.10 report the estimates under both sets of fixed effects; the outcome-specific sample restrictions are described in the table notes.

D Mathematical Appendix

D.1 Proof of Proposition 1 - Uniqueness of Debt Pricing Schedule

The debt pricing schedule is given by

$$q(b', y) = \frac{1}{1+r_f} [1 - \mathbb{E}(d(b' + o(y')m', y'))] \\ + \frac{1}{1+r_f} \mathbb{E}[d(b' + o(y')m', y')(1-\delta')q((1-\delta')(b' + o(y')m'), y')].$$

Let $B(\mathbb{R}^2)$ be the set of bounded functions $q : \mathbb{R}^2 \rightarrow \mathbb{R}$. Define the operator $T : B(\mathbb{R}^2) \rightarrow B(\mathbb{R}^2)$ as follows:

$$T(q)(b', y) = \frac{1}{1+r_f} [1 - \mathbb{E}(d(b' + o(y')m', y'))] \\ + \frac{1}{1+r_f} \mathbb{E}[d(b' + o(y')m', y')(1-\delta')q((1-\delta')(b' + o(y')m'), y')]. \quad (12)$$

We will use Blackwell's sufficient conditions to show that T is a contraction mapping with modulus $\frac{1}{1+r_f} < 1$.

Monotonicity. Suppose $q_1(b', y) \geq q_2(b', y)$ for all b' and y . Then,

$$T(q_1)(b', y) = f(b', y) + \frac{1}{1+r_f} \mathbb{E}[d(b' + o(y')m', y')(1-\delta')q_1((1-\delta')(b' + o(y')m'), y')] \\ \geq f(b', y) + \frac{1}{1+r_f} \mathbb{E}[d(b' + o(y')m', y')(1-\delta')q_2((1-\delta')(b' + o(y')m'), y')] \\ = T(q_2)(b', y)$$

where $f(b', y) = \frac{1}{1+r_f} [1 - \mathbb{E}(d(b' + o(y')m', y'))]$. The first inequality follows from the fact that $d(b' + o(y')m', y')(1-\delta') \geq 0$ for all (b', y') .

Discounting. We have that

$$\begin{aligned}
T(q+a)(b', y) &= f(b', y) + \frac{1}{1+r_f} \mathbb{E} [d(b' + o(y')m', y')(1-\delta') [q((1-\delta')(b' + o(y')m'), y') + a]] \\
&= T(q)(b', y) + \frac{1}{1+r_f} \mathbb{E} [d(b' + o(y')m', y')(1-\delta')] a \\
&\leq T(q)(b', y) + \frac{1}{1+r_f} a,
\end{aligned}$$

where the last inequality follows from $d(b' + o(y')m', y')(1-\delta') \in [0, 1]$ for all (b', y') . Therefore, T is a contraction mapping with modulus $\frac{1}{1+r_f}$. Hence, the debt pricing schedule $q(b', y)$ exists and is unique, given a delinquency policy function $d(\tilde{b}', y)$.

D.2 Moments of Income Process

Suppose that the income process takes the following form:

$$\ln y_t = \begin{cases} \rho \ln y_{t-1} + \epsilon_t & \text{w.p. } \lambda, \\ \ln y_{t-1} & \text{w.p. } 1 - \lambda, \end{cases} \quad (13)$$

where ϵ_t follows a Normal distribution with standard deviation σ . We calibrate the structural parameters $\{\rho, \sigma, \lambda\}$ to match the persistence of income, as well as the standard deviation and kurtosis of income changes. Since the PSID reports household-level income every two years, we match the 2-year autocorrelation of income, as well as the standard deviation and kurtosis of 2-year income changes.

Proposition 2. *The income process admits the following moments:*

$$\begin{aligned}
\text{acorr}_2(\ln y_t) &= \frac{\mathbb{E} [\ln y_t \ln y_{t+2}]}{\text{var}(\ln y_t)} = (1 - \lambda + \lambda\rho)^2, \\
\text{var}(\Delta \ln y_t) &= \mathbb{E} [(\ln y_{t+1} - \ln y_t)^2] = 2(1 - (1 - \lambda + \lambda\rho)) \text{var}(\ln y_t), \\
\text{var}(\Delta_2 \ln y_t) &= \mathbb{E} [(\ln y_{t+2} - \ln y_t)^2] = 2(1 - (1 - \lambda + \lambda\rho)^2) \text{var}(\ln y_t), \\
\text{kurt}(\Delta \ln y_t) &= \frac{3}{4} \frac{1}{\lambda} \left((1 - \rho)^4 + 2(1 - \rho)^2(1 - \rho^2) + (1 - \rho^2)^2 \right) \frac{(1 + \rho)^2}{(1 - \rho^2)^2}, \\
\text{kurt}(\Delta_2 \ln y_t) &= \frac{2(1 - \lambda) K_\Delta V_\Delta^2 + \lambda^2(3(\rho^2 - 1)^4 V_y^2 + 3(1 + \rho^4)\sigma^4 + 6\rho^2\sigma^4 + 6(\rho^2 - 1)^2(1 + \rho^2) V_y \sigma^2)}{V_{\Delta_2}^2},
\end{aligned}$$

where $V_y = \text{var}(\ln y_t)$, $V_\Delta = \text{var}(\Delta \ln y_t)$, $V_{\Delta_2} = \text{var}(\Delta_2 \ln y_t)$ and $K_\Delta = \text{kurt}(\Delta \ln y_t)$.

Proof. 1. **Proof of 2-year autocorrelation.**

$$\mathbb{E} [\ln y_t \ln y_{t+2}] = (1 - \lambda)^2 \text{var}(\ln y_t) + 2\lambda(1 - \lambda)\rho \text{var}(\ln y_t) + \lambda^2\rho^2 \text{var}(\ln y_t).$$

Rearranging we have

$$\frac{\mathbb{E} [\ln y_t \ln y_{t+2}]}{\text{var}(\ln y_t)} = (1 - \lambda)^2 + 2(1 - \lambda)\rho\lambda + \lambda^2\rho^2 = (1 - \lambda + \lambda\rho)^2.$$

2. Proof of 1-year income change variance.

$$\mathbb{E} [(\ln y_{t+1} - \ln y_t)^2] = 2 \text{var}(\ln y_t) - 2\mathbb{E} (\ln y_{t+1} \ln y_t).$$

Similarly to how we proved the 2-year autocorrelation, we have that

$$\mathbb{E} (\ln y_{t+1} \ln y_t) = (1 - \lambda + \lambda\rho) \text{var}(\ln y_t).$$

So we obtain

$$\mathbb{E} [(\ln y_{t+1} - \ln y_t)^2] = 2(1 - (1 - \lambda + \lambda\rho)) \text{var}(\ln y_t).$$

3. Proof of 2-year income change variance.

$$\mathbb{E} [(\ln y_{t+2} - \ln y_t)^2] = 2 \text{var}(\ln y_t) - 2\mathbb{E} (\ln y_{t+2} \ln y_t).$$

We have that

$$\begin{aligned} \mathbb{E} (\ln y_{t+2} \ln y_t) &= (1 - \lambda)\mathbb{E} (\ln y_{t+1} \ln y_t) + \lambda\rho\mathbb{E} (\ln y_{t+1} \ln y_t) \\ &= (1 - \lambda + \lambda\rho)\mathbb{E} (\ln y_{t+1} \ln y_t). \end{aligned}$$

Combining everything we have

$$\mathbb{E} [(\ln y_{t+2} - \ln y_t)^2] = 2(1 - (1 - \lambda + \lambda\rho)^2) \text{var}(\ln y_t).$$

4. Proof of 1-year income change kurtosis. Let's start from the unconditional variance, $\text{var}(\ln y_t)$.

From equation (13), we have

$$\text{var}(\ln y_t) = \lambda(\rho^2 \text{var}(\ln y_t) + \sigma^2) + (1 - \lambda)\text{var}(\ln y_t).$$

Rearranging we obtain

$$\text{var}(\ln y_t) = \frac{1}{1 - \rho^2} \sigma^2.$$

Now we derive the kurtosis:

$$\begin{aligned}
\mathbb{E}(\Delta \ln y_t)^4 &= \lambda \mathbb{E}((\rho - 1) \ln y_t + \sigma \epsilon)^4 \\
&= \lambda \left[(\rho - 1)^4 \mathbb{E}(\ln y_t)^4 + 6(1 - \rho)^2 \mathbb{E}(\ln y_t)^2 \sigma^2 + 3\sigma^4 \right] \\
&= \lambda \left[(1 - \rho)^4 + 2(1 - \rho)^2(1 - \rho^2) + (1 - \rho^2)^2 \right] \frac{1}{(1 - \rho^2)^2} 3\sigma^4
\end{aligned}$$

where we've used

$$\mathbb{E}(\ln y_t)^2 = \text{var}(\ln y_t) = \frac{1}{1 - \rho^2} \sigma^2,$$

and we can obtain $\mathbb{E}(\ln y_t)^4$ as follows

$$\mathbb{E}(\ln y_t)^4 = (1 - \lambda) \mathbb{E}(\ln y_t)^4 + \lambda \left[\rho^4 \mathbb{E}(\ln y_t)^4 + 6\rho^2 \frac{1}{1 - \rho^2} \sigma^4 + 3\sigma^4 \right]$$

so that

$$\mathbb{E}(\ln y_t)^4 = \frac{1}{1 - \rho^4} \left(\frac{2\rho^2}{1 - \rho^2} + 1 \right) 3\sigma^4 = \frac{1}{(1 - \rho^2)^2} 3\sigma^4$$

Finally we get that

$$\text{Kurt}(\Delta \ln y_t) = \frac{\mathbb{E}(\Delta \ln y_t)^4}{\text{var}^2(\Delta \ln y_t)} = \frac{3}{4\lambda} \left[(1 - \rho)^4 + 2(1 - \rho)^2(1 - \rho^2) + (1 - \rho^2)^2 \right] \frac{(1 + \rho)^2}{(1 - \rho^2)^2}$$

where we've used $\text{Var}(\Delta \ln y) = 2\lambda \frac{(1 - \rho)}{1 - \rho^2} \sigma^2 = 2\lambda \frac{1}{1 + \rho} \sigma^2$.

5. Proof of 2-year income change kurtosis.⁵³

$$\begin{aligned}
\mathbb{E}(\ln y_{t+2} - \ln y_t)^4 &= 2(1 - \lambda) \lambda \mathbb{E}((\rho - 1) \ln y_t + \sigma \epsilon)^4 + \lambda^2 \mathbb{E}((\rho^2 - 1) \ln y_t + \rho \sigma \epsilon_1 + \sigma \epsilon_2)^4 \\
&= 2(1 - \lambda) \mathbb{E}(\Delta \ln y)^4 + \lambda^2 \mathbb{E}((\rho^2 - 1) \ln y_t + \rho \sigma \epsilon_1 + \sigma \epsilon_2)^4
\end{aligned}$$

Let's expand the second term:

$$\mathbb{E}((\rho^2 - 1) \ln y_t + \rho \sigma \epsilon_1 + \sigma \epsilon_2)^4 = (\rho^2 - 1)^4 3V_y^2 + (1 + \rho^4) 3\sigma^4 + 6\rho^2 \sigma^4 + 6(\rho^2 - 1)^2 (1 + \rho^2) V_y \sigma^2,$$

where we've used $\text{kurt}(\ln y_t) = 3$. Combining this expression with the first term we obtain

$$\begin{aligned}
\mathbb{E}(\ln y_{t+2} - \ln y_t)^4 &= 2(1 - \lambda) K_{\Delta} V_{\Delta}^2 + \lambda^2 (3(\rho^2 - 1)^4 V_y^2 + 3(1 + \rho^4) \sigma^4 + 6\rho^2 \sigma^4 + 6(\rho^2 - 1)^2 (1 + \rho^2) V_y \sigma^2).
\end{aligned}$$

Together with the definition of kurtosis, $\text{kurt}(\Delta_2 \ln y_t) = \frac{\mathbb{E}(\ln y_{t+2} - \ln y_t)^4}{(\mathbb{E}(\ln y_{t+2} - \ln y_t)^2)^2}$, we proved the desired expression.

⁵³Since this moment involves extensive algebra, we confirmed via simulations that our formula is correct.

E Quantitative Analysis Appendix

E.1 Medical Expenditure Parameters by Income Quintile

Table E.1 reports the parameters of the medical expenditure process for each income quintile. For each quintile q , a medical expenditure shock arrives with probability λ_q^m , and conditional on arrival, expenditure is log-normally distributed with mean μ_q^m and standard deviation σ_q^m . The parameters are calibrated to the MEPS over 2000–2019, as described in Section 4.1. Both the arrival probability and the median expenditure rise with income, while dispersion falls.

Table E.1: Medical Expenditure Parameters by Income Quintile

	Q1	Q2	Q3	Q4	Q5
Arrival rate of medical shocks, λ_q^m	0.82	0.84	0.87	0.90	0.93
Mean log medical expenditure, μ_q^m	−3.32	−3.44	−3.33	−3.15	−3.02
Std. dev. of log medical expenditure, σ_q^m	1.80	1.62	1.54	1.51	1.46

Notes: Parameters of the log-normal medical expenditure process by income quintile, calibrated to the MEPS (2000–2019). Expenditure is measured in units of median household income.

E.2 Oligopolistic Competition in Credit Markets

Our baseline model assumes that lenders behave competitively. In this section, we extend the model to allow for oligopolistic competition among lenders. We show that the equilibrium price of debt equals the actuarially fair price minus a constant markdown. We calibrate this version of our model and show that the markdown has little impact on the welfare effects of the expansion of Medicaid. Throughout, we suppress the household type k to ease notation.

Credit Supply. We assume that there is a continuum of lenders, all ex-ante identical, risk neutral, and discount future cash flows using the risk-free interest rate. Each period, a household that wants to borrow is matched with a finite number, N , of lenders and has idiosyncratic preferences over them.⁵⁴ These preferences may reflect differences in convenience, rewards, customer service, or pre-existing banking relationships. The timing is as follows. First, the household chooses its desired level of promised repayment, b' . If $b' > 0$, it then meets N lenders and chooses one from which to borrow. Households that save, $b' \leq 0$, earn the risk-free rate as in the baseline model.

Let $q_j(b', y)$ denote the price of debt offered by lender j . Household i 's consumption-equivalent payoff from borrowing from lender j is

$$v_{ij}(b', y) = b' \left[q_j(b', y) + \frac{1}{\sigma - 1} (\varepsilon_{ij} - \bar{\varepsilon}) \right], \quad (14)$$

⁵⁴The assumption of a continuum of lenders is made for tractability. With a fixed set of lenders, a lender that expects to meet the same household again would have dynamic incentives. For example, a lender may choose to lower interest rates in order to deter future default, with the purpose of continuing to profit from lending to this household in the future.

where the ε_{ij} 's are i.i.d. Type-I extreme-value shocks and $\sigma > 1$ governs the degree of substitutability across lenders. We set $\bar{\varepsilon} \equiv \gamma + \ln N$, where γ is the Euler-Mascheroni constant. Since the expected maximum of N i.i.d. Type-I extreme-value shocks equals $\gamma + \ln N$, this normalization implies that when all lenders offer the same price q ,

$$\mathbb{E} \left[\max_j v_{ij}(b', y) \right] = b' \left[q + \frac{1}{\sigma - 1} (\gamma + \ln N - \bar{\varepsilon}) \right] = b' q. \quad (15)$$

Lender-specific preferences therefore affect which lender the household chooses, but do not alter its choice of b' . In a symmetric equilibrium, the household's problem is identical to the one in the baseline model, with the competitive pricing schedule replaced by the oligopolistic one.

Market Shares. The household chooses the lender that delivers the highest payoff. Since $b' > 0$ and $\bar{\varepsilon}$ is common across lenders, lender j is chosen if and only if $(\sigma - 1)q_j + \varepsilon_{ij} \geq (\sigma - 1)q_l + \varepsilon_{il}$ for all l . Using the distributional assumption on the taste shocks, the probability of this event, which is lender j 's market share, takes the standard logit form

$$s_j(b', y) = \frac{\exp \{ (\sigma - 1)q_j(b', y) \}}{\sum_{l=1}^N \exp \{ (\sigma - 1)q_l(b', y) \}}, \quad \text{with} \quad \frac{\partial s_j}{\partial q_j} = (\sigma - 1) s_j (1 - s_j). \quad (16)$$

Lenders' Problem. Let $\bar{q}(b', y)$ denote the actuarially fair price of debt. Given households' delinquency policy, $\bar{q}$ is the unique solution to the zero-profit condition (7),

$$\begin{aligned} \bar{q}(b', y)(1 + r_f) &= [1 - \mathbb{E} (d(b' + o(y')m', y'))] \\ &\quad + \mathbb{E} [d(b' + o(y')m', y')(1 - \delta') \bar{q}((1 - \delta')(b' + o(y')m'), y')]. \end{aligned} \quad (17)$$

A lender that extends $q_j(b', y)b'$ to the household acquires a claim worth $\bar{q}(b', y)b'$. Lender j takes the prices of the other lenders as given and chooses q_j to maximize its expected profits,

$$\Pi_j(b', y) = b' [\bar{q}(b', y) - q_j(b', y)] s_j(b', y). \quad (18)$$

Using (16), the first-order condition is

$$\frac{\partial \Pi_j}{\partial q_j} = b' s_j [-1 + (\bar{q} - q_j) (\sigma - 1) (1 - s_j)] = 0 \implies \bar{q}(b', y) - q_j(b', y) = \frac{1}{\sigma - 1} \frac{1}{1 - s_j(b', y)}. \quad (19)$$

Equilibrium Pricing. In a symmetric equilibrium, $s_j = 1/N$, and the price of debt is

$$q^{\text{olig}}(b', y) = \bar{q}(b', y) - \kappa, \quad \kappa \equiv \frac{1}{\sigma - 1} \frac{N}{N - 1}. \quad (20)$$

We assume that κ is sufficiently small so that $\bar{q}(b', y) > \kappa$ for all contracts offered in equilibrium. The equilibrium definition is identical to the one in the baseline model, except that the pricing schedule

satisfies (20) for $b' > 0$. When lenders are perfectly substitutable ($\sigma \rightarrow \infty$), the markdown vanishes and we recover the competitive pricing of the baseline model.

Imperfect competition therefore introduces a constant markdown relative to the actuarially fair price of debt. The markdown reflects the number of lenders and their substitutability, but it is independent of the borrower’s default risk. Since N and σ are unaffected by the expansion of Medicaid, κ is constant across policy regimes and

$$\Delta q^{\text{olig}}(b', y) = \Delta \bar{q}(b', y). \quad (21)$$

Imperfect competition raises borrowing costs, but it does not attenuate the improvement in the price of debt generated by the reduction in default risk.

Quantitative Results. Only the composite κ matters for allocations, so we calibrate it directly. We set $\kappa = 0.0238$, so that the average interest rate markup across borrowers equals the 3.4 percentage point excess spread on credit cards estimated by [Herkenhoff and Raveendranathan \(2025\)](#). All other parameters are held at their estimated values.

The markdown matters for the level of welfare in the stationary equilibrium. Relative to the competitive benchmark, households in the oligopoly economy are worse off by 0.09% of consumption on average, with patient households losing the most (0.38%). The markdown matters much less for the effect of the policy. The welfare gain from the expansion of Medicaid is 0.239% with oligopolistic lenders, compared with 0.245% in the competitive benchmark. The reduction in default risk passes through to the price of debt despite the presence of imperfect competition. So, while imperfect competition is an important determinant of the level of borrowing costs and welfare, it is largely orthogonal to the welfare effects of expanding social insurance.

E.3 Alternative ZIP-Code Designs in the Model

In the main text, we assign households to ZIP codes in every period based on their current income, so that each ZIP code’s income distribution is fixed over time. Here we compare this fixed-composition design with a cohort design, in which we assign households to ZIP codes once, at the policy date, and follow the same households over the subsequent five years. Because individual incomes mean-revert, the income distribution within each ZIP-code cohort drifts toward the national distribution over time, even absent the policy. As in the main text, we compute the change in each ZIP code’s outcomes relative to their level absent the expansion, which nets out this drift. Table E.2 reports the model-implied coefficients under both designs, alongside the empirical estimates.

The two designs deliver the same qualitative picture: borrowing and credit limits rise, utilization barely moves, and delinquency falls. Under the cohort design, the coefficients on borrowing and credit limits are smaller. As a ZIP-code cohort’s income mix converges toward the national distribution, the cross-sectional variation in treatment intensity dilutes over the five-year window, biasing the estimated coefficients toward zero. The fixed-composition design avoids this dilution and, as discussed in the main text, is the closer analog of the data, where ZIP-code income distributions are stable while residents turn over.

Table E.2: Model-Implied Responses under Alternative ZIP-Code Designs

	log(CC Rev. Bal.)	log(CC Lim.)	CC Util. (pp)	Delinq. (pp)	Collections (pp)
Data	0.383 (0.063)	0.313 (0.090)	0.011 (0.015)	-0.022 (0.010)	-0.119 (0.022)
Model: fixed-composition	0.422	0.251	0.041		-0.074
Model: cohort	0.342	0.171	0.028		-0.060

Notes: Each row reports the effect of a one percentage point increase in the share of a ZIP code’s population newly eligible for Medicaid. The data column reports the estimates of Section 2, with standard errors clustered by state in parentheses. The model columns report the model counterpart of the empirical DiD coefficient, computed on model-generated ZIP codes (Section 4.3.1), under the fixed-composition design of the main text and under the cohort design described above; both are converted from enrollment to eligibility units using the implied take-up rate of 58%.

F Numerical Appendix

F.1 Algorithm for the Stationary Equilibrium

Lenders observe each household’s type and fund themselves at the exogenous risk-free rate r_f . Each type therefore faces its own pricing schedule $q_k(b', y)$. We compute the stationary equilibrium type by type, and aggregate outcomes are population-weighted averages across types. For each type k , we use value function iteration on a discretized state space. The algorithm consists of the following steps, described in more detail below.

0. Start with a guess for the value function $V_k(\tilde{b}, y)$ and the pricing schedule $q_k(b', y)$. For the pre-expansion equilibrium, we set the price of all debt to the risk-free price, $q_k(b', y) = 1/(1 + r_f)$. For counterfactual equilibria, we start from the pre-expansion solution.
1. *Household problem.* Given the expected continuation value and the pricing schedule, solve the delinquency and borrowing problem at every grid point. Obtain the policy functions $d_k(\tilde{b}, y)$ and $b'_k(\tilde{b}, y)$ and an updated value function.
2. *Debt pricing.* Given the updated delinquency policy, update the pricing schedule using the zero-profit condition (7). Check the distance between the value functions and pricing schedules of consecutive iterations. If it is not sufficiently small, repeat from step (1).
3. *Stationary distribution.* With the converged policies and prices, compute the stationary distribution $\Lambda_k(\tilde{b}, y)$ and the model moments.

Grids. We discretize the log-income process with $N_y = 15$ equally spaced points spanning ± 3 unconditional standard deviations of its AR(1) component, using the Tauchen (1986) method with the tails of the distribution assigned to the endpoint states. Denote the resulting transition matrix by Π^{AR} . Because income remains unchanged with probability $1 - \lambda_y$, the transition matrix of income is $\Pi_y = \lambda_y \Pi^{AR} + (1 - \lambda_y)I$, where I is the identity matrix. We normalize income so that median income equals one; all debt levels below are in units of median income.

The grid for total obligations $\tilde{b}$, which is also the grid for borrowing b' , has $N_b = 1,924$ points. The debt segment, $\tilde{b} \in [0, 2]$, has 1,601 equally spaced points with a spacing of 0.00125. The fine spacing is needed because household debt is small relative to income, and it keeps the interest rate schedules and delinquency thresholds smooth. The savings segment has 299 equally spaced points on $(0, 6]$ and 24 additional points at $\{7, 8, \dots, 30\}$. We verify that no households accumulate at the top of the grid.

We map medical bills onto the grid of obligations. For a household with income y in income quintile $q(y)$, a medical expenditure shock arrives with probability $\lambda_{q(y)}^m$, in which case the out-of-pocket bill satisfies $\ln(o(y)m) \sim \mathcal{N}(\ln o(y) + \mu_{q(y)}^m, (\sigma_{q(y)}^m)^2)$. For each income state, we construct a transition matrix $\mathcal{G}_y(\tilde{b}' | b')$ from the chosen debt level b' to total obligations $\tilde{b}' = b' + o(y)m$. The probability of moving from b' to grid point $\tilde{b}'$ is the probability that the log bill falls between the log-midpoints of the distances from b' to $\tilde{b}'$ and to the two neighboring grid points. Bills that would take obligations beyond the grid are assigned to its largest debt level. Similarly, delinquent obligations follow the haircut transition matrix $\mathcal{H}(\tilde{b}' | \tilde{b})$. With probability $1 - \lambda_\delta$ the debt is unchanged. With probability λ_δ it receives a haircut drawn from the Power distribution with shape α , assigned to grid points using interval midpoints. Nonpositive positions (savings) receive no haircuts.

Step 1 – Solving the Household Problem. Denote by $\mathcal{W}_k(b', y)$ the expected continuation value of a household that enters next period with debt b' , given current income y . Next period's medical bill is drawn given next period's income:

$$\mathcal{W}_k(b', y) = \sum_{y'} \Pi_y(y' | y) \sum_{\tilde{b}'} \mathcal{G}_{y'}(\tilde{b}' | b') V_k(\tilde{b}', y'). \quad (22)$$

The value of delinquency requires no optimization:

$$V_k^d(\tilde{b}, y) = u(y) - \xi_k + \beta_k \sum_{\tilde{b}'} \mathcal{H}(\tilde{b}' | \tilde{b}) \mathcal{W}_k(\tilde{b}', y). \quad (23)$$

We obtain the value of repaying by grid search over b' :

$$V_k^r(\tilde{b}, y) = \max_{b'} u(y - \tilde{b} + q_k(b', y)b') + \beta_k \mathcal{W}_k(b', y), \quad (24)$$

subject to strictly positive consumption. If no feasible choice exists, the household is delinquent. Finally, $d_k(\tilde{b}, y) = \mathbb{1}[V_k^r(\tilde{b}, y) < V_k^d(\tilde{b}, y)]$ and $V_k = \max\{V_k^r, V_k^d\}$.

We exploit two monotonicity properties to reduce the cost of the grid search. First, the borrowing policy $b'_k(\tilde{b}, y)$ is nondecreasing in $\tilde{b}$. We therefore use the divide-and-conquer algorithm of [Gordon and Qiu \(2018\)](#): we solve the problem at the two endpoints of the grid and then at successive midpoints, restricting the search at each point to the interval between the policies of its already-solved neighbors. Second, the delinquency policy is monotone: if a household is delinquent at $\tilde{b}$, it is delinquent at every larger level of obligations with the same income. Once we find a delinquent point, we assign all larger obligations to delinquency without further search.

Step 2 – Updating the Pricing Schedule. Given the updated delinquency policy, the discretized zero-profit condition (7) is

$$q_k(b', y) = \frac{1}{1 + r_f} \sum_{y'} \Pi_{y'}(y' | y) \sum_{\tilde{b}'} \mathcal{G}_{y'}(\tilde{b}' | b') \left[1 - d_k(\tilde{b}', y') + d_k(\tilde{b}', y') \sum_{b''} \mathcal{H}(b'' | \tilde{b}') \frac{b''}{\tilde{b}'} q_k(b'', y') \right] \quad (25)$$

for $b' > 0$, and $q_k(b', y) = 1/(1 + r_f)$ for $b' \leq 0$. The last term is the value of the lender's claim on delinquent debt: after the haircut, each dollar of obligations becomes $b''/\tilde{b}'$ dollars of debt, valued at the price of that debt. Given the delinquency policy, the right-hand side of (25) is a contraction in q_k (Appendix D). We apply it once per iteration of step (1), so that values, policies, and prices converge jointly, and the converged pricing schedule is the fixed point of (25) given the converged delinquency policy. We iterate until the sup-norm distance between consecutive iterations of the repayment and delinquency value functions, their expected continuation values, and the pricing schedule falls below 10^{-7} .

Step 3 – Stationary Distribution and Moments. We compute the stationary distribution Λ_k by iterating on the discretized law of motion (8) until the sup-norm change falls below 10^{-10} . Several moments depend on households' histories: whether a debt was extended by a lender or is an unpaid medical bill (Section 4.1), and delinquency entry and persistence. We therefore compute moments from a long simulated panel, following a household of each type for 101,000 years and discarding the first 1,000. We use the same random draws for every economy we compare (common random numbers), so that differences across economies reflect policies rather than sampling noise.

Households' decisions depend only on total obligations $\tilde{b}$, but in the simulation we track credit card debt and medical debt separately. A new out-of-pocket bill adds to medical debt. A repaying household consolidates all of its obligations into its new credit card balance b' and clears its medical debt. In delinquency, the credit card balance and the medical debt both carry forward and receive the same haircut draw, and any savings are applied against medical debt first. Credit card debt in the model is then defined as in Section 4.1.

F.2 The Medicaid Expansion: Counterfactuals, Decomposition, and Welfare

Post-Expansion Equilibrium. The expansion replaces the out-of-pocket share $o(y)$ by its post-expansion counterpart $o^{ME}(y)$, which changes the bill transition matrices $\mathcal{G}_y$. We compute the post-expansion stationary equilibrium of each type using the algorithm of Appendix F.1. Because the risk-free rate is fixed and there is no aggregate state, the post-expansion pricing schedule and policy functions apply from the moment the expansion takes place. Only the distribution of households evolves over time, which we study in Appendix F.3.

Budget-Neutral Financing. In the budget-neutral expansion, all households pay a uniform income tax τ : repaying households face the budget constraint $c + \tilde{b} \leq (1 - \tau)y + q_k(b', y)b'$, and delinquent households consume $(1 - \tau)y$. The government pays the out-of-pocket bills that households no

longer pay and recovers the reduction in uncompensated medical care. The tax therefore solves

$$\tau \mathbb{E}[y] = \Delta \mathbb{E}[o(y)m] - \Delta \mathbb{E}[L], \quad (26)$$

where Δ denotes the change in stationary expectations between the pre- and post-expansion equilibria (pre minus post), aggregated across types. L is the uncompensated care generated by a new medical bill, $L = d_k(\tilde{b}, y) o(y)m [1 - (1 - \delta) q_k((1 - \delta)\tilde{b}, y)]$, where $\tilde{b}$ denotes obligations including the new bill and δ is the haircut the household receives at the end of the period ($\delta = 0$ if no haircut arrives). A bill generates uncompensated care only if the household is delinquent, in which case the loss is the unpaid bill net of the market value of the resulting medical debt: a share $1 - \delta$ of the obligation survives the haircut and is valued at the price $q_k((1 - \delta)\tilde{b}, y)$.

Decomposition. Denote by B the stationary mean of credit card debt aggregated across types. We decompose the change in credit card debt between the pre-expansion equilibrium, B_0 , and the post-expansion equilibrium, B_1 , using two intermediate economies. Each is solved with the post-expansion out-of-pocket share and, in the budget-neutral expansion, the tax τ :

1. *Direct effect*, $B_D - B_0$. Households' borrowing and delinquency policy functions and the pricing schedule are held at their pre-expansion values. Only the out-of-pocket share, and hence the bill process, changes. We compute the value function of this economy by iterating on the Bellman equations with policies held fixed.
2. *Credit demand channel*, $B_{DQ} - B_D$. Households re-optimize their consumption, borrowing, and delinquency, but face the pre-expansion pricing schedule. We solve this economy by iterating on step (1) with q_k held fixed.
3. *Credit supply channel*, $B_1 - B_{DQ}$. The pricing schedule adjusts to the post-expansion zero-profit condition.

All four economies are simulated with the same random draws.

Welfare. With log utility and an additive delinquency cost, scaling consumption by $1 + g$ in every period raises the value at every state by $\ln(1 + g)/(1 - \beta_k)$. For type k , the consumption-equivalent welfare gain from moving to economy s therefore has the closed form

$$CE_k^s = \exp \{ (1 - \beta_k) [\mathcal{V}_k^s - \mathcal{V}_k^0] \} - 1, \quad \mathcal{V}_k^s = \sum_{b,y} \bar{\Lambda}_k(b, y) \mathcal{W}_k^s(b, y), \quad (27)$$

where $\bar{\Lambda}_k(b, y)$ is the pre-expansion stationary distribution of households at the end of the period: after their delinquency and borrowing decisions and any haircut, and before next period's income and medical expenditure shocks. Because the post-expansion value is evaluated at households' pre-expansion positions, the measure accounts for the transition of the distribution to its new stationary level. The aggregate welfare gain is $\sum_k \omega_k CE_k^s$. We attribute welfare gains to channels using the

value functions of the intermediate economies: the direct effect is CE^D , the credit demand channel is $CE^{DQ} - CE^D$, and the credit supply channel is $CE^1 - CE^{DQ}$.

F.3 Algorithm for Transition Dynamics

The expansion is unanticipated and permanent. It takes place at the start of the first year of the transition, and the economy starts from the pre-expansion stationary distribution.

Baseline: Instant Repricing. In the baseline, lenders reprice credit immediately. Policies and the pricing schedule jump to their post-expansion stationary values in the first year, and the transition consists only of the evolution of the distribution. We compute it by simulation. We take the households of the pre-expansion simulated panel as the initial distribution and follow each of them for ten years under the post-expansion policies and bill process. A counterfactual panel follows the same households, with the same random draws, under the pre-expansion policies. Aggregate paths are population-weighted means across households at each horizon.

Sticky Credit Supply. In the sticky credit supply variant, the pricing schedule follows

$$q_{k,t}(b', y) = \lambda q_{k,t-1}(b', y) + (1 - \lambda) q_{k,t}^f(b', y), \quad q_{k,0} = q_k^0, \quad (28)$$

where $t = 1, 2, \dots$ indexes the years of the transition, q_k^0 is the pre-expansion stationary schedule, and $\lambda = 0.5$. The date- t fair schedule $q_{k,t}^f$ satisfies the zero-profit condition (25) given the delinquency policy of period $t + 1$, $d_{k,t+1}$, with delinquent claims valued at next period's market price, $q_{k,t+1}$. Households have perfect foresight of the price path. Profits and losses that lenders earn from off-fair pricing accrue to outside investors. We assume the economy is in its post-expansion stationary equilibrium after $\bar{T} = 16$ years, when the initial price gap has shrunk by a factor of $\lambda^{\bar{T}} \approx 1.5 \times 10^{-5}$. For each type, the algorithm consists of the following steps.

0. Start with a guess for the price path $\{q_{k,t}\}_{t=1}^{\bar{T}}$, given by the partial-adjustment rule with the post-expansion stationary schedule q_k^1 as target: $q_{k,t} = q_k^1 + \lambda^t(q_k^0 - q_k^1)$.
1. Solve the household problem backward. Starting from the post-expansion stationary value function in period $\bar{T} + 1$, for $t = \bar{T}, \dots, 1$ perform one Bellman update (step (1) of Section F.1) given $q_{k,t}$ and the continuation value of period $t + 1$. Store the policy functions $d_{k,t}$ and $b'_{k,t}$.
2. Compute the fair schedules $q_{k,t}^f$ from (25), using $d_{k,t+1}$ and valuing delinquent claims at $q_{k,t+1}$. In period $\bar{T}$, use the post-expansion stationary objects.
3. Iterate forward to obtain the implied price path, $\hat{q}_{k,t} = \lambda \hat{q}_{k,t-1} + (1 - \lambda) q_{k,t}^f$ with $\hat{q}_{k,0} = q_k^0$. If $\max_t \|\hat{q}_{k,t} - q_{k,t}\|_\infty < 10^{-5}$, stop. Otherwise, update the guess with damping, $q_{k,t} \leftarrow \frac{1}{2} q_{k,t} + \frac{1}{2} \hat{q}_{k,t}$, and repeat from step (1).

The damping prevents cycles in which delinquency decisions at states where households are nearly indifferent flip back and forth across iterations. The algorithm converges within 30 iterations for all types. We then simulate the transition as in the baseline, using the time-varying policies $\{d_{k,t}, b'_{k,t}\}$.